Three Tensor Isomorphism in Single Exponential Time over Finite Fields of Characteristic Two

Abstract

We give a deterministic algorithm for isomorphism of arbitrary three-tensors over a finite field $\mathbb{F}_q$ of characteristic two. For tensors of size $n_1 \times n_2 \times n_3$, the running time is $q^{O(n_1+n_2+n_3)}$. The algorithm returns either an empty transporter or one isomorphism together with generators of the full source automorphism group. It applies to singular and non-concise tensors and to every even field, including $\mathbb{F}_2$.

The algorithm combines contractions and common zero blocks with the structure of the current matrix group. Exact affine subquotients retain the dependencies created by invariant flags and previously matched observations. We develop complete affine algorithms for the resulting degree-two representations, using finite image algebras and a reduction to natural-module affine transport. In the orthogonal case this reduction handles injective modules with nonzero first cohomology. Cubic translations are confined to pairing insertions; cohomology of the quotient then permits a reduction to a literal tensor and one correlated lower-degree problem. A dimension-based analysis bounds the whole recursion tree and preserves complete transporter cosets throughout.

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1 Introduction

1.1 Three-tensor isomorphism

Let q be a prime power, and let $\mathbb{F}_q$ be the field of order q. For positive integers n_1, n_2, n_3, let

$$\mathrm{T}(n_1 \times n_2 \times n_3, q) = \mathbb{F}_q^{n_1 \times n_2 \times n_3}$$

be the space of three-way arrays of this format. An array $\mathbf{A} = (a_{ijk})$ defines the trilinear form

$$T_{\mathbf{A}}(x, y, z) = \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} \sum_{k=1}^{n_3} a_{ijk} x_i y_j z_k, \qquad x \in \mathbb{F}_q^{n_1},\ y \in \mathbb{F}_q^{n_2},\ z \in \mathbb{F}_q^{n_3}.$$

We say that $\mathbf{A}, \mathbf{B} \in \mathrm{T}(n_1 \times n_2 \times n_3, q)$ are *isomorphic* if there are matrices $P \in \mathrm{GL}(n_1, q)$, $Q \in \mathrm{GL}(n_2, q)$, and $S \in \mathrm{GL}(n_3, q)$ such that

$$T_{\mathbf{A}}(Px, Qy, Sz) = T_{\mathbf{B}}(x, y, z) \quad \text{for all } x, y, z. \tag{1}$$

The three-tensor isomorphism problem asks whether such a triple exists. We call $\ell = n_1 + n_2 + n_3$ the *length* of the tensor.

There is a direct algorithm with running time $q^{O(\ell^2)}$: enumerate the three change-of-basis matrices and test (1). The question addressed here is whether the exponent can be made linear in the length, for every input. Our main result gives such a bound in characteristic two.

Theorem 1.1. *Let q be a power of 2. Given $\mathbf{A}, \mathbf{B} \in \mathrm{T}(n_1 \times n_2 \times n_3, q)$ of length $\ell = n_1 + n_2 + n_3$, there is a deterministic algorithm using*

$$q^{O(\ell)}$$

bit operations that decides whether $\mathbf{A}$ and $\mathbf{B}$ are isomorphic. If they are, it returns one isomorphism and generators of the full automorphism group of $\mathbf{A}$. The tensors are given by their entries over the standard explicitly represented field $\mathbb{F}_q$.

We use the standard encoding $q = 2^e$ and $\mathbb{F}_q = \mathbb{F}_2[X]/(f)$, with a supplied monic irreducible polynomial f of degree e; field elements are binary coefficient vectors of degree less than e. The two dense arrays occupy $O(n_1 n_2 n_3 \log q)$ bits, and the field description has polynomial length in $\log q$. Since $n_1 n_2 n_3 \le (\ell/3)^3$, every fixed polynomial factor in the input length and $\log q$ is absorbed by $q^{O(\ell)}$. No additional size parameter or structural certificate is part of the tensor input.

The constant in the exponent is absolute. The theorem includes singular tensors and tensors whose slices are linearly dependent. Corresponding dimensions of the two inputs must agree; the three dimensions within either input need not be equal. Zero-dimensional arguments are immediate, since the tensor is then zero.

1.2 The matrix-space viewpoint

For a field $\mathbb{F}$, write $\mathrm{M}(a \times b, \mathbb{F})$ for the space of $a \times b$ matrices over $\mathbb{F}$, and write $\mathrm{M}(a, q) = \mathrm{M}(a \times a, \mathbb{F}_q)$. The frontal slices of $\mathbf{A}$ form a tuple

$$\mathbf{A} = (A_1, \ldots, A_{n_3}) \in \mathrm{M}(n_1 \times n_2, \mathbb{F}_q)^{n_3}, \qquad (A_k)_{ij} = a_{ijk}.$$

When the slices are independent, they are a basis of a matrix space $\mathcal{A} \le \mathrm{M}(n_1 \times n_2, \mathbb{F}_q)$. Changing the third basis changes the chosen basis of $\mathcal{A}$; changing the first two bases acts by left-right equivalence.

Thus tensor isomorphism can also be viewed as matrix-space equivalence. We keep the tensor formulation because it treats kernels and unequal dimensions uniformly.

The contraction and zero-block arguments below build on the viewpoint developed for alternating matrix-space isomorphism in [IMQSZ24]. A useful step in that approach is to put a low-rank matrix space into a form with a large common zero block. Figure 1 illustrates the corresponding tensor geometry. If the first a frontal slices vanish on a fixed $c \times e$ block, then the remaining information is concentrated in three smaller parts of the array. The transformations on these parts are related: the same change of basis acts wherever an argument is shared.

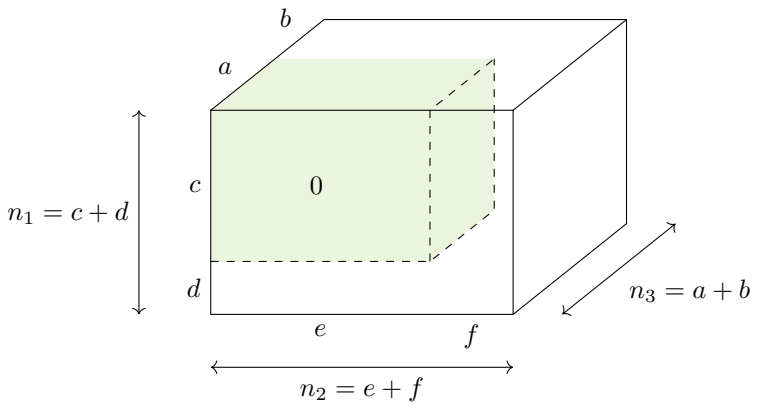


Figure 1: A tensor with a common zero block. The first a frontal slices have a zero $c \times e$ block; the remaining dimensions are $d = n_1 - c$, $f = n_2 - e$, and $b = n_3 - a$. The three changes of basis act on the entire array, so the transformations on the nonzero parts must be shared.

For the bound in Theorem 1.1, this compatibility is central. Solving the smaller parts independently would generally admit transformations that cannot be combined. We therefore compute a complete set of compatible transformations at each step. A nonempty such set is represented by one matrix triple and generators of its source stabilizer. This representation lets later conditions be imposed without listing all solutions.

1.3 Previous algorithms and the scope of the improvement

Tensor isomorphism captures a number of algebraic isomorphism problems. Grochow and Qiao developed the Tensor Isomorphism complexity framework and polynomial-time reductions among tensor, group, polynomial, and algebra isomorphism problems [GQ23]. Their linear-length reductions [GQ25] are especially relevant to exponential-time bounds, since a polynomial increase in length can change the exponent substantially.

In odd characteristic, the previous bound relevant to general three-tensor isomorphism is $q^{\widetilde{O}(\ell^{3/2})}$. More precisely, Ivanyos, Mendoza, Qiao, Sun, and Zhang [IMQSZ24, Theorem 3.2] prove this bound for equivalence of matrix spaces of length ℓ. Here $\widetilde{O}$ suppresses factors polynomial in $\log \ell$. This also gives the same bound for three-tensor isomorphism with dependent slices. Indeed, the slice map $\mathbb{F}_q^{n_3} \to \mathcal{A}$ is a surjection onto the span of the slices. Once the ranks of the two slice maps agree, an equivalence of their image spaces lifts to an invertible map on the third argument by choosing any compatible isomorphism of the kernels. Replacing the slice tuple by a basis of its span decreases the length. Thus the matrix-space result applies to every three-tensor input, after this elementary reduction.

The alternating matrix-space congruence problem has an additional relation between its two matrix arguments: the same matrix acts on both. Sun's algorithm [Sun23] introduced individualization and low-rank matrix-space methods in this setting and yielded an $N^{O((\log N)^{5/6})}$ algorithm for groups of class 2 and exponent p, for odd p, where N is the group order. The work of Ivanyos et al. [IMQSZ24] improved the matrix-space techniques, obtained the $q^{\widetilde{O}(\ell^{3/2})}$ equivalence algorithm, and applied it to groups of Frattini class 2. We distinguish these matrix-space and group statements from the unrestricted three-tensor problem in Theorem 1.1.

Theorem 1.1 gives a linear exponent in characteristic two. The additional ingredient is a recursive use of the structure of the current transformation group. The analysis keeps exact affine fibers and full stabilizers while charging every unresolved choice to a decrease in the original argument dimensions. The matrix-tuple approach based on algebras with involution [IQ19] motivates the simultaneous form step. Appendix A proves the characteristic-two norm and form algorithms needed here, including their radical obstructions. Constructive noncommutative rank [IQS18], finite-algebra algorithms, and primitive linear-group structure [JZP, HMPQ] provide the established algorithmic and structural inputs; their precise uses are given in the appendices.

1.4 Outline of the algorithm

We describe the main ideas before introducing the group and module notation used in their proofs.

Contractions reduce the number of large actions. Fixing a vector in one argument turns a trilinear form into a bilinear form. A family of such contractions can be compared by matrix-algebra methods. The cost of guessing the image of a vector in a d-dimensional argument is at most q^d. This is affordable when the resulting restriction decreases the remaining dimensions by a comparable amount. We use the structure of the current matrix group to prove that a suitable contraction either has this effect or leads to a common zero block.

Large classical groups identify the difficult part. We pass to the direct sum of the composition factors of the current argument representation, while retaining the original filtered action. The kernel on this graded space is the normal p-subgroup, where p is the characteristic, and the quotient acts faithfully and completely reducibly on the graded space. A large primitive linear group contains a quasisimple component acting naturally on a classical or deleted-permutation factor. The remaining factor is retained as multiplicity; the precise structural theorem is recalled in Appendix B. The algorithm treats bounded local actions by evaluation. What remains is a tensor with natural classical actions in some of its arguments. Problems with at most two such actions are completed by a separate module algorithm. Only the case with three large natural actions requires further tensor recursion.

A zero block gives smaller arguments. Suppose a tensor $T : C \times B \times V \to L$ vanishes on $C \times X \times U$, for subspaces $X \leq B$ and $U \leq V$. Its two adjacent parts descend to

$$C \times (B/X) \times U \quad \text{and} \quad C \times X \times (V/U).$$

After these parts have been matched, the last part is an affine problem on $C \times (B/X) \times (V/U)$. The affine translations record exactly the remaining choices of quotient lifts. Section 7 proves that the common zero block obtained from low rank gives a constant-factor decrease in an argument of every unresolved child.

Translations are retained as one subspace. The affine quotient occurring at a later stage can have dependencies between different entries, labels, or tensor summands. We retain these dependencies as one affine space modulo the subspace of changes induced by the transformations that actually remain in the original group. A sequential exact-sequence argument gives the new numerator and denominator after each condition. For a three-large-factor tensor, the translations have a particularly restricted form: they insert an invariant pairing in two arguments and a linear parameter in the third. We classify these insertions and reduce their quotient problem to an ordinary tensor together with one lower-degree parameter problem.

The running-time argument follows the entire recursion. Two combinatorial issues need care. First, permuted direct summands must be matched as a multiset under a common group action. A subset dynamic program does this in $2^{O(b)}$ local work for b components, after the component-pair problems have each been solved once. Second, a restriction of a classical factor may leave the argument dimension unchanged. Such restrictions are kept within the same call until an argument really becomes smaller. Their accumulated branching is then charged to this dimension decrease. A weighted count of the resulting tree proves the single-exponential bound.

Organization. Section 2 introduces the group actions and exact representations used throughout. Section 3 gives the recursive algorithm and its invariants. Sections 4–7 establish its reductions, and Section 8.3 completes the correctness and running-time proof. The appendices prove the algebraic subroutines: structural decomposition of supplied matrix groups, affine transport on matrix modules, the finite-group lifting arguments, and the detailed contraction calculations. Each subroutine returns transformations in the original group, together with its complete kernel.

2 Preliminaries

2.1 Spaces, matrices, and fields

For a positive integer n, put $[n] = \{1, \dots, n\}$. Vectors are columns. The ith standard vector in $\mathbb{F}^n$ is denoted by $\mathbf{b}_i$. If V and W are finite-dimensional spaces over a field $\mathbb{F}$, then $\mathrm{Hom}_{\mathbb{F}}(V, W)$ is the space of $\mathbb{F}$-linear maps from V to W, and $\mathrm{End}_{\mathbb{F}}(V) = \mathrm{Hom}_{\mathbb{F}}(V, V)$. We write $V^* = \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F})$ for the dual space. The subscript on Hom or End is omitted when its field has just been specified. A subspace is written $U \le V$; a quotient of invariant subspaces U/P, with $P \le U \le V$, is an *invariant subquotient*.

We also use $\mathrm{M}(a, \mathbb{F}) = \mathrm{M}(a \times a, \mathbb{F})$ for square matrices. For a matrix space $\mathcal{A} \le \mathrm{Hom}_{\mathbb{F}}(V, W)$ and a subspace $U \le V$, put

$$\mathcal{A}(U) = \mathrm{span}_{\mathbb{F}}\{Au : A \in \mathcal{A},\ u \in U\}, \qquad \mathrm{mrk}(\mathcal{A}) = \max_{A \in \mathcal{A}} \mathrm{rk}(A).$$

A basis of a vector space, subspace, or quotient is always part of its representation. Kernels, images, intersections, and quotient maps are computed by Gaussian elimination. We use reduced column-echelon bases when a basis must depend only on a subspace.

Write $q = 2^e$, where $e \ge 1$, and put $k = \mathbb{F}_2$. We use $p = 2$ in statements where the prime appears as an index. Some fields recovered from a group action need not contain $\mathbb{F}_q$. We therefore perform those steps over k. The original $\mathbb{F}_q$-linearity conditions are retained as equations requiring matrices to commute with the original scalar operators. If K/k is a recovered field extension, its degree is denoted by $f = [K : k]$ when no other subscript is needed. An additive map between K-spaces is

only k-linear unless K-linearity is explicitly stated. In particular, all additive translation spaces are initially k-spaces.

2.2 Group actions and complete sets of solutions

For a group H acting on a set Ω from the left and $x, y \in \Omega$, define

$$\mathrm{Stab}_H(x) = \{h \in H : hx = x\}, \qquad \mathrm{Trans}_H(x, y) = \{h \in H : hx = y\}.$$

If $h_0 x = y$, then $\mathrm{Trans}_H(x, y) = h_0 \,\mathrm{Stab}_H(x)$. Thus an entire transporter can be represented by one element and a generating set for its source stabilizer. We call this a *complete transporter*. The empty set is represented separately.

Why is a complete transporter needed? Suppose two conditions are to be imposed on the same matrix. A witness for the first condition and a witness for the second need not be the same matrix, even when both conditions can hold simultaneously. For example, in $\mathrm{GL}(2, 2)$ the matrices $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ and $\left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right)$ fix $\mathbf{b}_1$ and $\mathbf{b}_2$, respectively, but neither fixes both; the identity does. To combine the conditions, we solve the second one within the full solution set of the first. Every recursive call in this paper consequently returns a complete transporter, together with the maps needed to lift it to the original matrices.

Let $V_i = \mathbb{F}_q^{n_i}$ for $i \in [3]$. For the algorithm we use the left action of $G = \mathrm{GL}(V_1) \times \mathrm{GL}(V_2) \times \mathrm{GL}(V_3)$ on trilinear forms given by

$$(gT)(v_1, v_2, v_3) = T(g_1^{-1} v_1, g_2^{-1} v_2, g_3^{-1} v_3), \quad g = (g_1, g_2, g_3) \in G.$$

A transporter from $T_{\mathbf{A}}$ to $T_{\mathbf{B}}$ gives the matrices in (1) by inversion. This fixes the action and coset conventions used below.

2.3 Restriction of scalars for tensor equality

Let $\mathrm{tr} : \mathbb{F}_q \to k$ be the field trace. Replace a trilinear form $T : V_1 \times V_2 \times V_3 \to \mathbb{F}_q$ by $\mathrm{tr} \circ T$ on the restricted binary argument spaces. Its binary coefficient array has polynomial size in the original dense encoding. This preserves equality under the actual product of $\mathbb{F}_q$-linear groups. Indeed, if the trace of the difference Δ is zero at every triple, then for every $a \in \mathbb{F}_q$ its value at (av_1, v_2, v_3) is $\mathrm{tr}(a\Delta(v_1, v_2, v_3)) = 0$. Nondegeneracy of the field-trace pairing forces $\Delta(v_1, v_2, v_3) = 0$. All original scalar-commutation equations remain imposed, so the output maps are still $\mathbb{F}_q$-linear. No binary change of basis outside the original group is admitted.

2.4 Original dimensions and auxiliary representations

The exponential part of the running time is measured on the original argument spaces. After restriction of scalars, retain the faithful space

$$W_0 = \bigoplus_{i=1}^{3} (V_i \oplus V_i^*), \qquad D_0 = \dim_k W_0 = 2e\ell.$$

The formal duals realize adjoints and contragredient actions; adding a fixed number of further copies changes only absolute constants. Every current subgroup is represented by actual matrices on these original spaces, or by a stated extension of degree $O(D_0)$ with a specified projection back to them. Generator words are stored as straight-line programs: a previously computed product can be reused without expanding its word.

An *argument subquotient* is an invariant subquotient of an original argument, possibly obtained through successive invariant flags. An auxiliary matrix or tensor space is represented by its explicit linear maps; its dimension does not replace D_0 in exponential bounds. In the algebraic subroutines, s denotes the bit length of that subroutine's explicitly supplied data, including any coefficient family or module maps. This is local complexity notation, not an additional parameter or promise in Theorem 1.1. For the tensor algorithm, the original dense data have polynomial size in ℓ and $\log q$, and an auxiliary representation created recursively is not a new original input. Section 8.3 bounds all exported representations uniformly in D_0 and $\log p$.

An ordered family of tensors can be written as a single map $T : C \times B \times V \to L$, where L is its value space. Fixing a basis of L recovers the ordered scalar-valued tensors. If the current group also acts on L, we retain that action in the tensor equation. A value space is *represented on the original arguments* when explicit invariant numerator and denominator spaces realize it inside a tensor product of a fixed number of current argument subquotients, their duals, and scalar spaces. The total degree of the spaces used in this realization is $O(D_0)$. The degree of this realization is the number of its tensor factors, with a dual counting as one factor. A bounded-degree realization has this number bounded by an absolute constant. The requirement states where the action comes from and allows all its kernels to be retained.

2.5 Classical groups and large natural factors

We briefly recall the representation terminology used by the algorithm. A module is *simple* if it has no proper nonzero invariant subspace, and *semisimple* if it is a direct sum of simple modules. An isotypic component is the sum of the simple summands of one isomorphism type. Over a finite field, an isotypic component has the form $A \otimes_K M$, where A is a simple module, $K = \mathrm{End}_H(A)$ is its endomorphism field, and M is the multiplicity space. The action on A and the action on M are kept separately, with their scalar ambiguity and common field automorphism.

The classical groups used here are $\mathrm{GL}_a(K)$ and $\mathrm{SL}_a(K)$, the symplectic group $\mathrm{Sp}_a(K)$ of a nondegenerate alternating form, the orthogonal group $\mathrm{O}_a(K)$ and its derived subgroup $\Omega_a(K)$, and the unitary group $\mathrm{U}_a(K/K_0)$ and its special unitary subgroup $\mathrm{SU}_a(K/K_0)$ for a quadratic extension K/K_0. Their *natural module* is the stated a-dimensional space over K. A Frobenius twist replaces every matrix entry $x \in K$ by x^{p^j} for a specified integer j. A *natural type* records the acting simple factor, this twist, and the natural or dual choice. Consequently the natural and dual modules of one general-linear factor can be distinct types. If the same simple factor acts in several positions, its natural-type identifications are the computed intertwining maps, together with the field and duality maps, expressing those occurrences in common coordinates. They prevent the algorithm from choosing independent transformations on linked arguments.

Fix once an absolute natural-degree cutoff $a_* \geq 64$ exceeding every representation-theoretic and geometric threshold used below. Choose an absolute C_* large enough that the order threshold below forces every selected natural factor to have dimension at least a_*. A quasisimple classical component is called *large* when its simple central quotient has order greater than

$$p^{C_* D_0}. \tag{2}$$

A quasisimple group is a perfect group with simple central quotient. The threshold (2) is fixed for the entire computation. Its choice ensures stable natural dimension and, more importantly, ensures that a later large factor acts trivially on previously recorded bounded actions. Appendix B gives the construction and proves this hereditary property. We count *occurrences* of large natural modules in a tensor: the same factor may act in two or three arguments.

For each large derived classical group, the *associated full classical group* is, respectively, GL, Sp, O, or U on the same natural module. It acts simultaneously on all linked occurrences, with the given twists and dualities. Enlarging to it is used only where a theorem proves the enlargement valid and supplies an exact restriction back to the original group.

A pairing insertion is a tensor obtained by combining an invariant pairing on two arguments with a linear parameter on the third. For example, for finite-dimensional k-spaces A, B, C, an invariant bilinear form $b : A \times B \to k$, and $\lambda \in C^*$, it is the trilinear form $(a, b', c) \mapsto b(a, b')\lambda(c)$. The more general defining-field and twisted versions use the same construction with explicit trace and intertwining maps. Their exact formulas and allowed parameter spaces are proved in Appendix I.

2.6 Affine actions and a choice of origin

Let H act linearly on a k-space M through a homomorphism $\rho : H \to \mathrm{GL}(M)$. An affine action on M has the form

$$h \cdot m = \rho(h)m + c(h), \qquad h \in H, m \in M, \tag{3}$$

where $c : H \to M$ obeys $c(gh) = c(g) + \rho(g)c(h)$. This identity is exactly the group-action law; a map satisfying it is a *cocycle*. If $c(h) = \rho(h)b - b$ for some $b \in M$, then $m \mapsto m + b$ changes (3) to a linear action. Equivalently, the original affine action has the fixed point $-b$. Such a cocycle is a *coboundary*.

A fixed origin need not exist for a general affine action. For example, the additive group of k acts on k by translations $m \mapsto m + a$ for $a \in k$; no point is fixed by every translation. The linear system for a coboundary then has no solution. Our use of cohomology is to prove that the particular classical-module systems arising here do have solutions, after an explicit embedding. In this language, $H^1(H, M) = 0$ means that every cocycle is a coboundary. The proofs in Appendices D and F establish precisely the cases used by the algorithm.

2.7 Flags and elementary group structure

A flag in a vector space W is a chain $0 = W^{(0)} < W^{(1)} < \cdots < W^{(s)} = W$ of subspaces. Its graded space is $\mathrm{gr}(W) = \bigoplus_{i=1}^{s} W^{(i)}/W^{(i-1)}$. An invariant flag gives a homomorphism from its stabilizer to the linear group on $\mathrm{gr}(W)$; the kernel acts trivially on every quotient. For a finite group H, write $O_p(H)$ for its largest normal p-subgroup, H' for its commutator subgroup, and $Z(H)$ for its center. The socle $\mathrm{Soc}(H)$ is generated by its minimal nontrivial normal subgroups. A central product is generated by commuting subgroups whose overlaps lie in their centers.

A supplied linear group is primitive if it is irreducible and preserves no nontrivial direct-sum decomposition by permuting its summands. The structural theorem finds such primitive blocks inside invariant flag quotients. Every algorithmic kernel or quotient is accompanied by its map on supplied generators and by lifts to original matrices.

For a p-group R acting on a k-space M, its augmentation filtration is $M_0 = M$ and $M_{i+1} = \sum_{r \in R}(r - 1)M_i$. It terminates because the characteristic is p. The group R acts trivially on each successive quotient. The lower-degree algorithm uses this filtration to replace the unipotent action by a sequence of exact affine problems.

2.8 Algebras with involution

A finite-dimensional k-algebra E with involution has a k-linear map $a \mapsto a^*$ satisfying $(ab)^* = b^*a^*$ and $(a^*)^* = a$. Its unitary group is

$$\mathrm{U}(E) = \{u \in E^\times : u^*u = 1\}.$$

This notation includes bilinear isometry, symplectic, and unitary groups, as well as a general-linear group represented on a module and its dual. A quadratic orthogonal group in characteristic two is obtained by retaining the explicit quadratic-form equations on the polar-form carrier. The Jacobson radical $\operatorname{rad} E$ is the largest nilpotent ideal of E; the quotient $E/\operatorname{rad} E$ is semisimple. A Levi complement is a subalgebra mapping isomorphically onto this quotient. A genuine E-bimodule has compatible left and right E-actions, and $\mathrm{U}(E)$ acts on it by $v \mapsto uvu^{-1}$. An invariant subquotient for this group action need not itself be an algebra bimodule. The distinction is important in the affine subroutine.

3 The recursive algorithm

The input tensor has three arguments, but a recursive subproblem also carries conditions imposed by earlier steps. We first explain the extra data and then give the algorithm. The elementary principle is to retain the exact set of transformations satisfying all conditions already solved.

3.1 Why quotient problems arise

For this example, let C, B, V be finite-dimensional k-spaces and let $T : C \times B \times V \to k$ be a trilinear form. For a general value space L, the same calculation is applied simultaneously to all its scalar coordinates. Suppose invariant subspaces $X \le B$ and $U \le V$ have been matched and $T(C, X, U) = 0$. Put $x = \dim_k X$, $b_0 = \dim_k(B/X)$, $u = \dim_k U$, and $v_0 = \dim_k(V/U)$. For a fixed $a \in C$, choose complements to write its matrix as

$$M_a = \begin{pmatrix} 0 & A_a \\ B_a & D_a \end{pmatrix}, \quad A_a \in \mathrm{M}(x \times v_0, k), \quad B_a \in \mathrm{M}(b_0 \times u, k), \quad D_a \in \mathrm{M}(b_0 \times v_0, k),$$

Changing the two quotient lifts uses matrices

$$P_s = \begin{pmatrix} I_x & s \\ 0 & I_{b_0} \end{pmatrix}, \quad Q_t = \begin{pmatrix} I_u & t \\ 0 & I_{v_0} \end{pmatrix}, \qquad s \in \mathrm{M}(x \times b_0, k),\ t \in \mathrm{M}(u \times v_0, k).$$

Direct multiplication gives

$$P_s^t M_a Q_t = \begin{pmatrix} 0 & A_a \\ B_a & D_a + s^t A_a + B_a t \end{pmatrix}. \tag{4}$$

Thus the two adjacent blocks are unchanged and the last block changes by a linear correction. The same s and t occur for every $a \in C$, so all these corrections are correlated.

For a concrete example, take $x = b_0 = u = v_0 = 1$ and two coefficient vectors $a_1, a_2 \in C$. Set $A_{a_1} = A_{a_2} = 1$ and $B_{a_1} = B_{a_2} = 0$. Then the pair (D_{a_1}, D_{a_2}) changes by (s, s). Either coordinate can be changed arbitrarily on its own, but the pair can only be changed along the diagonal subspace $\{(s, s) : s \in k\}$. This is why separate quotient calculations for the two coordinates would be incorrect.

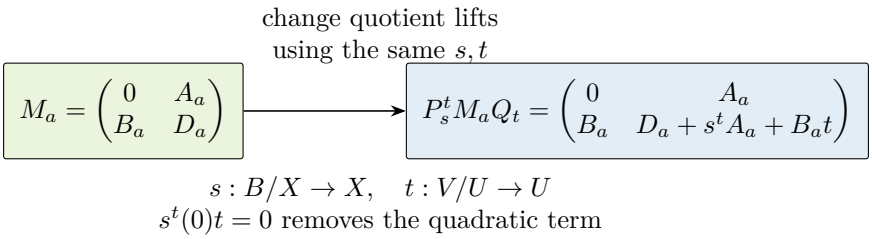


Figure 2: After the two adjacent blocks are matched, the top quotient block is an affine problem. The maps s and t are shared by every coefficient vector $a \in C$. Their admissible image is one subspace of the complete family of blocks.

In a recursive call, the admissible pairs (s, t) can themselves be restricted by earlier conditions. We compute their image from the actual remaining subgroup and retain both its kernel and lifts of an image basis. For its general version, let Z be the current coefficient space, let $t_0 \in Z$ be one lift of the matched coefficients, and let $P \leq M \leq Z$ be k-subspaces. The affine fiber is $t_0 + M/P \subseteq Z/P$: M is the direction of the remaining coefficient choices, and P is the subspace of changes induced by the actual transformations.

3.2 The recursive input and output

Definition 3.1 (Recursive state)**.** A recursive state consists of the following data over $k = \mathbb{F}_p$.

1. A subgroup H acting on the original spaces, or a coset of H, represented by matrices and compressed original generator words. The original field-linearity, forms, and all previously imposed conditions are included.

2. Three active argument subquotients Q_1, Q_2, Q_3, with $d_i = \dim_k Q_i$ for $i \in [3]$, and one simultaneous tensor family on these arguments. Its value space, if it moves, has an explicit bounded-degree realization on the original arguments.

3. An exact affine fiber $t_0 + M/P \subseteq Z/P$, where Z is the current coefficient space, $t_0 \in Z$, and $P \leq M \leq Z$ are invariant k-spaces. In a cubic layer, P is the image of the actual translation homomorphism of the current subgroup. All its correlations, original lifts, and its full kernel are retained.

4. The invariant flags already found, the conditions already matched, and the next coefficient layer to be processed. A later refinement is made inside the existing flag intervals. Every completed tensor family is kept with its complete induced action, including upper-triangular changes between its coefficients.

5. The structural description of H from Theorem B.1: its normal p-subgroup, large natural classical factors, fields, multiplicities, twists, diagonal identifications, and the explicit actions used for evaluation. The threshold (2) remains unchanged.

6. During a sequence of restrictions without a constant-factor dimension decrease, its initial argument dimensions and its current natural-type identifications. The state also records whether the present insertion quotient has already been reduced to a tensor and parameter pair.

The output is empty or a complete transporter in the original group, with all kernels and original lifts. A fresh structural description is constructed for the returned stabilizer before it is used by a

later call.

There are two restrictions on quotient representations in this definition. Coefficients are pulled back injectively when an argument is written as a tensor product over an extension field; the field-balancing equations remain numerator equations. Denominators from a deleted-permutation realization are used inside completed lower-degree calls. An outer cubic denominator always comes from actual transformations on an argument filtration. Sections 4 and I prove that exactly these denominators have the insertion form needed by the algorithm.

The initial state has $H = G$, the three original arguments, $P = 0$, and no matched conditions. A source-target dimension mismatch rejects immediately. A literal zero observation with zero target imposes no new condition. An affine zero observation still imposes that its cocycle vanish and is processed by the exact affine routines.

3.3 The algorithm and its order of operations

The following description is pseudocode for one complete recursive call. Each named subroutine is proved in the indicated section or appendix. A *complete local call* returns before any later step begins; its internal list of choices is never passed on as additional outer branching.

Algorithm 3.2 (Complete tensor transporter).

Input. Source and target states as in Definition 3.1

Output. Their complete transporter in the original group

1. Normalize an incoming coset by its representative. Prepare the current subgroup, refining only retained flags (Appendix B).

2. Process successive argument-filtration layers and compute each exact translation image and its kernel (Section 4). If a current value has at most two large natural occurrences, complete it by Theorem E.1 and continue to the next layer.

3. If an active argument is a group-permuted direct sum, compute every component-pair transporter once in the same parent group. Keep the unique possible pair larger than half the argument within the current restriction sequence; all other pair calls have smaller arguments. Combine the complete answers by Theorem 5.1.

4. If a natural factor has new multiplicity greater than one, enumerate its compatible-frame orbit. In each branch, replace the argument by its natural factor and keep the multiplicity coordinates in one value family (Lemma 5.2).

5. For an unresolved three-large-occurrence summand, choose a smallest argument and apply Proposition 6.2. A proper-subspace restriction continues in the returned subgroup. A field-value or insertion-only case is completed locally. An insertion quotient is replaced by its exact tensor and parameter pair. A high-rank or bounded-tuple restriction retains its chosen vector images. A common zero block is processed by Theorem 7.1.

6. Reapply the component and multiplicity tests after every restriction. Keep all constraints until the whole current tensor condition is solved or some corresponding argument has decreased by the factor $1 - 1/128$. Only the latter case exports an unresolved smaller tensor call.

7. Lift every complete child answer, correct its actual translation error, and restore the entire translation kernel. Join successful complete branches and compute their full source stabilizer.

8. Reconstruct the structural description of the returned original group. Return its complete transporter, with every kernel and original lift.

The tests for permuted components and new multiplicity are repeated after each subgroup restriction. This ensures that a smaller natural factor occurring with several copies is recognized as a genuine dimension decrease. In contrast, a change in the name or rank of a classical group, with no decrease in the argument dimensions, does not start an unrestricted new recursive call.

Three kinds of progress. A call can finish, make a strict restriction without branching, or create smaller tensor calls with a charged number of choices. The contraction analysis distinguishes these outcomes explicitly. A proper invariant subspace gives a strict subgroup restriction. An injective field-value map reduces to a completed lower-degree call. A high-rank contraction, a new multiplicity frame, or a common zero block produces a dimension decrease that pays for the choices it required. Figure 3 summarizes their use.

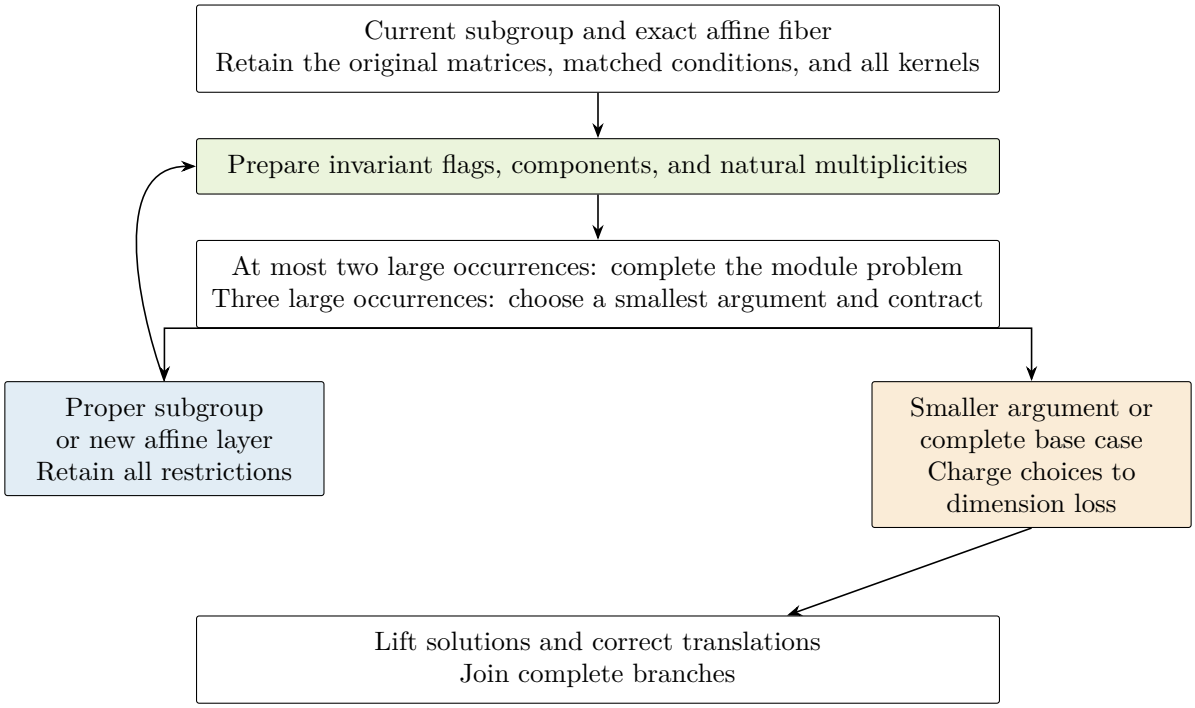


Figure 3: The roles of local restrictions and smaller recursive calls. A list is joined only after all conditions of its branch have been solved; the resulting answer is again one complete transporter.

What remains to prove. Correctness requires that these reductions preserve the complete transporter and that every generated state again satisfies Definition 3.1. Running time requires a bound for the whole recursion tree, since a single-exponential bound for one subroutine would not control repeated branching. The proof of Theorem 8.3 establishes both statements after the reductions have been developed.

4 Exact affine fibers

A common zero block leaves an affine problem because changing a quotient lift adds a contraction of a previously matched block. The allowed changes need not fill a whole matrix space, and different blocks can have correlated changes. This section explains how to preserve those correlations. Its second purpose is to separate the action on the original filtered arguments from the action on their associated graded spaces. In characteristic two these actions need not have a common semisimple subgroup.

4.1 Imposing observations

Let Z be a finite-dimensional k-space, let $P \le U \le Z$ be invariant subspaces for a group H, and let $z : H \to U/P$ be a supplied cocycle. Thus $z(gh) = z(g) + gz(h)$, and $h * x = hx + z(h)$ defines an affine action on U/P. An observation is an equivariant linear map $\epsilon : Z \to E$, where E is another finite-dimensional k-space. The same map is applied to the whole numerator and denominator.

Lemma 4.1 (Sequential affine fibers)**.** *Put $U_0 = U \cap \ker \epsilon$ and $P_0 = P \cap \ker \epsilon$. There is an exact sequence*

$$0 \longrightarrow U_0/P_0 \longrightarrow U/P \longrightarrow \epsilon(U)/\epsilon(P) \longrightarrow 0.$$

After matching the last observation, the remaining problem is an affine point transporter in U_0/P_0 under the complete returned source stabilizer.

Proof. The last map takes $u + P$ to $\epsilon(u) + \epsilon(P)$. If its value is zero, choose $p \in P$ with $\epsilon(p) = \epsilon(u)$. The same point is represented by $u - p \in U_0$. Two such representatives differ by P_0, proving exactness. Suppose a matches the observed source and target. Replace the target by $a^{-1} * y$. Its difference from the source belongs to this kernel. The subgroup stabilizing the observed source preserves its fiber and acts there by the restriction of the original affine action. Hence solving the kernel condition inside that entire subgroup imposes exactly the original equality, and no further condition. □

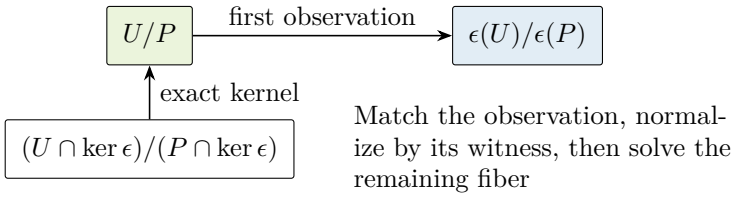


Figure 4: A sequential quotient calculation restricts both the numerator and the denominator. Keeping only the smaller denominator would generally add solutions.

For instance, if $U = \{(a, a, t) : a, t \in k\}$, $P = \{(a, a, 0) : a \in k\}$, and $\epsilon(a, b, c) = a$, then the remaining quotient is the one-dimensional space with coordinate t. Restricting only the denominator would incorrectly leave two dimensions.

4.2 The filtration on a quotient

Let $0 = F_0 \le F_1 \le \cdots \le F_s = Z$ be an invariant filtration of the coefficient space. The filtration induced on U/P is

$$W_i = (P + (U \cap F_i))/P \qquad (0 \le i \le s). \tag{5}$$

Writing $\bar{U}_i = ((U \cap F_i) + F_{i-1})/F_{i-1}$ and $\bar{P}_i = ((P \cap F_i) + F_{i-1})/F_{i-1}$ as subspaces of F_i/F_{i-1} gives $W_i/W_{i-1} \cong \bar{U}_i/\bar{P}_i$. Indeed, the kernel of $U \cap F_i \to W_i/W_{i-1}$ is $(U \cap F_{i-1}) + (P \cap F_i)$.

For tensor coefficients, the filtration is induced by invariant flags in their argument spaces. Order the finitely many tensor cells by a linear extension of the componentwise order. An element acting trivially on all argument grades then acts trivially on every coefficient grade. If the subgroup is later restricted, refine inside the existing argument intervals and coefficient cells. Do not replace this order by a new total-weight order: an already matched cell must remain matched. For a fixed tensor arity r, at most $O_r(D_0^r)$ original cells occur. The new affine actions are computed from the original matrices and the already fixed coefficients.

4.3 Translations and actual quotient groups

Let $R \trianglelefteq H$ be the exact kernel of the action on the argument grades. On one coefficient grade Q, its linear action is trivial. Center the affine action at the source $x \in Q$ and write

$$c_x(h) = h * x - x \in Q, \qquad \eta(r) = c_x(r) \quad (r \in R).$$

The image of η is an actual binary subspace, regardless of any extension-field structures on Q.

Lemma 4.2 (Translation support)**.** *The map $\eta : R \to Q_+$ is a homomorphism to the additive group of Q, and*

$$\eta(hrh^{-1}) = h\eta(r) \qquad (h \in H, r \in R).$$

Every composition factor of $\eta(R)$ is a composition factor of a Hom space between original argument grades. The image, a lift of each image-basis vector, and the full kernel are computable in $2^{O(D_0)}$ time times polynomial explicit linear-algebra work.

Proof. The homomorphism and conjugation identities follow by substituting the cocycle identity and using the trivial linear action of R. Let I be the nilpotent algebra of endomorphisms strictly lowering the original argument flags. Then $R \leq 1 + I$. Set $R_j = R \cap (1 + I^j)$. The map $r \mapsto r - 1$ embeds R_j/R_{j+1} into the additive space I^j/I^{j+1}, equivariantly under H. Conjugation by R is trivial on this quotient. Each matrix block in I^j/I^{j+1} is a Hom space between argument grades, so its factors have at most two natural occurrences. Passing to the images $\eta(R_j)$ proves the assertion. The constructive image and kernel statement is Lemma A.1; it uses the original faithful group action and does not enumerate the image subspace. □

The relevant classical groups are normal subgroups of the actual graded quotient. A constructive lift of a quotient element is an original word, and is not asserted to be a homomorphic section. The following exact lifting procedure is used at every affine grade.

Lemma 4.3 (Lifting an affine transporter)**.** *Let $\pi : H \twoheadrightarrow L$ be an effective quotient with kernel K acting trivially on the linear direction Q. Put $B = c_x(K)$ and $N = \ker(c_x|_K)$. The action on Q/B factors through L. A complete transporter in that action lifts to a complete original transporter in Q by correcting its witness and every stabilizer generator through $c_x|_K$, and adjoining N.*

Proof. Conjugation equivariance makes B invariant. Changing a lift of an element of L changes an affine value by B, so the quotient action is well defined. Put $t = y - x$. If $\bar{a}$ carries $0 + B$ to $t + B$ and a is any original lift, choose $r \in K$ with $c_x(r) = t - c_x(a)$. Since K acts linearly trivially, ra carries 0 to t in source-centered coordinates. For each generator $\bar{s}$ of the quotient source stabilizer, lift it to $s \in H$ and choose $r_s \in K$ with $c_x(r_s) = -c_x(s)$. The corrected element $r_s s$ fixes the source. These elements project onto the entire quotient stabilizer; their kernel is exactly N. They therefore generate the whole original stabilizer together with N. □

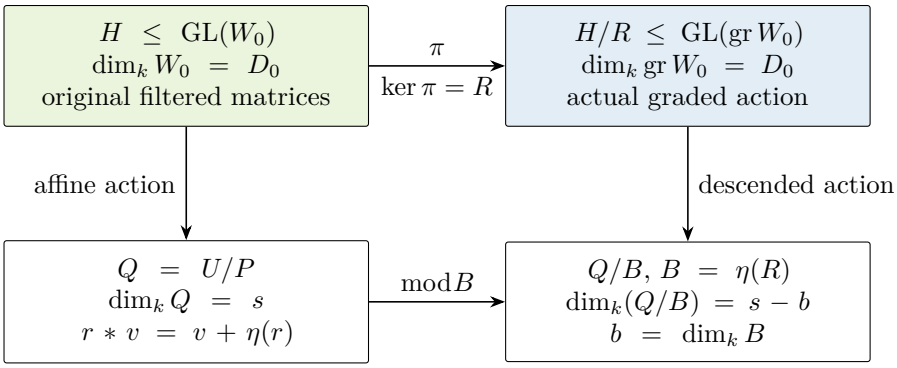


Figure 5: A coefficient grade is solved through the actual graded quotient. The image B has binary dimension $b \le s$; it need not be linear over a larger defining field. Every quotient witness and stabilizer generator is corrected through η, and $\ker \eta$ is restored. No section of π is needed.

4.4 Removing the cubic part of the cocycle

Let M_3 be the sum of coefficient grades with three large natural occurrences. Let $J \le M_3$ be the sum of all pairing insertions: an invariant pairing is placed in two positions and a linear parameter in the remaining position. Every permitted field, dual, and Frobenius signature is included. The construction and the representation-theoretic proofs are in Appendix F.

Proposition 4.4. *The image of the actual translation subspace in M_3 is contained in J. For the current normal product S of large graded classical cores,*

$$(M_3/J)^S = 0, \qquad H^1(S, M_3/J) = 0.$$

These conclusions hold in both orthogonal signs and over every even field. All ranks below the fixed absolute cutoff are processed as bounded factors.

Here $H^1(S, V) = 0$ means that every cocycle $c : S \to V$ has the form $c(s) = (s-1)b$ for some $b \in V$. This is precisely the linear system used below. The proposition is a statement about a submodule of the actual cubic representation. It does not assert that every degree-at-most-two composition factor of a cubic module is an insertion.

Lemma 4.5 (Cubic affine descent)**.** *After a computable affine change of coordinates on the entire current numerator, the cocycle in M_3 is J-valued. The denominator and the transporter in the original group are unchanged.*

Proof. Quotient first by J. Proposition 4.4 kills the translations of the kernel on argument grades, so the affine action factors through the actual graded quotient. Solve $c(s) = (s-1)b$ on generators of its normal core S. The proposition guarantees consistency, and its fixed-vector conclusion gives uniqueness. After translating coordinates by b, the cocycle vanishes on S. If h belongs to the whole graded group, normality and the cocycle identity imply that its new cocycle value is fixed by S. It is therefore zero.

Choose a lift of b in the original coefficient ambient Z/P. This lift need not belong to U/P. Translate the whole affine numerator and both points by that lift. Under the convention $v \mapsto v + b$, the new cocycle is $z(h) + (1-h)b$. This is a conjugacy of affine actions, and retains the original denominator P. The translation is a coordinate change, not an additional allowed group transformation. □

4.5 A literal tensor and one parameter quotient

On the scalar-value case of the contraction analysis, Appendix I constructs a parameter extraction. Let S_0 be the finite-dimensional k-space of insertion parameters modulo the kernel of insertion, and let $J_0 : S_0 \hookrightarrow M_3$ be the injective insertion map. Write $\sigma(t) \in S_0$ for the extracted parameter of a tensor t. The extraction has the identities

$$\sigma(ht) = h\sigma(t), \qquad \sigma(t + J_0 s) = \sigma(t) + s$$

whenever the specified insertion shift is in the allowed scalar-value class. For a linear action modulo a translation subspace $P_3 \leq J$, the exact replacement is

$$(t - J_0\sigma(t),\ \sigma(t) + J_0^{-1}(P_3)). \tag{6}$$

Matching these two entries is equivalent to matching $t + P_3$: subtracting the first equality leaves exactly the second equality inside $J_0^{-1}(P_3)$. Both directions of this argument apply to every group element, so the entire transporter is preserved.

In the actual affine problem the insertion-valued cocycle is retained in the second entry. If M_2 denotes the simultaneous lower-degree coordinates, use

$$F : M_2 \oplus S_0 \longrightarrow M_2 \oplus M_3, \qquad F(v, s) = (v, J_0 s),$$

and the one denominator $F^{-1}(P)$. In coordinate-free terms, with $E' = \ker(Z \to M_3/J)$, the direction of this fiber is

$$[U \cap (P + E')]/P.$$

Thus correlations between high and low coordinates survive. The literal tensor is processed first; the one affine degree-two residual is then processed under its full source stabilizer by Theorem E.1. Finally Lemma 4.3 corrects all outputs through the actual translation map and restores its kernel.

5 Permuted components and multiplicities

An invariant direct-sum decomposition of one argument decomposes the tensor into a family indexed by those summands. The current group may permute the summands. We must therefore match two multisets of component tensors, while using the same transformations on the other arguments and on the value space. This section gives a subset dynamic program for that task. Its important cost property is that each component-pair problem is solved once, before the subset calculation.

A second decomposition, $A \otimes_K M$, has a different role. It describes several isomorphic copies of one natural module. Fixing a compatible basis of the multiplicity space M turns those copies into an ordered tensor family on the smaller argument A. We bound the cost of choosing this basis by the decrease in the argument dimension.

Theorem 5.1 (Fixed-source-prefix multiset transport)**.** *Let Γ act on a set Ω, let $b \geq 1$ be an integer, and let $x_i, y_i \in \Omega$ for $i \in [b]$, allowing repetitions. Suppose every pair transporter $C_{ij} = \mathrm{Trans}_\Gamma(x_i, y_j)$ is supplied as empty or a complete coset, and exact coset intersection is available. The full multiset transporter can be computed using 2^b states, at most $b2^{b-1} - b$ intersections, and $O(b2^b)$ additional group operations.*

Proof. For $J \subseteq [b]$ define

$$D(J) = \mathrm{Trans}_\Gamma(\{x_1, \ldots, x_{|J|}\}, \{y_j : j \in J\}),$$

where braces denote multisets. Set $D(\varnothing) = \Gamma$ and $D(\{j\}) = C_{1j}$. For $k = |J| \geq 2$,

$$D(J) = \bigcup_{j \in J} (C_{kj} \cap D(J \setminus \{j\})). \tag{7}$$

A solution maps the last source occurrence to some target occurrence. Removing that occurrence proves one inclusion; adding it proves the other, including repeated values.

Let H_k stabilize the source prefix multiset and let G_k stabilize x_k. Every nonempty branch in (7) is $g_j K_k$ with $g_j \in \Gamma$, where

$$K_k = G_k \cap H_{k-1}$$

is independent of j and J. Choose a nonempty branch representative $g_0 \in \Gamma$ and let

$$L = \langle K_k,\ g_0^{-1} g_j : \text{nonempty branches} \rangle.$$

Every generator fixes the source prefix, so $L \leq H_k$. Conversely $g_0 h \in D(J)$ for every $h \in H_k$, hence $g_0 h = g_j u$ for some $u \in K_k$, and $h \in L$. Thus $L = H_k$ and the answer is the full coset $g_0 L$. This generation is justified for the entire successful union, not a partial union of unrelated probes.

The number of intersections is at most $\sum_{k=2}^{b} k\binom{b}{k} = b2^{b-1} - b$. Only one branch subgroup list is needed at a state. All nonempty level-k states have the same H_k, so its generators may be reused. □

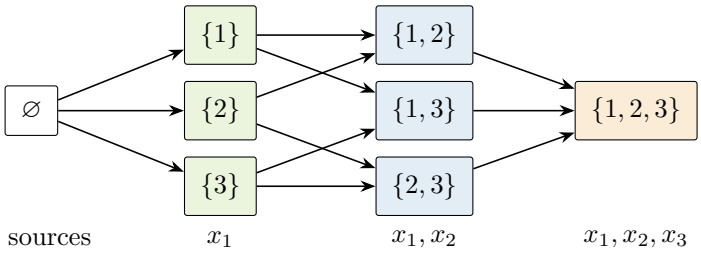


Figure 6: The subset recurrence for three components. At level k, the source is always the fixed prefix $x_1, \ldots, x_k$, and the node records the chosen target subset. An edge uses a previously computed pair transporter and an exact coset intersection.

For example, with two source occurrences x_1, x_2 and two targets y_1, y_2, the last state joins the two complete sets $C_{21} \cap C_{12}$ and $C_{22} \cap C_{11}$. Repeated values cause no difficulty because target occurrences, rather than only distinct target values, are indexed. The common action of Γ is essential to the conclusion that the complete union is a coset.

5.1 Pair calls in the original group

Let I be the finite set of invariant blocks in an active argument and write $Q_1 = \bigoplus_{i \in I} A_i$. Let Γ be the current subgroup permuting these blocks. For $i \in I$, write M_i for the component coefficient space and $t_i \in M_i$ for its value. If $b : \Gamma \to \bigoplus_i M_i$ is the actual affine cocycle, with component $b_i(g)$, write $\rho_i(g) : M_i \to M_{gi}$ for the induced linear map. The action on indexed fibers is

$$g(i, t_i) = (gi, \rho_i(g) t_i + b_{gi}(g)).$$

Normalize an incoming coset once, so both source and target use this same action of Γ. For a pair (i, j), first compute an index transporter r_{ij} and the full index stabilizer Γ_i. Solve the whole fiber equation

$$h(i, t_i) = r_{ij}^{-1}(j, t'_j), \qquad h \in \Gamma_i.$$

The child retains the full ancestor representation with original group words, the two unchanged arguments, all labels, and every invisible evaluation kernel. If its complete result is $h_{ij}S_i$, return $r_{ij}h_{ij}S_i$ in the original Γ.

For example, a global translation space $P = \langle (1,1) \rangle$ has a full one-dimensional image on either coordinate. The two pair calls must still return cosets in that same diagonal group. Replacing them by independent marginal shift groups would permit incompatible translations.

The exact group-coset intersection used by (7) is a consequence of the degree-two subroutine, not a permutation-group primitive. For an integer $d \geq 1$, subgroups $G_1, G_2 \leq \mathrm{GL}_d(k)$, and a matrix $t \in \mathrm{GL}_d(k)$, the product action on $\mathrm{Hom}(k^d, k^d)$ is

$$(g_1, g_2)X = g_1 X g_2^{-1}.$$

The complete transporter of I to t projects onto $G_1 \cap tG_2$. Its source stabilizer is the diagonal copy of $G_1 \cap G_2$. Preparation of the actual product group and the degree-two solver therefore give exact intersection in $p^{O(d)}$; Appendix E establishes the required noncircular dependency order.

There are at most b^2 pair calls, each computed once. Their costs add. The 2^b dynamic-program states use these completed results and add local work; they do not multiply every child cost. There is at most one source block larger than half its argument, and at most one dimension-compatible target block of that size. Hence there is at most one heavy pair. Keep it in-node with the inherited marks and ties. All other pair calls shrink that argument by at least half.

5.2 Fixing a multiplicity basis

A basis and its evaluation-dual basis must be moved together. This preserves the natural–dual identifications and the scalar kernel in the group of tensor factorizations. The next lemma gives the exact orbit to enumerate and explains why it is small enough.

Lemma 5.2 (The cost of fixing a multiplicity basis)**.** *Suppose a large isotypic original row is $A \otimes_K M$, where $K = \mathbb{F}_{p^f}$, $\dim_K A = a$, $\dim_K M = d \geq 2$. Supply the actual natural–multiplicity group of factorizations, its scalar kernel, field and duality records, and original projection words. A compatible-frame orbit has at most p^{fd^2} elements. Within each frame branch the active row becomes A, with all multiplicity coordinates retained in one label family. If the fixed threshold for selecting large factors implies $a \geq cd$ for an absolute $c > 0$, then*

$$fd^2 = O(fa(d-1)),$$

so the branching exponent is paid by the actual row loss. The row at least halves.

Proof. Choose a reference basis e of M and its evaluation-dual basis e^*. Compatible frames are

$$\mathcal{F} = \{(eB, e^*B^{-T}) : B \in \mathrm{GL}_d(K)\}.$$

The formal contragredient is used even if an original dual row carries an additional character; that character stays separate. A factorization-group element acts on frame coordinates by

$$B \longmapsto m\theta(B),$$

where $m \in \mathrm{GL}_d(K)$ is the multiplicity matrix and θ is a field map or a field map followed by inverse transpose. The semidirect multiplication makes this a genuine action and $\theta(I) = I$. It is not a homomorphism obtained by forgetting the outer record.

Breadth-first orbit search and Schreier generators on the orbit of I use at most $|\operatorname{GL}_d(K)| \le p^{fd^2}$ points. The stabilizer has multiplicity matrix part I, while compatible field and exchange records remain. Keep the full central factorization kernel until complete original returns are projected.

Expand M in the fixed frame. Its d coordinates are one entire tensor family on A, with common natural maps, field-balancing equations, old frames, and explicitly realized label action. Conversely these retained factorization equations are exactly the lifting conditions to the original row. The new natural row is realized as $A \otimes e_1$ inside the current grade, with its paired dual component index when an exchange occurs. Its ancestry is therefore still an original invariant subquotient under the frame stabilizer. The new degree is fa, the old degree fad, and their difference is $fa(d-1)$. Since $d/(d-1) \le 2$ and $a \ge cd$, the displayed bound follows. □

Selection of large factors is measured against original degree: $fa^2 > C_* D_0$ and original mass $fad_{\text{total}} = O(D_0)$, where d_{total} is the original total multiplicity of the linked type, imply $a > C'd$, for $d \le d_{\text{total}}$. Only the new local multiplicity is enumerated. Prior frame conditions remain imposed. When $d = 1$, no matrix-frame list is opened; scalar, field, duality and similitude records remain in the current subgroup.

6 Contractions and invariant subspaces

We have reached a tensor summand with one large natural occurrence in each argument. Permuted components and new multiplicities have already been processed. Our next task is to find a condition whose possible images are inexpensive to enumerate and whose stabilizer is substantially simpler. A contraction in a smallest argument provides such a condition unless all the resulting maps have low rank or nearly scalar action on a large subspace.

Write the simultaneous tensor as $T : C \times B \times V \to L$ over k, with $c = \dim_k C \le \min(\dim_k B, \dim_k V)$ and $m = \dim_k L$. A choice of basis of L gives scalar components T_j for $j \in [m]$. For $x \in C$, the full contraction is

$$M_x : B \to V^* \otimes_k L, \qquad M_x(y)(z) = T(x, y, z), \quad y \in B, z \in V.$$

The component maps are denoted by $M_j(x)$. Their simultaneous rank is $\operatorname{rk}(M_x)$; it is not the sum of their separate ranks.

6.1 Different natural types

Let $E \le \operatorname{End}_k(W)$ be the current semisimple algebra with involution for the natural groups, represented on the direct sum W of their natural modules and the formal duals required for adjoints. In this representation each ordinary simple-algebra type is counted once. If $E \cong \prod_i \mathrm{M}(a_i, K_i)$ as an ordinary algebra, its natural degree is $\sum_i [K_i : k] a_i$. This degree measures the remaining independent natural actions. The actual subgroup, field records, and any restrictions on its matrices remain part of the state.

When the two operator arguments belong to distinct ordinary simple components, the contraction occurs in an off-diagonal matrix block. A large-rank map in that block forces a decrease in the natural degree of its centralizer.

Lemma 6.1 (A high-rank contraction)**.** *Suppose the two operator arguments have distinct ordinary simple types. If $x \in C$ has $\rho = \operatorname{rk}(M_x) \ge c/64$, there are at most p^c possible images to test for x. After its simultaneous bilinear contraction has been matched completely, the natural degree decreases by at least $\rho/2$.*

Proof. Let $B_x \le \mathrm{End}_k(W)$ be generated by $\mathrm{End}_E(W)$, all the off-diagonal maps $M_j(x)$, and their adjoints. The projections onto the distinct simple components lie in B_x. Consequently its commutator ideal $I = B_x[B_x, B_x]B_x$ contains every $M_j(x)$ and its adjoint. The sum of the adjoint images has dimension ρ, so $\dim_k IW \ge \rho$.

Write the Krull–Schmidt decomposition of the B_x-module as $W = \bigoplus_i Q_i^{m_i}$, where each Q_i is indecomposable. Let $K_i = \mathrm{End}_{B_x}(Q_i)/\operatorname{rad}\mathrm{End}_{B_x}(Q_i)$, put $f_i = [K_i : k]$, and write $\dim_k Q_i = f_i j_i$. The semisimple quotient of the centralizer has factors $\mathrm{M}(m_i, K_i)$ and natural degree $\sum_i f_i m_i$. If $j_i = 1$, the action on Q_i factors through a field, so I annihilates it. Therefore

$$\rho \le \dim_k IW \le \sum_{j_i \ge 2} f_i j_i m_i \le 2 \sum_i f_i (j_i - 1) m_i.$$

This is twice the decrease in natural degree. The complete bilinear transporter retains the actual current subgroup, its radical actions, and all kernels. Every subsequently selected large factor acts trivially on the old bounded complementary actions, so its remaining representation lies in this centralizer. Further invariant-subspace conditions can only decrease the simple-module degrees. Finally x has at most $|C| = p^c$ possible images. □

If all contractions have rank at most $r < c/64$, Proposition G.2 supplies a canonical pair $X \le B$, $E_0 \le V^* \otimes L$ with $M_x(X) \le E_0$ for every $x \in C$ and $\operatorname{codim}_B X + \dim E_0 \le 4r$. Section 7 turns this pair into smaller tensor problems. Both alternatives use the simultaneous contraction, so neither introduces the number of labels into the exponent.

6.2 The same natural type in both operator arguments

When both operator arguments have the same natural type $A = K^a$, with $[K : k] = f$, the contractions can act like scalar operators on a large subspace. A single contraction need not distinguish this possibility. We compare the algebras generated by tuples of at most three contractions.

For a tuple S of at most three vectors in C, define

$$B_S = \mathrm{alg}_k(K, I, \{M_j(x) : x \in S,\ j \in [m]\}) \le \mathrm{End}_k(A),$$

where alg_k means the unital k-algebra generated by the indicated operators. The copy of K is the distinguished scalar-operator subalgebra. It is not assumed to be central in B_S.

For $a < 64$, the bounded-matrix-size procedure in Appendix C completes the problem locally. Otherwise put $r_K = \lfloor a/32 \rfloor$. A simple centralizer factor of prime-field natural degree greater than $f(a - r_K) > fa/2$ must come from an indecomposable module of dimension one over its residue field. Indeed, two such residue-field dimensions would exceed the total dimension fa. This yields a unique field quotient of B_S acting on an annihilator subspace of K-dimension at least $a - r_K$.

It follows that one of two cases occurs. Either some tuple has every centralizer natural degree at most $f(a - r_K)$, or every tuple has this unique large field quotient. The first case gives a useful restriction: enumerate the images of the at most three vectors and match all their bilinear contractions simultaneously. There are at most $p^{3c} = p^{O(fa)}$ choices, and the resulting natural components of both operator arguments have degree at most $f(a - r_K)$. The constant-fraction decrease pays for these choices. Radicals and multiplicities are retained and processed before recursion continues.

The second case gives a coherent field-valued map. The pair and triple algebras determine compatible embeddings of their field quotients. Appendix H proves that they all embed in one field F/K with

$$[F : K] \le \mathrm{lcm}(1, \dots, a) = \exp(O(a)).$$

The quotient values of the contractions define a k-linear map

$$\lambda : C \to \mathrm{Hom}_k(L^*, F) \cong F \otimes_k L. \tag{8}$$

It is determined up to one common Frobenius automorphism and is equivariant for the actual field and value-space actions. The dual L^* indexes scalar observations of the L-valued tensor. The construction and its arithmetic cost $p^{O(c+fa)}$ times a polynomial in the labels; it does not enumerate the elements of F.

6.3 Using the field-valued map

There are three possibilities for (8). Each has a different algorithmic role.

1. If $0 < \ker \lambda < C$, put $W = \ker \lambda$. This is a proper nonzero invariantly defined k-subspace of the original coefficient argument. Match it by the complete subspace subroutine. The Hom-frame equation in Appendix E gives its complete conductor in the actual group. The current graded natural core acts irreducibly on this coefficient argument, so the nonzero proper kernel gives a strict restriction. Every original kernel and lift is retained, and the new actual subgroup is prepared again. This creates no outer list and requires only binary linearity of the subspace.

2. If λ is injective, list the admissible common field automorphisms. For each one, match the whole map (8) by a degree-one or full Hom-module call, treating the coordinates of F as auxiliary multiplicity. A remaining coefficient transformation g_C fixing all other recorded actions satisfies $\lambda g_C = \lambda$, so $g_C = I$. The hereditary bound on complementary actions excludes a new large factor hidden in those actions. Thus the remaining tensor has at most two large natural occurrences. Complete it before considering the next field automorphism, and join all these complete answers locally.

3. If $\lambda = 0$, every labelled contraction annihilates its common large subspace. The simultaneous rank is at most fr_K. The canonical boundary pair then satisfies

$$\mathrm{codim}_B X + \dim E_0 \le 4fr_K \le fa/8.$$

   This is small relative to the two operator arguments, even when c is much smaller. The geometric reduction in Section 7 applies.

6.4 Contractions modulo pairing insertions

A cubic affine quotient can contain the translations classified by Proposition 4.4. Ordinary contractions then change when the tensor representative changes. Appendix I constructs projected contractions for which this change is a controlled scalar term. The same tuple argument either gives a constant-fraction decrease, a proper invariant subspace, or a nonzero quotient field value. It uses at most six marked vectors, so its exponent is still linear in the corresponding argument degree.

If all quotient values are zero, the scalar extraction from Section 4 gives the exact tensor and parameter pair (6). Apply the ordinary contraction analysis to its literal tensor entry, then match its one affine parameter quotient under the entire returned source stabilizer. This is a change of representation within the current call. Its insertion denominator is processed once.

For one paired occurrence and a larger independent argument, a different construction avoids enumerating vectors in the larger argument. It appends the pairing value to the contraction, making

insertion translations into output shears. The resulting small-boundary pair supplies a proper argument subspace, a proper shear-fixed value subspace, or an insertion-only parameter problem. The full calculation is Proposition I.3; its augmented tensor is used only for these alternatives.

Proposition 6.2 (Complete contraction alternatives)**.** *For a tensor summand with three large natural occurrences, the preceding procedures produce one of the following: a high-rank restriction whose branching is paid by natural-degree loss; a bounded-tuple restriction giving smaller natural arguments; a strict subgroup restriction without outer branching; a completed field-value or insertion-only base case; the exact tensor and parameter pair; or a common-zero-block reduction. A literal zero condition with zero target is complete immediately; an affine zero condition still imposes its cocycle equation.*

Proof. The two operator types are equal or distinct. The distinct-type case is exhausted by Lemma 6.1 and the small-boundary alternative. The equal-type tuple test is exhausted by smaller centralizer factors and the unique field quotient. A linear field-value map has proper nonzero kernel, is injective, or is zero. Proposition 4.4 and Appendix I supply all pairing, duality, twist, and unequal-dimension cases for translation quotients. Every imposed lower-degree or subspace condition is matched completely in the current subgroup. Images of marked vectors remain imposed until the original tensor condition is solved. Hence every genuine transporter occurs in a returned branch, and every branch still retains the exact original condition to be completed. □

7 Smaller tensors from a common zero block

Low simultaneous rank produces subspaces on which every contraction has a small image. We turn this information into invariant subspaces of the two operator arguments. The resulting zero block makes two quotient tensors intrinsic, and makes the last quotient problem affine rather than quadratic in the choice of lifts. This is the geometric step that reduces argument dimensions.

Let C, B, V, L be finite-dimensional k-spaces and let the current tensor be $T : C \times B \times V \to L$. Put $c = \dim_k C$, $b = \dim_k B$, and $v = \dim_k V$. For $x \in C$, its contraction is $M_x : B \to V^* \otimes_k L$. Suppose a canonical boundary pair $X \leq B$, $E \leq V^* \otimes_k L$ has been matched completely and satisfies $M_x(X) \leq E$ for every $x \in C$. The current group still retains the full value-space action and all previously matched conditions.

Theorem 7.1 (The zero-block reduction)**.** *Assume*

$$h = \operatorname{codim}_B X + \dim E \leq \frac{1}{8}\min(b, v). \tag{9}$$

Assume the current graded large natural cores act trivially on the retained label grades, as guaranteed by the hereditary complementary bound. All argument and label filtrations and their actual lifts are retained. Then the full tensor transporter reduces, with shared changes of basis and exact affine lifts, to completed degree-two calls and three-large-factor children for which at least one corresponding natural slot has degree at most h. The other slots do not increase. No condition involving c is needed.

Proof. Set

$$V_0 = \operatorname{im}(E \otimes L^* \longrightarrow V^*), \qquad U = \operatorname{ann}_V(V_0).$$

Evaluation is equivariant for the whole boundary-pair stabilizing parent. Hence X and U are actual invariant original subspaces, and

$$T(C, X, U) = 0. \tag{10}$$

One must not infer $\dim V_0 \le \dim E$. Instead filter the equivariant evaluation map by the retained argument and label flags. On their associated grades, every newly large quotient core is trivial on the label factors. Each composition factor of V_0 carrying that core therefore occurs in E tensored with trivial multiplicity, and has natural degree at most $\dim E \le h$. The same bound holds for the dual factors of V/U. This uses only graded actions and does not split either original filtered module. Also $\dim(B/X) \le h$.

Figure 7 displays the decomposition. The two adjacent parts are intrinsic tensors

$$T_1 : C \times (B/X) \times U \to L, \qquad T_2 : C \times X \times (V/U) \to L.$$

Changing either quotient lift changes its value only by (10). Match T_1 completely and then T_2 inside its full returned parent coset. Afterward the remaining top on $C \times (B/X) \times (V/U)$ is an exact affine fiber. If quotient-lift changes are $s : B/X \to X$ and $t : V/U \to U$, its change is

$$T_2(-, s(-), -) + T_1(-, -, t(-));$$

the prospective quadratic term $T(-, s(-), t(-))$ is zero. Use the current subgroup image and kernel of these shears, rather than all formal Hom parameters. Lemma 4.1 supplies the exact current numerator and denominator at each step.

The first and top arms already have an original argument B/X of degree at most h. For the second arm, prepare the actual V/U row. Its bounded-factor pieces have at most two remaining large positions and are complete local calls. A large isotypic piece is $A_\tau \otimes_{K_\tau} M_\tau$ with $\dim_k A_\tau \le h$. Match its genuine one-axis component family by Theorem 5.1, with all C and X maps shared. Apply Lemma 5.2 immediately if needed. The resulting active natural row is A_τ, not its full isotypic original mass. The pullback along

$$A_\tau \otimes_k M_\tau \twoheadrightarrow A_\tau \otimes_{K_\tau} M_\tau$$

is injective on tensor coefficients, with unmatched-type and field-balancing relations imposed exactly. Thus each unresolved child has the required small row. Its degree is at most one eighth of the corresponding old operator row by (9).

Finally complete the affine top, correct by actual original translation lifts, and restore all invisible kernels. The two arms and top determine the entire original tensor, so the resulting coset is exact. □

7.1 The rank bounds needed by the reduction

The geometric proof compares the boundary with the two operator arguments. It makes no comparison with the coefficient argument C. This allows both rank alternatives from Section 6 to use the same theorem. For an off-diagonal low branch, the smallest coefficient dimension gives maximum stack rank $r < c/64$. The canonical boundary pair of Appendix G has $h \le 4r < c/16$, which implies (9). For a same-row field branch, let both operator rows be K^a, where $[K : k] = f$, and put $r_K = \lfloor a/32 \rfloor$. Zero field value means that every original labelled operator annihilates the same core of codimension at most fr_K over k. Consequently the simultaneous stack has rank at most fr_K, and

$$h \le 4fr_K \le fa/8 = \tfrac{1}{8}\min(b, v),$$

even when $c \ll fa$. Requiring $h \le c/16$ here would not be supplied by the field argument and is unnecessary for either arm.

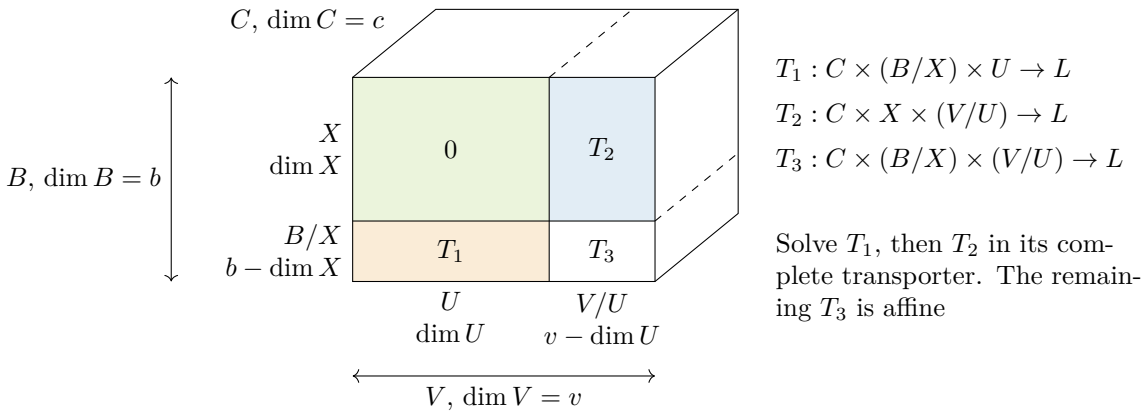


Figure 7: The common zero block and its three remaining parts. Here $c = \dim_k C$, $b = \dim_k B$, and $v = \dim_k V$; all displayed subspace and quotient dimensions are over k. The tensor takes values in the same space L throughout. Quotient-lift changes in T_3 have cross term $T(C, X, U) = 0$. The bound on V/U in the proof is a bound on each large natural constituent; its entire dimension need not be small.

The zero identity (10) is retained while the smaller original tensor summands are formed. A correction $T - J(s)$ with s outside the actual allowed translation-parameter space can refill this zero block. Therefore canonical reduction to a tensor is performed inside a child only after that child's smaller original format has been certified.

7.2 The augmented contraction requires a separate argument

A small-boundary pair for an insertion-covariant augmented contraction family does not imply (10). Its output contains an anchor and may contain a nonzero graph representing a pure insertion. Appendix I proves separate alternatives: a proper original domain, a proper shear-fixed original output subspace, or an insertion-only degree-one terminal. Theorem 7.1 is applied only to an ordinary tensor or a tensor made ordinary by an exact prior correction.

8 Correctness and running time

The preceding reductions each have a single-exponential local cost. To obtain the same bound for the complete algorithm, we must distinguish choices whose costs add from choices whose numbers multiply along a recursive path. Component-pair calls and completed field-automorphism cases add. Images of marked vectors remain unresolved while smaller tensor problems are solved, so their branching must be charged to a decrease in argument dimension.

8.1 Retaining restrictions until a dimension decrease

Fix the dimensions (d_1, d_2, d_3) at the beginning of a call, and put $c_0 = \min_i d_i$. Set

$$\epsilon = 1/128.$$

Continue the current restriction sequence while every corresponding argument has dimension greater than $(1 - \epsilon)d_i$. Components and newly created multiplicities are processed before this test. An

argument that becomes smaller than this threshold gives a *geometric decrease*. The constant ϵ is below the 1/64 loss supplied by the tuple analysis after bounded-size cases are removed, and below the losses from components, multiplicities, and the zero-block reduction.

There are at most six ordinary natural types among the three arguments and their paired duals. We retain their identifications throughout the restriction sequence. This matters because, for example, a natural and a dual general-linear module can become the same orthogonal or symplectic type after restriction.

Lemma 8.1 (Charging a sequence of restrictions)**.** *Suppose a vector or bounded-tuple image list has at most p^{β} choices, with β bounded by a constant times its proved loss of distinct ordinary natural degree. Suppose its cost is also at most $p^{C_0 c_0}$ for an absolute constant C_0. Include the direct argument-dimension charges for tuple decompositions and new multiplicity frames. If a retained sequence ends with dimensions (d_1', d_2', d_3') and a geometric decrease, then*

$$\sum \beta \le C \sum_{i=1}^{3} (d_i - d_i')$$

for an absolute constant C. If it ends by completing the tensor condition, then $\sum \beta = O(d_1 + d_2 + d_3)$.

Proof. Before the final geometric decrease, an old natural type can have at most one constituent of more than $(1 - \epsilon)$ of its original dimension: two would exceed the original argument dimension. Its identifications with other retained types can therefore only merge along this continuation. Among at most six types there are at most five such mergers.

Between mergers, the loss of natural degree counted with paired duals is at most twice the sum of actual argument-dimension losses. The charges consequently telescope. Each merger contributes at most $C_0 c_0$, so their combined contribution is $O(c_0)$.

The final restriction can split an old type; its last vector-image charge is bounded directly by $O(c_0)$. At the geometric decrease, some argument loses at least ϵ of its entry dimension. The total dimension loss is thus at least ϵc_0, which pays all the bounded merger and final charges. A tuple decomposition already loses a fixed fraction of an operator argument. Lemma 5.2 pays each new frame by its actual argument loss. Proper-subspace restrictions have no outer branching, and injective field-value cases are completed locally. If the sequence instead ends in a complete solution, its whole entry dimension pays these same bounded charges. □

For example, restricting $\mathrm{SL}_a(k)$ to a diagonal copy of $\mathrm{SL}_{a/2}(k)$ on two equal summands leaves the isotypic dimension a unchanged. The multiplicity test immediately replaces this with a natural argument of dimension $a/2$ and two value coordinates. The algorithm therefore records a geometric decrease at this point.

8.2 Bounding work between geometric decreases

Every new invariant flag refines old intervals. For a new normal p-subgroup, intersect its augmentation flag with these intervals and retain all old inter-interval shears in the exact translation kernels. Along a path, at most $O(D_0)$ argument intervals can be created, and their induced tensor filtration has polynomially many layers. Previously matched conditions remain matched under the full triangular action on finer coordinates.

A step without geometric decrease or outer branching must do at least one of the following: strictly restrict the current original subgroup, properly refine a retained argument flag, match a previously unfinished condition, or complete the current reduction from an insertion quotient to its

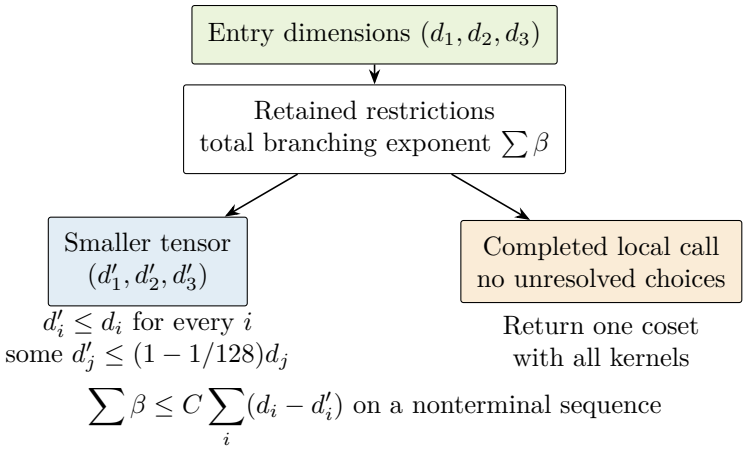


Figure 8: Branching is charged to an actual argument decrease at the end of a retained restriction sequence. At a target-choice node with at most p^β children, assigning weight $p^{-\beta}$ to each edge bounds the total weight by one. This gives a bound for the complete recursion tree.

tensor and parameter pair. The last reduction cannot recur without one of the earlier changes. A strictly descending chain of subgroups of $\mathrm{GL}_{D_0}(p)$ has length at most

$$\log_2 |\mathrm{GL}_{D_0}(p)| = O(D_0^2 \log p).$$

Hence these sequences have polynomial length. Long auxiliary additive filtrations are handled within a completed degree-two call and do not create one outer call per coordinate.

In a component decomposition there is at most one dimension-compatible pair containing more than half the argument on each side. Its continuation is retained in the same call and its integer dimension strictly decreases. All other component-pair calls have a geometric decrease. The retained continuation therefore contributes a polynomial-length sequence with additive side calls.

8.3 Counting the complete recursion tree

The following statement isolates the combinatorial argument. A target-choice node is one that opens a list of possible images which remains imposed on unresolved descendants. An additive side call is a complete subproblem whose cost is added to the current computation.

Theorem 8.2 (Weighted recursion bound)**.** *Suppose a recursive procedure has the following properties.*

1. *A target-choice node has at most p^β children, and the sum of its branching exponents on any root-to-leaf path is $O(D_0)$.*
2. *A geometric decrease reduces one of three corresponding argument dimensions by a fixed factor and increases none. Between such decreases there is a polynomial-length sequence with at most one retained large-component continuation and polynomially many additive side calls.*
3. *Each component-pair call is computed once. Every other local list is completed in $p^{O(D_0)} \operatorname{poly}(D_0, \log p)$ time before it returns.*
4. *The number of labels does not create outer branching, and all polynomial exponents are uniform.*

Then the total running time is $p^{O(D_0)} \operatorname{poly}(D_0, \log p)$.

Proof. At a target-choice node, assign weight $p^{-\beta}$ to every outgoing edge. The sum of these weights is at most one. By induction on a finite rooted tree, the sum of the products of edge weights over any prefix-free set of paths is at most one. Fixing the choices of additive side calls, each terminal target-choice path has weight at least p^{-CD_0} for an absolute constant C. There are therefore at most p^{CD_0} such paths.

Each positive integer argument dimension can decrease by a fixed factor only $O(\log(D_0 + 2))$ times. Contract the intervening polynomial-length sequences. Let $P(D_0, \log p)$ be a fixed polynomial bounding the number of additive side choices at one contracted node. There are at most

$$P(D_0, \log p)^{O(\log(D_0+2))} = p^{O(D_0)}$$

possible sequences of such choices. Here $(\log(D_0 + 2))^2 = O(D_0)$ and $\log(D_0 + 2) \log\log p = O(D_0 \log p)$. One may fix this finite sequence before exposing the target choices; a choice not present on a particular branch produces no leaf. Thus adaptive target lists do not add a further factor.

The polynomial lengths of the contracted sequences bound the internal nodes as well. Multiplication by the complete local cost $p^{O(D_0)}$ changes only the constant in the exponent. The uniform polynomial dependence on the labels contributes one polynomial input factor. □

Lemma 8.1 telescopes across geometric decreases and supplies the first property, including the final completed restriction sequence. The component-pair precomputation and the preceding polynomial event count supply the second and third properties.

8.4 Uniform representation size

For each original argument, rebuild the current factorization using one direct sum of natural spaces and one direct sum of multiplicity or evaluation spaces. More precisely, a sum of pieces $A_\tau \otimes_{K_\tau} M_\tau$ is a quotient of

$$\left(\bigoplus_\tau A_\tau\right) \otimes_k \left(\bigoplus_\tau M_\tau\right)$$

by the relations eliminating unmatched types and imposing K_τ-balancing on matching types. Pullback is injective on tensor coefficients. For three original arguments this requires at most six factor positions. A scalar or homogeneous coordinate and one temporary subspace frame bring the maximum to eight. The total degree of their original-space realizations remains $O(D_0)$.

The representation is rebuilt from the original rows and their current subquotient maps, so an exported child has coefficient and label description length $\mathrm{poly}(D_0, \log p)$. Previously expanded auxiliary spaces are not tensor-multiplied again. The lower-degree algorithm permits any fixed arity r, with cost $p^{O_r(D_0)}$, provided at most two positions have large natural action. Exponentially many coordinates of the glued field occur only inside completed local calls; their polynomial arithmetic is included in those calls and they are not exported as growing recursive value spaces.

For a complete successful union, choose a representative b_0 and generate its source stabilizer from the branch stabilizers and the ratios $b_0^{-1} b_j$ of successful representatives. Retain a generator only if it enlarges the generated subgroup. Each addition at least doubles its order, so $O(D_0^2 \log p)$ generators suffice. Constructive membership uses the original vector action, and all words remain shared straight-line programs. Reapplying the structural theorem to this actual returned group gives the data required by the next call.

8.5 Proof of the main theorem

Theorem 8.3 (Correctness of the recursive construction)**.** *Algorithm 3.2 is exhaustive, returns the complete original transporter, and satisfies the running time in Theorem 1.1.*

Proof. We induct on geometric argument decreases, using the finite subgroup, flag, and completed-condition progression for the intervening work. First we verify the recursive invariants.

At the initial state, the arguments are the original vector spaces, the translation denominator is zero, and the product group has its standard representation. Every completed return is an actual subgroup on these original spaces, so Theorem B.1 applies within the retained flags. The exact quotient filtration (5) produces coefficient layers, and Lemma 4.2 gives their actual translation images without a physical section. Lemma 4.5 translates the whole affine numerator to remove the cubic quotient cocycle, retaining its exact denominator. Proposition 4.4 applies to precisely these cubic denominators. Deleted-permutation denominators remain within completed lower-degree calls. The fixed hereditary threshold ensures that a newly selected large factor acts trivially on saved bounded value-space actions, as required by the field-value and zero-block reductions.

A component-pair problem is solved under the actual index stabilizer with its full original preimage. It therefore retains the shared affine action and all kernels acting on other components. A multiplicity frame uses the genuine group of factorizations, realizes the natural argument as $A \otimes e_1$ inside the current argument quotient, and pulls coefficients back with all field-balancing equations. Both reductions retain the entire ordered value family.

After these reductions, each contraction is a complete supported lower-degree observation under the current subgroup. A proper-kernel restriction is its actual subspace stabilizer, and every chosen vector image remains a condition on unresolved tensor children. Scalar extraction is applied only after the insertion lemmas establish its permitted shift law. The resulting pair retains the joint lower-degree residual and original translation lifts. The augmented one-pair construction is used only for its proved invariant-subspace or insertion-only alternatives. In the zero-block reduction, the zero identity is retained while the two intrinsic quotient tensors and the affine final part are formed. Their smaller-argument certificates and actual shear fibers therefore satisfy Theorem 7.1. Every generated state again has the form in Definition 3.1.

Proposition 6.2 exhausts the contraction cases. Theorem E.1 completes every value with at most two large occurrences. Every other exported child has a geometric decrease, except the unique large component continuation which retains its restrictions within the call. Complete pair transporters, the exact fiber sequence, actual translation correction, and complete-branch union show inductively that the output is the full original transporter with every kernel. Structural preparation of its actual source stabilizer completes the induction.

Finally Lemma 8.1, the polynomial event count, component-pair precomputation, and the uniform representation bound verify all hypotheses of Theorem 8.2. The total cost is $p^{O(D_0)}\,\mathrm{poly}(D_0, \log p)$. Since $D_0 = 2e\ell$ and $q = p^e$, the exponential factor is $q^{O(\ell)}$. The remaining fixed polynomial in D_0 and $\log p$ is absorbed in the same bound, as is the standard input encoding from Section 1. This proves the stated $q^{O(\ell)}$ bit bound. □

A Affine natural modules and simultaneous form equations

The degree-two algorithm will ultimately reduce to two elementary types of problems: affine transport on copies of the original natural modules, and simultaneous equations for matrices and forms. In characteristic two the latter requires care with the radical of an algebra with involution. Its

obstruction is a quotient of symmetric matrices by alternating matrices, whose diagonal coordinates form a Frobenius-twisted natural module. This observation lets us solve the obstruction under the actual current subgroup.

Throughout this appendix, $k = \mathbb{F}_2$. The same arguments over $\mathbb{F}_{2^e}$ are implemented after restriction of scalars. Let W be an original faithful k-space of dimension d; all exponential group operations are measured on W, or on a direct sum of a bounded number of such spaces. The explicit coefficient spaces may have larger polynomial dimension.

A.1 Elementary group and algebra operations

A matrix group $H \le \mathrm{GL}(W)$ acts faithfully on the 2^d vectors of W. The deterministic permutation-group algorithms for membership, point stabilizers, specified quotient kernels, normal closures, normal-subgroup centralizers and constructive lifting therefore cost $2^{O(d)}$ times a polynomial in the supplied data [Seress03, KL90]. We store original words as straight-line programs, so a repeated subword need not be expanded. Simultaneous actions on a bounded number of vector sets still have degree $2^{O(d)}$. No general set stabilizer or general subgroup intersection is assumed here.

For a represented finite-dimensional algebra, we use deterministic algorithms for its radical, semisimple quotient and module isomorphism [CIK97, Ron90]. A polynomial factor in the size of the underlying field is sufficient for our bound. Over k this is constant. When a center field K must be searched, $[K : k]$ is bounded by the degree of its faithful natural representation and hence $|K| \le 2^{O(d)}$. We never enumerate the whole algebra, whose dimension can be quadratic in d.

Lemma A.1 (An additive image and its full kernel)**.** *Let $K \le \mathrm{GL}(W)$ be generated by $g_1, \dots, g_t$, and let $\eta : K \to E_+$ be a supplied homomorphism, where E is a finite-dimensional binary space and values on original words are computable. One can compute a basis of $B = \eta(K)$, lifts $b_1, \dots, b_s \in K$ of that basis, and generators of $\ker \eta$, without enumerating B.*

Proof. The vectors $\eta(g_i)$ span B. Gaussian elimination selects a basis and its lifts. Express each $\eta(g_i)$ as $\sum_j \lambda_{ij}\eta(b_j)$, where $\lambda_{ij} \in k$. Let N be the normal closure in K of

$$b_j^2, \qquad [b_j, b_l], \qquad g_i\Big(\prod_{j=1}^{s} b_j^{\lambda_{ij}}\Big)^{-1}.$$

Every displayed element has zero image. Conversely K/N is generated by the commuting involutions b_jN, so has order at most 2^s. Its map onto B is surjective and $|B| = 2^s$. Thus $N = \ker \eta$. Normal closure and word recovery use the original faithful vector action. The number of defining elements and all coefficient computations are polynomial in the supplied data. □

A.2 Arbitrary affine sections of natural modules

The next lemma does not require any cohomology vanishing. Its important hypothesis is that every simple constituent occurs in a small original module. It accepts arbitrary invariant denominators and arbitrary subgroups.

Theorem A.2 (Affine degree-one transport)**.** *Let $H \le \mathrm{GL}(W)$ be supplied faithfully, let L be a finite-dimensional k-space on which H acts trivially, and let $P \le U \le W \otimes_k L$ be invariant subspaces. Given a cocycle $z : H \to U/P$ and points $x, y \in U/P$, the complete affine transporter is computable in $2^{O(d)}$ times a fixed polynomial in the explicit data. The statement also holds for a subquotient of a direct sum of such modules, including natural duals and Frobenius twists, with d replaced by their total distinct faithful degree.*

Proof. Construct a composition flag of W by enumerating nonzero vectors in each current subquotient and closing their spans under the group generators. A proper invariant subspace, if one exists, contains a nonzero seed that generates a proper submodule. There are at most $2d-1$ recursion nodes, each with at most 2^d seeds and polynomial linear algebra. Let V_i/V_{i-1} be the resulting simple factors.

Induce the flag on U/P by

$$Q_i = (U \cap (P + V_i \otimes L))/P.$$

Each Q_i/Q_{i-1} is a subquotient of a direct sum of copies of the simple module V_i/V_{i-1}; it is therefore semisimple and isotypic. Refine it to a composition flag by solving the generator equations for $\mathrm{Hom}_H(V_i/V_{i-1}, Q_i/Q_{i-1})$. Every nonzero map from the simple source is injective. Repeatedly take its image and pass to the quotient. Thus we obtain

$$0 = W_0 < W_1 < \cdots < W_u = U/P$$

with every quotient W_j/W_{j-1} of binary dimension at most d. This construction uses linear equations even when the multiplicity coordinates in U/P are correlated.

Process the flag downwards. Maintain the coset aK of all solutions modulo W_j, with K stabilizing the source modulo W_j. On the remaining fiber the centered action is

$$g \circ v = g * (x + v) - x \qquad (g \in K, v \in W_j).$$

The orbit of $0 + W_{j-1}$ in W_j/W_{j-1} has at most 2^d points. Breadth-first search finds a word g_0 carrying it to $a^{-1} * y - x + W_{j-1}$, or proves this impossible. Schreier generators give the entire source stabilizer $K' = \mathrm{Stab}_K(0 + W_{j-1})$ for this centered action. Replace a by ag_0, K by K', and continue with W_{j-1}. The source convention means that K' stabilizes the original source modulo the new direction. At the end aK is exactly the desired transporter.

Each step retains every element in the kernel of the quotient action. Generator pruning takes place on the original 2^d-point vector action, not on the larger space U/P. The polynomial number of sequential calls adds their costs. This proves both the time bound and the complete stabilizer assertion. □

A Frobenius twist of a natural K-row is an actual binary representation. Coordinatewise Frobenius is a binary intertwiner, so repeated isomorphic twists do not have to be counted repeatedly in the faithful degree. This observation is useful when a quotient is only binary linear.

A.3 The obstruction in one radical layer

Let A be a semisimple finite-dimensional k-algebra with an involution $a \mapsto a^*$, and let B be an A-bimodule with a compatible involution $s : B \to B$, so

$$s(axb) = b^* s(x) a^* \qquad (a, b \in A, x \in B).$$

The additive quotient

$$\widehat{B} = \ker(1 + s)/\operatorname{im}(1 + s)$$

is well defined because $(1 + s)^2 = 0$. It carries the conjugation action of $\mathrm{U}(A) = \{g \in A^\times : g^* g = 1\}$.

Lemma A.3 (Natural form of the obstruction)**.** *The module $\widehat{B}$ is a direct sum of passive copies of Frobenius-natural and Frobenius-dual modules of the represented simple factors of A. Its nonzero simple constituents have dimensions bounded by the original faithful degree. The decomposition and all maps are constructive over k.*

Proof. Finite fields are separable, so $A \otimes_k A^{\mathrm{op}}$ is semisimple. Decompose B into simple central bimodule types. The involution permutes them. Two exchanged types contribute zero: every fixed pair (x, sx) is $(1+s)(x, 0)$.

A fixed type pairs star-partner algebra factors of equal natural dimension. In compatible coordinates write one factor as $\mathrm{M}(a, K)$, with

$$u^* = H^{-1} u^{\sigma t} H, \qquad H^{\sigma t} = H,$$

where K/k is a finite extension and σ is either the identity or the center involution. A simple central bimodule has action $u \cdot X \cdot v = uX\tau(v)$ for a field automorphism τ of K. Comparing both free matrix arguments in $s(X) = PX^{\mu t}Q$ gives $\mu\tau = \sigma$ and $\mu = \tau\sigma$. Hence $\tau^2 = 1$ and $\mu = \sigma\tau$.

If $\mu \neq 1$, choose $c \in K$ with $c + \mu(c) = 1$. For every fixed x, $(1+s)(cx) = x$, so the quotient is zero. If $\mu = 1$, then $\tau = \sigma$; replacing X by HX makes s ordinary transpose and turns unitary conjugation into natural or dual congruence. Consequently

$$\ker(1+t)/\operatorname{im}(1+t) = \{\text{symmetric matrices}\}/\{\text{alternating matrices}\}.$$

Its diagonal coordinates transform by the entrywise-squared natural or dual action. A passive multiplicity involution has one-dimensional trivial blocks and two-dimensional regular blocks. A regular block again contributes zero, while a trivial block contributes one such natural row. Exchanged algebra factors can have fixed cross-corners; the same transpose calculation applies to those corners. Thus none is discarded merely because the two algebra factors are exchanged.

All decompositions are finite-algebra module calculations and generator intertwiner equations. Alternatively, their output maps can be verified directly against the supplied actions. The k-dimension of a natural K-row is $a[K : k]$, bounded by the degree of a faithful representation containing its algebra factor. □

A.4 Constructing unitary groups and solving norm equations

Let $E \leq \mathrm{End}_k(W)$ be an algebra with involution, supplied by a basis and the matrices of its involution. Put $J = \operatorname{rad} E$. The radical is star-stable, but we do not require a star-stable complement to it. The nilpotence length of J is at most d in a faithful representation.

Theorem A.4 (Characteristic-two norm algorithm)**.** *One can construct generators of* $\mathrm{U}(E)$ *and solve, for an invertible star-symmetric* $a \in E$*, the equation* $c^*ac = 1$ *with its full solution coset in* $2^{O(d)}$ *times polynomial explicit work.*

Proof. First solve the semisimple problem in $\bar{E} = E/J$. Its simple star-orbits are exchanged pairs of matrix algebras or matrix algebras with adjoint involution. Exchanged pairs give a general-linear group. Fixed factors give a nonsingular bilinear or Hermitian form. Formed-space elimination classifies and compares the forms and constructs their full isometry groups. Over a finite field, the one- or two-dimensional scalar equations can be solved by enumerating the actual center field. In characteristic two alternating and nonalternating symmetric bilinear forms are distinguished. A symmetric bilinear form is not replaced by an unspecified quadratic refinement.

To see explicitly that generating these groups does not require enumerating all matrices, choose a nondegenerate one-space when appropriate and otherwise a nondegenerate two-plane. Enumerate its isometric image frames, at most $|K|^{2a}$ for a natural K-space of dimension a. Retain only image frames compatible with the canonical norm-vector flag and with the required complement isometry type, tested by formed-space elimination, and complete each retained frame to an isometric basis.

Every actual isometry supplies such a frame. Recursively generate the pointwise frame stabilizer once on the complement. The transversal and that stabilizer generate the whole group. The costs add over complement dimensions, giving $|K|^{O(a)}$. Nonalternating bilinear forms have their canonical fixed-vector flag, with symplectic quotient; the same elimination on that flag yields the group and its full unipotent kernel.

An ordinary algebra complement to J lifts the semisimple generators. Together with generators of $1 + J$ this constructs

$$H_1 = \{g \in E^\times : g^* g = 1 \pmod{J}\}.$$

The complement is used only to lift units. Define inductively

$$H_i = \{g \in E^\times : g^* g = 1 \pmod{J^i}\}, \quad B_i = J^i/J^{i+1}, \quad n_i(g) = g^* g - 1 \in B_i.$$

For $g, h \in H_i$, multiplication gives

$$n_i(gh) = h^{-1} n_i(g) h + n_i(h), \qquad n_i(1 + x) = x + x^* \quad (x \in J^i). \tag{11}$$

The replacement of h^* by h^{-1} in the first identity is valid modulo J^{i+1} because the error lies in $J^{2i} \subseteq J^{i+1}$. Also $n_i(g)$ is star-fixed. The group $1 + J^i$ has precisely $\mathrm{im}(1 + *)$ as its translation image on this layer. Thus the remaining condition is affine stabilization of zero in

$$\widehat{B}_i = \ker(1 + * : B_i \to B_i)/\,\mathrm{im}(1 + *).$$

The action factors through the actual semisimple image of H_i. Lemma A.3 and Theorem A.2 solve it under H_i itself, even when this image is a proper subgroup. Right actions in (11) are converted to left actions by inversion.

Let K_i be the full preimage of the resulting zero stabilizer. For each generator g of K_i, solve $x + x^* = n_i(g)$ in B_i and correct by an element of $1 + J^i$ on the appropriate side. Adjoin the entire kernel of the additive map $1 + J^i \to \mathrm{im}(1 + *)$ from Lemma A.1. These corrected lifts and the kernel generate exactly H_{i+1}: they surject onto the allowed quotient and contain its full kernel. Repeating until $J^{i+1} = 0$ gives $\mathrm{U}(E)$.

For the norm equation, first obtain a semisimple solution and lift it to a unit $c \in E$. Suppose $c^* a c = 1 + \Delta \pmod{J^{i+1}}$, with $\Delta \in B_i$. Right multiplication by $g \in H_i$ changes its error to

$$g^{-1} \Delta g + n_i(g).$$

Transport the class of Δ to zero in $\widehat{B}_i$, then remove its remaining $\mathrm{im}(1 + *)$ part by an actual $1 + J^i$ correction. If no transporter exists, there is no lift. Otherwise the full solution set at the next precision is the new cH_{i+1}. Hence only one complete coset is carried through the radical filtration. This proves correctness and retains all possible lifts.

The radical filtration, obstruction coordinates and linear equations have polynomial size. There are at most d layers, and every exponential operation uses a faithful degree $O(d)$ representation. In particular neither J nor E is used as a new faithful vector set for group membership. □

A.5 Complete simultaneous bilinear and quadratic transport

The norm algorithm applies to degenerate tuples as well as nondegenerate forms. The following doubled algebra is useful because it never asks for one invertible input matrix.

Theorem A.5 (Simultaneous form transport)**.** *For ordered tuples of bilinear forms and quadratic forms on a total original space of dimension d, complete simultaneous transport under prescribed full block general-linear or classical groups is computable in $2^{O(d)}$ times polynomial explicit work. Baseline forms, field-commutation equations, fixed blocks and cross-block bilinear observations can all be imposed simultaneously.*

Proof. We first treat congruence of bilinear tuples $\mathbf{B} = (B_1, \dots, B_m)$ and $\mathbf{C} = (C_1, \dots, C_m)$ in $\mathrm{M}(d,k)^m$. Set

$$E = \mathrm{M}(d,k) \times \mathrm{M}(d,k), \quad (P,R)^* = (R^t, P^t),$$

faithfully represented by $\mathrm{diag}(P,R)$ in degree $2d$. Its unitary elements are (g, g^{-t}). On $\mathrm{M}(d,k)^m$, use the bimodule actions $(P,R)B_i = PB_i$ and $B_i(P,R) = B_iR$. Unitary conjugation is $B_i \mapsto gB_ig^t$.

For the whole tuple define the star-subalgebra and its right module

$$F = \{a \in E : a\mathbf{B} = \mathbf{B}a, a^*\mathbf{B} = \mathbf{B}a^*\},$$
$$X = \{c \in E : c\mathbf{B} = \mathbf{C}c, c^*\mathbf{C} = \mathbf{B}c^*\}.$$

These are linear equations. If $c \in X$ is invertible, substitution gives $X = cF$. A module-isomorphism algorithm tests $F_F \cong X_F$ and returns a module generator. If a unit exists, every module generator is that unit times an F-unit and is therefore invertible; conversely an invertible returned generator is the required unit. Thus this test and an invertibility check decide whether X contains a unit without searching its vector space.

For a unit $c \in X$, the element $a = c^*c$ lies in F and satisfies $a^* = a$. Theorem A.4 solves $d^*ad = 1$ in F. If d exists, $u = cd$ is unitary and carries the tuple; all solutions are $u\,\mathrm{U}(F)$. Projecting to the first block gives the entire congruence transporter. The proof includes common radicals and zero tuples.

Block restrictions are imposed by retaining the corresponding full matrix subalgebra and its block idempotents in the equations. An original extension field is imposed by its scalar operators and intertwiners. A baseline alternating, quadratic or Hermitian form is simply an additional observation. Distinct argument spaces are placed in separate diagonal blocks; a bilinear map between them occupies its corresponding off-diagonal block. Natural duals and Frobenius twists are represented by their actual binary matrices, so the same simultaneous equations apply.

For quadratic forms, first impose all their polar forms together with the bilinear observations. Normalize a nonempty polar transporter. Under its whole source stabilizer, the difference of two matched-polar quadratic forms has zero polar, and hence is uniquely

$$x \longmapsto \sum_{j=1}^{d} a_j x_j^2.$$

Its coefficient row has a Frobenius-dual affine action. Impose these rows successively by Theorem A.2, always keeping the whole current stabilizer. No enumeration of quadratic refinements is necessary. This proves the claimed algorithm for the whole tuple. □

The preceding theorem does not solve arbitrary-subgroup transport on arbitrary degree-two quotients. Its full block groups and simultaneous literal equations are used first. The exact actual-subgroup reduction in Appendix E is proved separately and only afterwards supplies general coset intersection.

B Decomposing a supplied matrix group

The tensor recursion needs to know which part of its current transformation group can still have a large natural action. This section starts with generators of that subgroup and constructs the required representation data. Its two essential outputs are explicit natural coordinates for the large components and a bound that remains valid under every later restriction of the complementary actions. All original matrices are retained, so the decomposition can be used without discarding extensions between flag quotients.

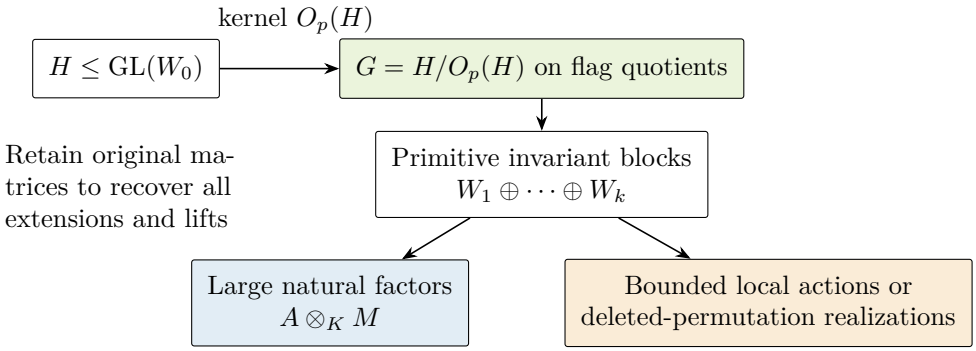


Figure 9: The decomposition of a supplied matrix group. The original flag and its graded action are retained together. All dimensions are charged to $\dim_k W_0 = D_0$, and the criterion for a large factor is fixed before recursion begins.

This section concerns a *supplied* matrix group. It does not search for an unknown tensor stabilizer. All its kernels, components, and quotient maps are computed from the supplied generators. In particular, none of the constructions below uses the group-intersection algorithm that is subsequently obtained from the tensor algorithm.

Write $q = 2^e$ and $p = 2$, and restrict scalars to $k = \mathbb{F}_2$ when necessary. A faithful representation of degree N over $\mathbb{F}_q$ then has degree $D = eN$, and $p^{O(D)} = q^{O(N)}$. Original scalar operators, forms, and other equations are retained as equations on the original matrices. Restriction of scalars does not authorize their removal.

B.1 Statement and external inputs

A *simple section* means a simple quotient of a subgroup; it need not be a composition factor of the entire group. A bounded-action block below means an actual row together with the image of its actual stabilizer, its ancestor projection words, and its component indices. Bounds on different such images are not multiplied unless this is explicitly justified.

Theorem B.1 (Structural decomposition of a supplied group)**.** *Fix the original prime-field degree bound D_0. There are absolute constants c and C_* with the following property. Given generators of an actual group $H \le \mathrm{GL}(V)$, where $\dim_{\mathbb{F}_p} V = D \le D_0$, one can construct, in $p^{cD_0}\mathsf{s}^{O(1)}$ deterministic time, the following certificate.*

1. *An H-invariant original flag, the actual group $R = O_p(H)$, and a faithful completely reducible image $G = H/R$ on its grades, of total degree D. The original flag, matrices, kernels, and lift words remain available.*
2. *A G-permuted direct sum $V_{\mathrm{gr}} = \bigoplus_{i=1}^{k} W_i$, with $\sum_i d_i = D$, $d_i = \dim_{\mathbb{F}_p} W_i$, such that the image*

P_i of G_{W_i} on W_i is irreducible and linearly primitive. The block permutation homomorphism and its full kernel N are supplied.

3. *Actual normal subgroups $\overline{D} \le C \le N \trianglelefteq G$. The group C is a central product of quasisimple components, all diagonal links between their local actions are supplied, and*

$$|N/C| \le p^{cD}.$$

The group $\overline{D}$ is the product of precisely the classical components of C whose simple central quotients have order greater than

$$T = p^{C_* D_0}. \tag{12}$$

Every nontrivial local occurrence of $\overline{D}$ has verified stable-rank natural classical coordinates, with its actual field, multiplicity, duality, and original projection maps.

4. *On a block where $\overline{D}$ is trivial, either its whole local image is retained as a bounded-action block of order at most p^{cD_0}, or there is a deleted-permutation realization with a genuine action of its full block stabilizer. These realizations induce to a genuine G-representation. All numerators, denominators, evaluation kernels, and subquotient maps are supplied. Original rows, grades, and added realizations have total degree $O(D)$.*

5. *The truly small quotient N/C, the block permutation action, the individual bounded block images, and the actions on natural and multiplicity coordinates are retained separately, with their correlations. Each complementary representation has the hereditary property that every nonalternating simple section of each of its subgroups has order at most T.*

The construction respects already retained invariant parent flags and block component indices, by refinement within their grades and induction over their actual orbits. It requires invariance under the current supplied group, not canonicity under an unrestricted ambient change of basis. The same cutoff (12) *is used after every later restriction. For any fixed number of copies of the original rows, the statement holds after enlarging the absolute constants.*

Here are the external inputs, in the exact forms needed.

(a) The constructive permutation and quotient-group library of Kantor–Luks [KL90, §§3–4, P1–P3, P6, P10–P16; §§6–9] computes orders, membership, specified homomorphism kernels, normal-subgroup centralizers, series, simple-factor names, prime cores, and socles, with straight-line words, in polynomial permutation degree. We use point stabilizers and pointwise kernels, never an arbitrary set stabilizer or an arbitrary intersection of two supplied permutation groups.

(b) There is an absolute c_0 such that a primitive irreducible $P \le \mathrm{GL}_d(p)$ of order greater than $p^{c_0 d}$ has a normal quasisimple subgroup L with $|P/L| \le p^{5d}$. On an appropriate extension-field tensor decomposition, L acts on one factor as a natural classical group over its defining subfield, or as an alternating group on its deleted natural module. This is [JZP, Proposition 5.7]; the prime-field tensor statement and its completely reducible formulation are [HMPQ, Theorems 3.1–3.3]. The tensor field is the field specified in those theorems, not an arbitrarily chosen older field. For an alternating factor the correct degree statement is $m \le d + 2$, since deletion can remove two dimensions.

(c) For $A \le \mathrm{Sym}(t)$, the product of the orders of its composition factors other than alternating groups of degree greater than 4 is at most $24^{(t-1)/3} < 3^t$ [GMP, Proposition 6.1, with parameter 4]. We apply this assertion to subgroups as well as to A itself.

The natural-module content in (b) is essential: abstract simple-factor names alone would not give the certificate. The representation gap may also be read through the base-size theorem [LL, Theorem 1]; it is not used as a constructive recognition oracle here. All further constructive reductions are given below.

B.2 Original flags and the exact prime kernel

We first separate the unipotent part of the supplied action. A composition flag is enough: its diagonal quotients are irreducible, and the kernel on their direct sum is exactly the normal p-subgroup. Enumerating cyclic vector spans finds this flag within the original-degree bound, while retaining every off-diagonal extension in the original matrices.

Lemma B.2. *An invariant composition flag of a supplied matrix group on a d-dimensional space over $\mathbb{F}_p$ can be found in $p^{O(d)}$ time without enumerating subspaces. The kernel on its grades is exactly its normal p-core.*

Proof. Enumerate nonzero vectors v. Close $\langle v\rangle$ under the supplied matrices by linear-span operations; equivalently, enumerate its vector orbit and take its span. The result $\mathbb{F}_pHv$ is an invariant subspace. If the module has a proper nonzero submodule, a seed in that submodule produces a proper span. Thus exhaustive seeds either find a proper submodule or certify irreducibility. Recurse on the submodule and quotient, retaining bases and quotient maps. The sum of dimensions at each split is unchanged, and there are at most $2d-1$ recursion nodes. At a node there are at most p^d seeds and each closure costs polynomial work in p^d and the supplied input. This proves the claimed bound.

Let $\gamma : H \longrightarrow \mathrm{GL}(V_{\mathrm{gr}})$ be the diagonal action on the resulting grades. Its kernel is block unitriangular and hence a normal p-group. Conversely a normal p-subgroup $A \trianglelefteq H$ fixes a nonzero vector on every nonzero characteristic-p module on which it acts: for example this follows by induction through a central subgroup of order p. On an irreducible H-module its fixed space is H-invariant, so it is the whole module. Hence every normal p-subgroup lies in $\ker\gamma$, proving $\ker\gamma = O_p(H)$.

For the computational assertion, represent H simultaneously on the vector sets of V and V_{gr}. Their disjoint union has at most $2p^d$ points. The pointwise kernel of the graded part has generators whose same words are evaluated on the original matrices. It is exactly $\ker\gamma$, not merely an abstract isomorphic group. The diagonal image is faithful and completely reducible by construction. Refining a supplied invariant flag uses the same procedure inside each of its grades. □

B.3 Primitive blocks from orbit matroids

An irreducible module may still split as a direct sum whose summands are permuted by the group. To find such a decomposition, we examine vector orbits and their minimal linear dependencies. This uses only $p^{O(d)}$ vectors in dimension d, in place of enumerating all subspaces.

Lemma B.3. *For an irreducible supplied group $A \le \mathrm{GL}(U)$, linear imprimitivity can be detected and a block system found in $p^{O(\dim U)}$ time. Refinement gives primitive block stabilizers with actual transversal words.*

Proof. For each nonzero u, form $\Omega = u^A$. It spans U. Regard its distinct vectors as the elements of a represented vector matroid. Choose a vector basis $B \subseteq \Omega$. For each $x \in \Omega \setminus B$, join all the elements in its fundamental circuit, as read from the expansion of x in B. The connected components $\Omega_1, \ldots, \Omega_s$ of the resulting graph are the matroid components. To see this directly, expansions show

that the spans belonging to distinct components use disjoint subsets of B, so their sum is direct; conversely each fundamental circuit must lie in one direct-sum component. Thus

$$U = \bigoplus_{j=1}^{s} \operatorname{span}(\Omega_j).$$

An automorphism permuting Ω preserves its circuits and hence this decomposition. The action on the summands is transitive because the action on Ω is transitive.

Suppose a genuine nontrivial block system $U = \bigoplus_{j=1}^{t} U_j$ exists. Take a seed $u \in U_1 \setminus 0$. Its orbit lies in $\bigcup_j U_j$. A minimal dependence between orbit vectors cannot meet two distinct U_j: projection onto one summand would give a strictly smaller nonempty dependence. Thus every matroid component lies in one true block, and this seed gives a nontrivial decomposition. Conversely any nontrivial matroid decomposition found above is valid. Exhaustive seeds therefore certify primitivity if none succeeds.

The stabilizer of a found block is a point stabilizer in the explicit action on at most $\dim U$ blocks. If its action on that block had a proper nonzero invariant subspace, inducing that subspace by actual block transversals would give a proper nonzero A-submodule of U. Its action is therefore irreducible. Apply the same test to it and induce every refinement to its block orbit. There are at most $\dim U$ final blocks and only linearly many nodes in the dimension-additive refinement tree. Orbits and circuit graphs have size at most $p^{\dim U}$; reduce generator lists by stabilizer chains at each step. The total cost is $p^{O(\dim U)}$. □

Apply the lemma to each graded irreducible summand. Compute the action on $\{W_1, \ldots, W_k\}$ directly from the matrices, and let N be its pointwise kernel. The restriction maps embed N in $\prod_i P_i$. The chosen blocks are G-invariant as a family; a known incoming coset representative transports the entire chosen family to the target. No independently chosen target decomposition has to be guessed. Existing block component indices are handled by starting the refinement inside their stabilizers and inducing it over their actual orbits; invariant flags are refined inside their grades. Such data must be actual current-group invariant data, as stipulated in the input, rather than incompatible decompositions from discarded branches.

B.4 Extracting the actual layer

The primitive-group theorem describes a large quasisimple subgroup, but the algorithm needs its actual matrices and projection maps. We recover all quasisimple components from the Fitting subgroup and the socle of a specified quotient. The calculation below uses only normal-subgroup operations on the supplied group.

For a finite group A, write $F(A)$ for its Fitting subgroup and $E(A)$ for the product of its subnormal quasisimple components.

Lemma B.4 (Layer extraction)**.** *The actual components of a supplied matrix group of degree at most D, with their centers, simple quotients, and original words, can be computed in $p^{O(D)}$ time using the quotient-group library above.*

Proof. Compute $F = F(A)$ as the product of the prime cores, then

$$B = C_A(F), \qquad Z = Z(F).$$

The group $F(B)$ is characteristic in $B \trianglelefteq A$, and is normal nilpotent in A. Consequently $F(B) \leq F \cap B = Z$; the reverse inclusion holds because Z is central in B. A nilpotent normal subgroup of

B/Z would lift to a nilpotent normal subgroup of B, since the kernel Z is central. It follows that $F(B/Z) = 1$. Its solvable radical is also trivial: a nontrivial solvable normal subgroup would have a nontrivial characteristic Fitting subgroup.

Compute $\mathrm{Soc}(B/Z)$. It is a direct product of nonabelian simple groups. Its individual simple factors can be obtained by a minimal-normal-subgroup computation *inside this direct product*, followed by its centralizer there, and iteration. All these are normal-subgroup operations in specified quotients. For one factor S_j let A_j be its full preimage in B and put $Q_j = A_j'$. Since A_j/Z is perfect,

$$A_j = A_j' Z, \qquad A_j' = [A_j' Z, A_j' Z] = [A_j', A_j'].$$

Thus Q_j is perfect, with simple central quotient S_j, and is subnormal in A. Distinct such groups commute: their commutators are central and a central commutator map out of a perfect group is trivial. Conversely every component centralizes $F(A)$, and its image in the radical-free group B/Z is a simple factor of its socle. Hence these are exactly the components.

Use the faithful action on at most p^D vectors for the quotient library. There is no need to factor a generic integer of magnitude p^{D^2} by trial division. All prime divisors of $|A|$ divide $p \prod_{j=1}^{D} (p^j - 1)$; these D integers each have magnitude at most p^D. Factoring them by exhaustive division, or a sieve of this size, is already within $p^{O(D)}$. The number of group operations is polynomial in D and $\log p$, apart from the permitted polynomial dependence on permutation degree. Every subgroup and quotient retains compressed original words. □

B.5 Natural coordinates for the dominant local component

Once a dominant component is known as an actual subgroup, its representation must be identified. The key point is homogeneity: on a primitive block, all its simple constituents are isomorphic. The endomorphism field and the evaluation map then provide the natural and multiplicity coordinates without guessing a tensor-product basis.

Lemma B.5 (Homogeneous original coordinates)**.** *Let P act irreducibly and primitively on W, and let $L \trianglelefteq P$. The restriction $W|_L$ is semisimple and homogeneous. If L is quasisimple, its faithful irreducible constituent X, the actual field $K = \mathrm{End}_L(X)$, and the evaluation isomorphism*

$$X \otimes_K M \longrightarrow W, \qquad M = \mathrm{Hom}_L(X, W), \qquad x \otimes f \longmapsto f(x) \tag{13}$$

can be found in $p^{O(\dim W)}$ time.

Proof. Let $A = \mathbb{F}_p[L] \subseteq \mathrm{End}_{\mathbb{F}_p}(W)$. Its radical J is invariant under conjugation by P, so JW is a P-submodule. Irreducibility and nilpotence force $JW = 0$, hence $J = 0$ because A is represented faithfully. The isotypic summands for A are permuted by P; primitivity forces there to be just one. Find a simple submodule X by the vector-span procedure, descending until irreducible. Compute its commutant by linear equations. Schur's lemma and the finite-division-ring theorem identify this commutant as a field K. It is an actual algebra of operators, so arithmetic, a K-basis of X, and a K-basis of M can be computed and verified inside the representation. Homogeneity and semisimplicity prove that (13) is an isomorphism; its rank is checked as part of the certificate. The kernel on X equals the kernel on every isomorphic copy and hence the kernel on W. Thus X is faithful.

The commutant field has at most $p^{\dim X}$ elements. Even constructing its entire addition and multiplication tables or finding a field generator by exhaustive enumeration is permissible. All Hom spaces and represented algebras are obtained by polynomial-size linear equations and matrix-span

closure, not by enumeration of their elements. Finally the center of a quasisimple L acts faithfully by scalars on X, giving

$$|Z(L)| \le |K| - 1 < p^{\dim W}. \tag{14}$$

□

Enlarge c_0 so that $c_0 > 12$ and the structural input applies. A block with $|P_i| \le p^{c_0 d_i}$ is declared locally bounded. Otherwise extract all components of P_i by Lemma B.4. The structural component L satisfies

$$|L/Z(L)| > p^{(c_0-6)d_i} > p^{5d_i}.$$

Every other component Q commutes with L and intersects it centrally. Its simple central quotient is therefore a section of P_i/L, of order at most p^{5d_i}. There is exactly one component above this last bound; call it L_i. This criterion is intrinsic. In particular $L_i \trianglelefteq P_i$, and conjugation carrying W_i to W_j carries L_i to L_j. One may construct it on one block of an orbit and transport it. Its normality makes the resulting subgroup independent of the chosen block transversal.

Apply Lemma B.5 to L_i. The field K_i thereby recovered is the field of its irreducible original representation. In the classical case its constituent is the natural module over the actual matrix-entry defining field; any extension field and tensor factor in the existence theorem have become multiplicity in (13). For the alternating case the constituent is the deleted module over $\mathbb{F}_p$, and the entire remaining scalar and tensor action belongs to M_i. This removes any need to search for a maximal semilinear field or for an unspecified tensor basis.

Verifying classical natural type. Work on X over its computed K, of degree a. In these coordinates test the candidate groups

$$\mathrm{SL}_a(K), \qquad \mathrm{Sp}_a(K,B), \qquad \Omega_a(K,Q), \qquad \mathrm{SU}_a(K/K_0,h).$$

Compute invariant bilinear forms by the equations $g^t B g = B$, and, when K has an involution σ, invariant Hermitian forms by $g^{\sigma t} B g = B$, $B^{\sigma t} = B$. These are linear equations over K or its fixed field. Irreducibility makes every nonzero invariant form nondegenerate; verify rank and the required symmetry explicitly. For the quadratic candidate solve directly for homogeneous quadratic polynomials $Q(gx) = Q(x)$ in the $a(a+1)/2$ coefficients. Polarization is injective on this invariant space: a zero-polar quadratic is λ^2 for a linear functional $\lambda \in X^*$, and its invariance would make λ invariant on the nontrivial irreducible module. Thus the invariant quadratic, if present, is unique up to scalar. Check its polar rank and its defining equations. An odd-dimensional quadratic natural representation has an invariant polar radical and is handled through the argument flag, with its symplectic quotient; it is not an irreducible quadratic candidate here. In the unitary case retain the actual involution and its fixed field.

The candidate standard groups can be generated deterministically within $p^{O(d_i)}$. For SL use elementary matrices. For the isometry groups, construct a standard orthogonal, symplectic, or Hermitian basis by elimination. For Hermitian or nonalternating bilinear forms one may use a one-space with nondegenerate restricted bilinear form; for symplectic and quadratic orthogonal forms always use a nondegenerate polar two-plane. Enumerate its possible image frames, and complete each image to an isometric basis. Generate the pointwise frame stabilizer recursively once on the complement, not once per image frame. There are at most $|K|^{2a}$ choices at any stage. Basis completion is ordinary form elimination, with field enumeration for norm equations if needed. The resulting transversals, together with the recursively generated pointwise frame stabilizer, generate the full isometry group: every isometry first has its initial frame matched by a transversal and

then fixes that frame. The symplectic group is already the required group. In stable dimension the derived groups of the orthogonal and full unitary groups are respectively Ω and SU, so the quotient library computes the required derived generators. Equivalently, in the orthogonal case one can use all orthogonal symmetries in nonsingular vectors and then derive their group. These constructions have degree at most $|K|^a \leq p^{d_i}$ and never enumerate the whole classical group.

For each candidate, test its generators in the actual image of L_i by constructive membership and compare exact orders. Inclusion and equal order prove equality. The structural theorem guarantees that one of the natural candidates succeeds. Failure of a representation check is not an isomorphism rejection: it would contradict the promised structural premise and must not be used to discard a tensor branch. Small exceptional ranks can be tested by these same group computations and are not retained as large components.

Repeated constituents and complementary maps. Compute the actual algebra $\mathbb{F}_p[\overline{D}]$ on the *whole* graded original representation, after $\overline{D}$ has been selected below. It is semisimple: apply the radical argument separately to every irreducible G-summand, using normality of $\overline{D}$. Its simple modules, commutant fields, isotypic projectors, and evaluation maps give all repeated occurrences. Duality is tested by the actual Hom equations against the contragredient matrices. When comparing coordinates over isomorphic finite fields, enumerate their field isomorphisms and solve the corresponding intertwiner equations; this retains the Frobenius record rather than assuming that a merely additive map is linear over an older field. The actual maps from a global component to its local groups are retained even when two presentations use different natural coordinates.

Completeness of these linked-occurrence tests uses the classical automorphism theorem, in its explicit inner–diagonal–field–graph form [Steinberg60, §3, statements 3.2–3.6]. In the selected stable ranks, its natural projective actions differ only by a field map, a change of basis, and the natural/dual choice; the exceptional orthogonal triality case has already been excluded by the permanent cutoff. There is no additional central ambiguity in the actual intertwiners. Indeed two representations of the same perfect component whose projective matrices have been made equal differ by a scalar homomorphism from that component to the multiplicative field group. Such a homomorphism is trivial. Thus the finite field/dual tests and linear intertwiner equations recover all its linked natural occurrences, with their actual central kernels. For a self-dual occurrence, the computed invariant form supplies the adjoint; for a non-self-dual pair, evaluation gives the exchange adjoint. The associated full classical group consequently acts on every occurrence by the recovered natural, dual, and field-twisted matrices, with the same diagonal links. It does not act independently on two linked rows.

For clarity, these computations also give a concrete normalizer *group of factorizations*, without requiring chosen factorizations to multiply. On one homogeneous row $X \otimes_K M$, a normalizer element induces a field automorphism σ of the center of $\mathrm{End}_K(X) \otimes 1$. After removing σ, its induced automorphism of $\mathrm{M}_a(K)$ is inner. To compute an implementing matrix u, solve

$$uA_j = \alpha(A_j)u$$

for a basis A_j of this matrix algebra. A nonzero solution is invertible by simplicity of its natural module. Removing $u\sigma$ from the original map leaves $1 \otimes v$, which is read by linear equations. Thus the map has the form $u \otimes v$ with their common field automorphism. Its factorization ambiguity is exactly

$$(u, v) \sim (cu, c^{-1}v), \qquad c \in K^\times. \tag{15}$$

Retain the group of *all* such factorizations, with its scalar kernel, rather than choosing a spurious homomorphic section. On different isotypic rows the same construction is performed with the actual

permutation of the simple algebra factors. Formal dual rows carry inverse-transpose maps. All determinant, form, field, scalar, and ancestor-group conditions remain conditions in this cover. For the paired star realization use a row and its formal dual, retain the induced adjoint, and fix the augmentation on an unrelated scalar coordinate to one. This is the normalizer data used by the subsequent transfer and graded-quotient arguments; it is not an enlargement of the exact returned group.

This cover itself has an original-degree implementation. For every supplied parent generator choose one computed factorization; add generators of all scalar kernels in (15). The resulting group is the full inverse image of the parent: any other factorization differs from a lift word by one of those kernels. Represent it on the original row and on the direct sums of its natural and multiplicity spaces. If their K-degrees are $a, b \geq 1$, then $a + b \leq 2ab$, so the added prime-field faithful degree is at most twice the row degree. The action on these factor spaces, together with the original ancestor matrix, is faithful; field automorphisms act on them as actual prime-field linear maps. Thus its original projection and every scalar kernel are specified homomorphisms on an $O(D)$-degree representation. A pointwise kernel or a full lift is again a permitted original-vector permutation computation. This is also a constructive implementation of the cover used to act on compatible natural–multiplicity frames.

B.6 A specified small-quotient kernel gives the diagonal links

Local components on different blocks may be occurrences of the same global simple factor. Their transformations must then remain linked. We detect these links by taking the kernel of explicitly represented small quotient actions. Passing to its derived subgroup removes central ambiguity and exposes the diagonal factors.

For a locally bounded block set $L_i = 1$ and $B_i = P_i$; for every other block set $B_i = P_i/L_i$. Then

$$|B_i| \leq p^{c_0 d_i}, \qquad \prod_i |B_i| \leq p^{c_0 D}. \tag{16}$$

Enumerate each quotient by coset breadth-first search, using membership in the actual L_i for coset equality and retaining representative words. For a bounded block this is enumeration of its whole bounded image. The regular quotient actions have total degree at most $\sum_i p^{c_0 d_i} \leq D p^{c_0 D}$. Restricting an element of N to the blocks gives the specified homomorphism

$$\rho : N \longrightarrow \prod_i B_i, \qquad M = \ker \rho. \tag{17}$$

On the disjoint union of those quotient domains, compute its pointwise kernel with the original faithful action retained. Equivalently enumerate its at most $p^{c_0 D}$ image tuples and use Schreier relators. Thus M is constructed exactly. The descriptive equality $M = N \cap \prod_{i:L_i \neq 1} L_i$ does *not* invoke a generic intersection primitive. Covariance of L_i makes $M \trianglelefteq G$.

Lemma B.6 (Exact central product and links)**.** *The group $C = M'$ is a normal central product of quasisimple groups and*

$$[N : C] \leq p^{(c_0+1)D}. \tag{18}$$

Its components and their local diagonal links can be constructed using only the specified quotient kernels and derived-subgroup operations.

Proof. The projection M_i of M to P_i is normal in P_i and contained in L_i. A normal subgroup of a quasisimple group is either central or the entire group: if its image in the simple central quotient is

full, then $L_i = M_i Z(L_i)$ and perfectness gives $L_i \le M_i$. Hence M_i is full or central. Put

$$Z_0 = M \cap \prod_i Z(L_i).$$

This is central in M. After ignoring central projections, M/Z_0 is a subdirect product of nonabelian simple groups.

Such a subdirect product is a direct product of full diagonal factors. Here is an induction proof. Project onto all but the last simple factor; by induction that image is a product of simple diagonal factors. The kernel of this projection, viewed in the last factor, is normal there and is either the whole factor or trivial. In the first case the last factor splits off. In the second case the last coordinate is a surjective homomorphism from the preceding product to a simple group. The images of its direct factors commute and are normal; precisely one is nontrivial. It is an isomorphism on that factor, giving a diagonal link. This proves the assertion, including links through automorphisms.

In particular M/Z_0 is perfect. Thus $M = M'Z_0$ and

$$M' = [M'Z_0, M'Z_0] = [M', M'].$$

So $C = M'$ is perfect with central quotient the same product of simple diagonal factors. It is normal in G because M is. Moreover

$$|M/C| \le |Z_0| \le \prod_i |Z(L_i)| \le p^D.$$

Together with (16), this proves (18).

To find links concretely, use the natural projective action of each classical L_i, whose kernel is its scalar center; its degree is at most $|K_i|^{a_i} \le p^{d_i}$. For an alternating L_i use its faithful letter action, constructed in the next subsection (the constructions are independent). For every nontrivial simple image of C, compute

$$J_i = \ker\bigl(C \longrightarrow L_i/Z(L_i)\bigr)$$

as a pointwise kernel on this explicit domain. Two indices are in the same diagonal class exactly when $J_i = J_j$. Mutual containment is tested by generator membership in the original faithful action.

For a diagonal class I, let A_I be the kernel of all the actions belonging to one representative from each *other* class, and set $Q_I = A_I'$. Modulo $Z(C)$, this group has exactly the one simple factor indexed by I. The same central-extension calculation proves Q_I is quasisimple. Different Q_I commute, since their commutators are central and each is perfect. Their product surjects onto $C/Z(C)$; its quotient in C is abelian, and is trivial because C is perfect. Therefore

$$C = \prod_I Q_I.$$

All the kernels here are kernels of displayed actions on disjoint unions of domains of size $p^{O(D)}$. Their derived generators are original words. Conjugation by G permutes the actual kernels J_i and hence the groups Q_I. This supplies exact complementary permutations and all diagonal correlations, without guessing isomorphisms of abstract simple groups. □

B.7 Deleted-permutation realizations of block stabilizers

The alternating case is handled by evaluation on a permutation representation. The original deleted module can be a quotient of the zero-sum subspace, so the realization must retain its numerator and denominator. It must also extend to the entire block stabilizer, including the multiplicity action, to be useful after a later restriction.

Lemma B.7 (Constructive alternating cover)**.** *Suppose the dominant local component is alternating. One can construct its natural letter action and a deleted-permutation realization of its entire isotypic row, extending to the full actual block stabilizer. The degree inflation is at most $m/(m-2)$, where m is its alternating degree.*

Proof. The structural input gives $L \cong A_m$ on its deleted natural constituent X. Increase c_0 if necessary so that every locally large alternating case has $m \geq 8$; the finitely many smaller alternating groups have bounded order and are locally bounded. Put

$$P = \mathbb{F}_p^m, \quad P_0 = \{x : \textstyle\sum_j x_j = 0\}, \quad U_m = \begin{cases} P_0, & p \nmid m, \\ P_0/\langle \mathbf{1} \rangle, & p \mid m. \end{cases}$$

Write $r = \dim U_m = m - \delta$, with $\delta \in \{1, 2\}$. The ordinary dot product induces a nondegenerate alternating form on U_m: its radical on P_0 is exactly $\langle \mathbf{1} \rangle$ when $p \mid m$ and is zero otherwise. The constituent is absolutely irreducible and defined over $\mathbb{F}_p$, so $W = X \otimes_{\mathbb{F}_p} M$ in the previously computed coordinates.

Here is an explicit recognition that checks this representation rather than relying on an abstract factor name. Try the candidates $m = \dim X + 1, \dim X + 2$ compatible with the characteristic and $|L| = m!/2$. Solve for a nonzero invariant symmetric form on X and check rank. Enumerate projective-line seeds and their L-orbits. Retain orbits of size $\binom{m}{2}$; join two distinct orbit lines when their invariant-form pairing is nonzero. A successful orbit exists: on U_m it consists of

$$\langle e_i - e_j \rangle \quad (i < j).$$

These lines remain nonzero and distinct after the possible quotient by $\mathbf{1}$. An alleged equality modulo that line would equate a vector explicitly realized on at most four coordinates to a constant vector, impossible for $m \geq 8$ unless that constant is zero. Their pairings are nonzero exactly when the two unordered pairs meet, also over $\mathbb{F}_2$. The resulting graph is $L(K_m)$.

Recover its letters without a graph-isomorphism oracle. For adjacent vertices ij, ik, the common-neighbor graph consists of a clique of the other $m - 3$ edges through i and the isolated vertex jk. The large component, with ij, ik, gives the star at i. Collect these stars for all adjacent pairs, discard duplicates, and check that there are m stars, each of size $m - 1$, every vertex lies in two stars, and adjacency is exactly intersection of the corresponding two-element endpoint sets. Check also that the supplied generator permutations preserve the stars. They now give a homomorphism $\pi : L \to \mathrm{Sym}(m)$. Require even generators and image order $m!/2$. Together with the checked order of L, this certifies a faithful action onto A_m.

Build the standard matrices of U_m from these actual permutations and solve the intertwiner equations with X. Keep a candidate only if an invertible intertwiner is found. For the promised irreducible modules, a nonzero intertwiner is invertible, which is also rank-checked. This last test rules out any accidental orbit configuration. There are at most $p^{\dim X}$ seeds and at most that many orbit points per seed, so all the work costs $p^{O(\dim X)}$.

Let P_W be the full actual block stabilizer image, in which L is normal. For $h \in P_W$, conjugation gives an automorphism of A_m. For $m \geq 8$, $\mathrm{Aut}(A_m) = S_m$, and $C_{S_m}(A_m) = 1$. Thus there is a unique permutation π_h such that

$$\pi_h \pi(s) \pi_h^{-1} = \pi(hsh^{-1}) \qquad (s \in L). \tag{19}$$

It can be found by orbit calculations on $[m] \times [m]$: for each choice of the image of one letter, the orbit of the corresponding pair under the two generator actions is the graph of an intertwining

bijection exactly when both projections are bijective and the orbit has m points. Test the resulting bijection in (19). Uniqueness implies $\pi_{hh'} = \pi_h \pi_{h'}$.

Write $d(\pi_h)$ for the actual deleted-module matrix on X. The map $(d(\pi_h)^{-1} \otimes 1)h$ centralizes L. Absolute irreducibility gives a unique $c_h \in \mathrm{GL}_{\mathbb{F}_p}(M)$ with

$$h = d(\pi_h) \otimes c_h. \tag{20}$$

Read c_h from the matrix entries. Since the two tensor actions commute, $c_{hh'} = c_h c_{h'}$. The full parent action therefore lifts genuinely to

$$\widetilde{W} = \mathbb{F}_p^m \otimes M, \qquad \widetilde{h} = \mathrm{perm}(\pi_h) \otimes c_h.$$

It preserves $P_0 \otimes M$ and, if $p \mid m$, its subspace $\langle \mathbf{1} \rangle \otimes M$. Their indicated subquotient is exactly W, with action (20). On an axis $e_j \otimes M$, the stabilizer acts by c_h. Since $c_s = 1$ for $s \in L$, this image factors through P_W/L and has order at most $p^{5 \dim W}$. Scalars and field actions on multiplicity are part of this actual $\mathbb{F}_p$-linear map, so none has been discarded.

Finally

$$\dim \widetilde{W} = m \dim M = \frac{m}{m - \delta} \dim W \leq \frac{4}{3} \dim W \qquad (m \geq 8). \tag{21}$$

No quadratic refinement of the binary deleted module is required. In particular, when $m \equiv 2 \pmod 4$, the usual refinement does not descend through the all-ones line, but the alternating pairing and the genuine permutation cover above remain valid.

The cover is made only for this single dominant alternating factor. No product of degree inflation factors from smaller tensor factors is introduced. □

For completeness, induction over a block orbit is ordinary. Fix transversals t_i from its representative block to W_i. If $gW_i = W_j$, then $t_j^{-1} g t_i$ belongs to the representative stabilizer. Define the map from the ith copy of its cover to the jth by the just constructed representation of $t_j^{-1} g t_i$. In a product $g'g$, these stabilizer expressions multiply with the middle transversals cancelling. Thus the induced matrices satisfy the group law. This also proves compatibility of all subquotient maps. Across distinct original blocks the degree bounds (21) add, giving total covered degree at most $4D/3$ and at most $4D/3$ coordinate axes.

B.8 A fixed threshold under later subgroup restrictions

A component placed in the evaluation part should stay there under subsequent subgroup restrictions. A bound on the present composition factors alone would not guarantee this: a subgroup can have new simple sections. We prove the stronger bound for simple sections of every subgroup and use the same order threshold throughout the computation.

Choose C_* once, larger than $c_0 + 1$, the fixed permutation-cover constants, and a fixed natural-degree cutoff $a_{\mathrm{st}} = a_*$ as fixed in Section 2, enlarged to at least 80. All retained classical ranks then satisfy the representation-theoretic hypotheses below. If a classical simple quotient has natural matrix degree a over $K = \mathbb{F}_{p^f}$, then

$$|S| < |K|^{a^2} = p^{f a^2}, \qquad f a \leq d_i.$$

Thus bounded a implies $|S| \leq p^{a D_0}$; increasing C_* ensures that every component selected by (12) lies in stable rank. Exceptional small classical–alternating coincidences and orthogonal triality do not occur in the selected set.

Let $\overline{D}$ be the product of the classical Q_I in Lemma B.6 with $|Q_I/Z(Q_I)| > T$. Its definition is conjugation-invariant, so $\overline{D} \trianglelefteq G$. The following point ensures that it captures every genuinely large local classical action.

Lemma B.8. *If L_i is classical and $|L_i/Z(L_i)| > T$, then some selected component Q_I projects onto L_i.*

Proof. Let N_i be the image of N in P_i; it is normal there. If it does not contain L_i, then $N_i \cap L_i \leq Z(L_i)$, so the simple quotient of L_i is a section of P_i/N_i. But P_i/N_i is an image of G_{W_i}/N, a subgroup of the block permutation group $G/N \leq S_k$, $k \leq D$. A nonalternating simple section of any subgroup of S_k has order at most $24^{(k-1)/3} < 3^k \leq 2^{(\log_2 3)D}$: apply the GMP bound to the subgroup in which that section occurs. This contradicts $|L_i/Z(L_i)| > T$.

Hence $N_i \geq L_i$. If the projection of C to L_i were central, its simple quotient would be a section of N/C, whose order is at most $p^{(c_0+1)D} < T$. This is again impossible. The projection of C is therefore all of L_i. In the diagonal decomposition exactly its class component Q_I projects onto it. The simple central quotients of Q_I and L_i are isomorphic, so that Q_I is selected. □

A perfect component has either full or trivial projection to any local L_i: its only other possible image is central and therefore trivial. It has trivial projection to a locally bounded block because M does. A selected component cannot project to an alternating local component in stable rank, or to a classical local component with smaller simple quotient. Consequently all of its nontrivial local representations are exactly the already certified natural occurrences, with their actual multiplicities and diagonal links. Distinct components have disjoint active local supports. In particular we have not taken an algebraic envelope of several independent tensor factors acting on the same row, which could create an unjustified full linear group.

A locally bounded block is retained with its entire image of order at most $p^{c_0 d_i}$. A classical local block not supporting $\overline{D}$ has $|L_i/Z(L_i)| \leq T$ by Lemma B.8. Its full image satisfies

$$|P_i| \leq p^{5d_i}|L_i| < p^{6d_i}T \leq p^{(C_*+6)D_0}. \tag{22}$$

Retain this *one* bounded image as one argument block. Formula (22) is not a bound on the product of all evaluation nodes. Alternating local blocks are handled by Lemma B.7.

Lemma B.9 (Hereditary complementary bound)**.** *With the same fixed threshold T, no nonalternating simple section of a subgroup of an evaluation-node image, an alternating-cover complementary representation, the bounded quotient, or a retained component index permutation image has order greater than T.*

Proof. First observe an elementary extension fact. If $B \trianglelefteq A$ and $J \leq A$, then $J \cap B \trianglelefteq J$ and $J/(J \cap B)$ embeds in A/B. Every composition factor of J occurs in one of these two groups. Applying this observation also to subgroups of J shows that a simple section of A comes from a simple section of B or of A/B. A central kernel contributes only abelian factors.

For a locally bounded node its whole order is at most $p^{c_0 D_0} < T$. For a bounded classical block used in evaluation apply the extension fact to $Z(L_i) \trianglelefteq L_i \trianglelefteq P_i$. Sections coming through the central kernel are abelian, those through $L_i/Z(L_i)$ have order at most $|L_i/Z(L_i)| \leq T$, and those through P_i/L_i have order at most $p^{5d_i} < T$. This uses the small simple quotient, rather than the slightly larger full-image bound (22).

For the sum of alternating covers, the permutation of axes has degree at most $4D/3$. Its every subgroup has nonalternating simple sections of order less than $3^{4D/3}$ by GMP. The kernel fixing these axes has local coordinate images contained in the commuting multiplicity images $c(P_i)$, each of

order at most p^{5d_i}. The product of these particular bounds is at most p^{5D}, since their exponents are charged to disjoint original blocks. The induced axis kernel embeds in that product, with all original correlations preserved. The extension fact now proves the claim for the permutation realization and every subgroup of it. The same GMP argument applies to the original block permutation image and any retained component index permutation image of linear degree. Finally the whole order of N/C is at most $p^{(c_0+1)D}$. The permanent choice of C_* dominates all these constants. □

The assertion concerns the recorded complementary representations individually and extensions assembled from their actual correlated maps. It does not assert small order for their arbitrary product, nor does it pass to evaluation the active natural components themselves. Because the bound applies to every subgroup and quotient encountered, a later restriction cannot revive a large nonalternating factor in a representation already assigned to evaluation. Resetting T to the new local dimension would lose precisely this property and is not permitted.

B.9 Normalizer reduction and the large commuting image

The natural factors can be very large, but their commuting multiplicity action is small because each selected factor already occupies a large part of the original degree budget. We use this observation only in a completed local call. The normalizer calculation also states exactly when a product with the full classical group is valid.

The following refinement supplies the exact terminal and frame-normalization data. Here L denotes the full classical group on the selected rows (with its paired dual realization where needed), whereas $\overline{D}$ denotes the derived core. Full classical group and derived subgroup are kept distinct.

Lemma B.10 (Constructive normalizer and commuting-image reduction)**.** *Let V_{crit} be the direct sum of the selected nontrivial isotypic original rows, written in the computed coordinates as $\bigoplus_j X_j \otimes_{K_j} M_j$. Put $f_j = [K_j : \mathbb{F}_p]$, $a_j = \dim_{K_j} X_j$, and $b_j = \dim_{K_j} M_j$. Here j runs over the distinct irreducible $\overline{D}$-module types in the entire retained representation V_{crit}, including its formal dual rows. All copies of the same type, even when they occur in different retained rows, are aggregated in the single multiplicity space M_j; inequivalent twists or dual types remain distinct. Original row labels and their stabilizing conditions are retained separately and can only restrict the ambient centralizer. Then*

$$\left|C_{\mathrm{GL}(V_{\mathrm{crit}})}(\overline{D})\right| \le p^{\sum_j f_j b_j^2} = p^{O(D_0)}. \tag{23}$$

The genuine group of factorizations constructed above supplies all natural, multiplicity, field, and exchange actions. After its actual finite field, type, exchange, and form-similitude records have been fixed, the normalizer of the full classical group has the form L times its actual centralizer. More precisely, if A_0 is the resulting actual normalizer subgroup and $L \le A_0$, then

$$A_0 = LC_{A_0}(L). \tag{24}$$

The centralizer's complete joint image on V_{crit} can be enumerated within $p^{O(D_0)}$, with its full kernel and original lifts retained.

Proof. Schur's lemma and the evaluation coordinates identify the commuting algebra on V_{crit} with $\prod_j \mathrm{End}_{K_j}(M_j)$. Its unit group is $\prod_j \mathrm{GL}_{b_j}(K_j)$, proving the first inequality. Correlations, forms, or fixed scalar conditions only shrink the actual commuting image.

For every selected row the large simple quotient satisfies $p^{C_* D_0} < |S_j| < p^{f_j a_j^2}$. Also the mass of these rows is bounded by the original mass, or by a fixed multiple κD_0 if formal duals have been

included:

$$\sum_j f_j a_j b_j \le \kappa D_0, \qquad \frac{b_j}{a_j} = \frac{f_j a_j b_j}{f_j a_j^2} < \frac{\kappa}{C_*}.$$

Consequently

$$\sum_j f_j b_j^2 < \frac{\kappa}{C_*} \sum_j f_j a_j b_j \le \frac{\kappa^2}{C_*} D_0.$$

This proves the exponential bound with the original degree; an artificial multiplicity tensor factor has not been charged as fresh original mass.

We give the normalizer calculation on each factor. Fixing the permutation of simple algebra factors and its induced field action reduces its algebra automorphism to an inner one. The implementing u is computed by the Skolem–Noether equations already displayed, with the scalar ambiguity retained in the factor cover. In a linear factor the full natural group is $\mathrm{GL}(X)$, so u is already in that group. In a symplectic, orthogonal, or unitary factor, normalizing the derived natural group preserves its unique invariant form up to a scalar: indeed the pullback form is another invariant form, and the appropriate intertwiner space has dimension one. The factorization-group matrix u therefore has its explicit multiplier λ. Once that multiplier has been fixed to one, u belongs to the full natural isometry group. The residual factor $u^{-1}h$ centralizes the natural group by the commutant equations. Paired linear factors carry (u, u^{-t}); fixed exchange records give the same assertion there. For linked occurrences choose u on one representative occurrence of the component and transport it using the verified field/dual intertwiners. These transported maps implement the same conjugation on the actual component as h, so their residual quotients centralize the component on every linked row. They therefore centralize its represented algebra and the full classical group contained in that algebra. Thus one obtains one ℓ per diagonal component, not independent natural matrices on linked rows. These verifications prove that each element of A_0 is ℓc in the ambient representation. If $L \le A_0$, then $c = \ell^{-1}h \in A_0$, proving (24).

For a nonsingular quadratic space in even dimension, the index-two orthogonal record is

$$\mathrm{Dickson}(g) = \mathrm{rank}_K(g - I) \pmod 2 \in \mathbb{F}_2.$$

Its kernel is Ω in the stable dimensions used here; the determinant is identically one. We retain this record explicitly whenever a full orthogonal group is used to solve literal equations [Tay92].

All records here are actual actions or scalar multipliers, not labels assigned from abstract group names. The cover is useful because a multiplier can change on changing a scalar factorization; the scalar kernel is retained when taking its multiplier kernel. If a derived SL, SU, or Ω is used in place of the full natural group, also fix respectively the determinant and the applicable unitary-determinant or Dickson quotient records before the same conclusion. No such extra condition is silently inferred from a similitude multiplier.

There are only $O(\sqrt{D_0})$ selected isotypic rows: each has $f_j a_j \ge \sqrt{f_j C_* D_0} \ge \sqrt{C_* D_0}$, whereas their total mass is $O(D_0)$. Hence their permutation-image order is at most $O(\sqrt{D_0})! = p^{O(D_0)}$. The products of field-automorphism orders, twofold exchange orders, and scalar-record orders are also $p^{O(D_0)}$, since $\sum_j f_j \le O(D_0)$ and $\prod_j f_j \le 3^{\sum_j f_j/3}$. Thus, when these records must be listed in a terminal call, their complete joint image is small enough to list. Their kernels are specified homomorphism kernels, computed successively without imposing unrelated conditions on their images.

The group L is normal in A_0. Compute $C = C_{A_0}(L)$ by the normal-subgroup centralizer algorithm, then its restriction homomorphism $C \to \mathrm{GL}(V_{\mathrm{crit}})$. Its image has the bound (23), so breadth-first

image search and Schreier relations compute its complete image and $C_{\mathrm{fix}} = \ker(C \to \mathrm{GL}(V_{\mathrm{crit}}))$, with original words. Since L is faithful on these rows, $L \cap C_{\mathrm{fix}} = 1$, and $[L, C_{\mathrm{fix}}] = 1$. Therefore

$$LC_{\mathrm{fix}} = L \times C_{\mathrm{fix}} \longrightarrow L, \qquad \ell c \longmapsto \ell \tag{25}$$

is an exact homomorphism with full kernel C_{fix}. Coset representatives for C/C_{fix} cover A_0 by cosets of LC_{fix}; duplicates caused by $L \cap C$ may be retained or removed by ordinary coset membership. This is the complete joint commuting-image normalization, rather than independent choices on separate value rows. □

If the current subgroup does not contain the full classical group L, one retains the displayed factorization only as representation data and uses the focused actual quotient of Appendix E. Its kernel and bounded outside records give the complete return without intersecting with an arbitrary overgroup transporter. It is false in general that an arbitrary subgroup of LC equals L times its own centralizer. No use of (24) is made without its containment hypothesis. The lists in the lemma are terminal lists, to be reunited before a complete result is returned. Their mere small size does not authorize opening one at every recursive node. For compatible multiplicity-frame normalization one instead uses its actual frame orbit and the row-loss charge of Lemma 5.2.

The same bounds extend Lemma B.9 to active-row commuting images and their finite outer records: each has total order $p^{O(D_0)}$ with a fixed constant dominated by the permanent C_*.

B.10 Original representations, lifting, and complexity

We finish by collecting the representation maps and their costs. The direct sum of the flag quotients is used to describe the diagonal action, while the original filtered module is retained to recover lifts. Keeping both descriptions avoids a splitting assumption on the original module.

The covers above represent G on its *grades*. Pulled back along $H \to G$, their new coordinates have trivial $O_p(H)$-action. They are not asserted to give an H-invariant complement in the original V, or a full-module cover splitting its possibly nonsplit flag extension. Keep the original V and its flag in parallel with all grades and covers. Their total degree is at most $D + D + 4D/3$, before any fixed number of formal dual or scalar coordinates, and hence remains $O(D)$. The exact numerator and denominator maps identify the covered grades, while the original matrices record all extensions between grades.

Likewise, $\overline{D}$ is initially an actual subgroup of the diagonal image G. Its generator words have preimages in H, but arbitrary preimages need not define a homomorphic section. No later step requires a physical section. On each exact coefficient grade, Lemma 4.3 quotients by the actual kernel translations, solves in the actual focused graded image, and corrects every lift. This applies to a nonsplit extension and retains the entire original kernel.

Every group operation used in this section takes place on the vector action of the original rows or of the displayed $O(D)$-degree group of factorizations, on a specified quotient of that action, or on a disjoint union of explicit quotient/projective/letter domains. All these domains have total size $p^{O(D)}$. Vector searches have at most p^D seeds. Represented algebra and Hom computations have polynomial dimension, and their field enumerations have at most p^D entries. The number of primitive blocks, local components, diagonal classes, and refinement stages is polynomial in D and $\log p$. Schreier generators are compressed and reduced between stages. These observations prove the stated deterministic $p^{O(D_0)}\mathsf{s}^{O(1)}$ bound, including construction of all kernels, quotient maps, and original words.

The proof of Theorem B.1 is now complete. Its permanent threshold and hereditary bound concern supplied groups and original representations. At a completed branch reunion one first

constructs the actual returned stabilizer, with its original words, and then applies this theorem to that supplied group while preserving the incoming parent data. The accounting for creation of new bounded-action blocks and for complete branch reunions belongs to the recursive algorithm; it is not a consequence of multiplying the individual local-order bounds.

C The degree two image algebras

The numerator and denominator of a module problem can be arbitrary binary invariant subspaces. A decomposition into composition factors would lose the extensions between them. We instead compute the algebra spanned by the actual group matrices and describe injective modules for that algebra. Every invariant quotient embeds into a direct sum of these modules. Their point equations are ordinary matrix and quadratic-form equations, with a small number of scalar conditions.

Let S be a product of the stable classical groups arising from Appendix B. Let R be the direct sum of its distinct natural argument modules and required duals, with $\dim_k R = O(D_0)$. A universal coefficient representation is

$$C_2 = k \oplus R \oplus (R \otimes_k R).$$

Its dimension is $O(D_0^2)$. Starting with the identity and the supplied generator matrices, repeatedly multiply a span basis by the generators and retain a matrix exactly when it is linearly independent of those already kept. After at most $(\dim_k C_2)^2$ additions this constructs

$$A = k[\rho(S)] \le \mathrm{End}_k(C_2).$$

Here $k[\rho(S)]$ denotes the image algebra, not an enumerated group algebra. To transfer its action to a supplied section Q, perform the same span closure on $C_2 \oplus Q$. Projection of this simultaneous algebra onto A has zero kernel because Q is an invariant section of passive copies of C_2. This verifies, constructively, the unique induced A-action on Q. The hypothesis follows from the representation maps retained by the algorithm, rather than being a promise imposed on the tensor input.

C.1 Embedding by all equivariant maps

A finite-dimensional A-module C is a cogenerator if every simple A-module embeds into C and each of its injective envelopes occurs in C. We use the following explicit construction rather than an implicit search for an invertible module map.

Lemma C.1. *Suppose C is an explicitly represented injective cogenerator of A. If $f_1, \dots, f_t$ is a k-basis of $\mathrm{Hom}_A(Q, C)$, then*

$$i : Q \longrightarrow C^t, \qquad i(v) = (f_1(v), \dots, f_t(v))$$

is injective. It is computable by linear equations and has target dimension at most $\dim_k Q(\dim_k C)^2$.

Proof. A nonzero common kernel contains a simple submodule X. Embed X in its injective envelope inside C, and extend this map to Q by injectivity. The resulting map is nonzero on X, contradicting the common-kernel assumption. Equivariance and the dimension bound are immediate. Generator intertwiner equations suffice because they imply commutation with their whole span algebra. A direct rank test verifies the resulting embedding. □

Conceptually, $v \mapsto (a \mapsto av)$ embeds Q in $\mathrm{Hom}_k(A, Q)$, where $(bf)(a) = f(ab)$. Thus the duals of the primitive right ideals of A form a cogenerator. The formulas below identify those duals in familiar tensor coordinates.

C.2 General linear squares

Let K be an even finite field, let $V = K^a$ with $a \geq 3$, and put $T = V \otimes_K V$. Write $\tau(v \otimes w) = w \otimes v$ and $x = 1 + \tau$. Then $x^2 = 0$, $\operatorname{rank} x = h = a(a-1)/2$, and

$$K[\mathrm{GL}(V)]|_T = \mathrm{End}_{K[x]}(T). \tag{26}$$

Here is a direct proof that includes the binary field. A homogeneous polynomial of degree two in matrix entries that vanishes on all invertible binary matrices becomes, on binary inputs, a square-free polynomial of degree at most two with no constant term. For two entry positions $\alpha = (i, j)$ and $\beta = (l, t)$, choose a permutation matrix P so that $P^{-1}E_\alpha$ and $P^{-1}E_\beta$ are off-diagonal matrix units whose two directed edges form no cycle. Such a choice exists for $a \geq 3$: for a common row choose its new index outside the two columns; for a common column choose two different new indices outside it; otherwise choose one outside both columns and the other equal to the first column. Complete these row assignments to a permutation. The four matrices $P + sE_\alpha + tE_\beta$, $s, t \in \mathbb{F}_2$, are invertible. Summing the four polynomial values isolates the mixed coefficient. All mixed coefficients therefore vanish. The pair $P, P + E_\alpha$ similarly kills each remaining linear coefficient. Consequently the coordinate functions of $g \otimes g$ have their full rank $a^2(a^2+1)/2$.

Since the image commutes with x, it lies in its centralizer. In coordinates consisting of the bottom and top of the h two-dimensional Jordan blocks and the a singleton blocks, that centralizer is

$$\left\{ \begin{pmatrix} A & B & C \\ 0 & D & E \\ 0 & 0 & A \end{pmatrix} : \begin{array}{l} A, C \in \mathrm{M}(h, K), \quad D \in \mathrm{M}(a, K), \\ B \in \mathrm{M}(h \times a, K), \quad E \in \mathrm{M}(a \times h, K) \end{array} \right\}. \tag{27}$$

Its dimension is $(h+a)^2 + h^2 = a^2(a^2+1)/2$, proving (26) first over $\mathbb{F}_2$ and then over K. Scalar matrices in $\mathrm{GL}(V)$ have all square scalars, so its binary image on the K-square contains KI and equals the same algebra.

The two indecomposable $K[x]$-modules in T are $K[x]/(x^2)$ and K. Taking duals of their primitive right ideals gives the injectives

$$T, \qquad T/\operatorname{im} x = \mathrm{Sym}^2 V.$$

The first is a bilinear-tensor module and the second a quadratic-form module after dual identification. Their ordinary and dual versions therefore have a simultaneous literal transporter by Theorem A.5.

The special-linear image agrees in bounded degree with the general-linear image. Indeed, on a sum of tensor words of degree at most r, the difference of two diagonal-torus weights has at most $2r$ nonzero positions. A character trivial on the determinant-one torus is constant in every position modulo $|K| - 1$. If $a > 2r$, a nonzero constant vector is impossible. Over a splitting field the actual weight restrictions to the determinant-one torus T_0 are distinct. Hence its projector is $e_\chi = |T_0|^{-1} \sum_{s \in T_0} \chi(s)^{-1} \rho(s)$, and for t in the full torus, $\rho(t) = \sum_\chi \chi(t) e_\chi$ lies in the special-linear image algebra. Since $\mathrm{GL} = \mathrm{SL}\ T$, this proves equality of the image algebras; linear descent returns it to the original field. Include a trivial coordinate whenever the common augmentation character is needed. For $K = \mathbb{F}_2$ the determinant quotient is trivial. This also preserves coincidences between a Frobenius-natural constituent of the square and an explicitly supplied natural row.

C.3 Symplectic squares

Let V have dimension $d = 2m$ over K with nondegenerate alternating form b. Let $e_1, f_1, \ldots, e_m, f_m$ be a symplectic basis. On $T = V \otimes_K V$ define

$$x = 1 + \tau, \quad \omega = \sum_{i=1}^{m} (e_i \otimes f_i + f_i \otimes e_i), \quad y(v \otimes w) = b(v, w)\omega.$$

Then $x^2 = y^2 = xy = yx = 0$. Set $C = K[x,y]/(x^2, y^2, xy)$, acting by these operators.

Lemma C.2 (Finite symplectic image)**.** *For $m \geq 5$, the K-image of $\mathrm{Sp}(V)$ on T is $A_K = \mathrm{End}_C(T)$. If $\alpha : A_K \to K$ is the scalar action on $K\omega$, its exact binary image is*

$$A_k = kI \oplus \ker \alpha.$$

Proof. Use the binary Siegel Levi $\mathrm{GL}_m(2)$ on $V = E \oplus E^*$. The three whole tensor grades EE, $EE^* \oplus E^*E$, and E^*E^* have disjoint composition-type sets. The square has exterior and Frobenius-natural types; its dual has their duals; the middle grade has trivial and adjoint-heart types. For $m \geq 5$ the respective restricted highest weights are distinct, including over $\mathbb{F}_2$, by the defining-characteristic restriction theorem [Ste68, §13, Theorem 43]. Thus the annihilator ideals of these three whole summands are comaximal, and their projections P_2, P_1, P_0 belong to the finite Levi image algebra. Internal nonsplit extensions inside a grade are not separated.

For a binary symmetric matrix $B \in \mathrm{M}(m,2)$, the upper unipotent $u(B) = \left(\begin{smallmatrix} I & B \\ 0 & I \end{smallmatrix}\right)$ acts on T as $I + L_B + Q_B$, where L_B raises grade by one and Q_B by two. Hence

$$L_B = P_2\rho(u(B))P_1 + P_1\rho(u(B))P_0, \quad Q_B = P_2\rho(u(B))P_0$$

belong to the finite image. Its scalar extension contains $I + tL_B + t^2Q_B = \rho(u(tB))$ for every algebraic scalar t. Binary symmetric matrix units span all symmetric matrices; upper unipotents add their parameters. The opposite construction gives all lower unipotents. Their root subgroups generate the algebraic symplectic group. Over the algebraic closure, every symplectic similitude is a scalar times a symplectic matrix, since its multiplier has a square root. The two groups therefore have the same tensor-square span. The group-algebra-image direction of [DDH08, Proposition 1.3] identifies that span with the commutant of the Brauer action. The displayed coefficient separation proves that the finite image has the same scalar extension, so equality descends by linear algebra.

For the binary image over K, the same projectors applied to $\rho(u(tB))$, $t \in K$, give tL_B and t^2Q_B. Squaring permutes K. These root coefficients generate the augmentation ideal $\ker \alpha$: their unital algebra is A_K, they annihilate ω, and their quotient is the scalar algebra. Every K-coefficient in a word containing a root coefficient can be placed on that one factor. Thus the entire K-valued ideal belongs to the binary image. Conversely every group element has α-value one, proving the formula. □

Put $A = \mathrm{im}\, x$, $h = \dim_K A = d(d-1)/2$, and define $\ell : A \to K$ by $\ell(v \otimes w + w \otimes v) = b(v,w)$. Thus $y(t) = \ell(xt)\omega$ and $\ell(\omega) = m$ in K. Choose a complement of A in $\ker x$, of dimension d. The remaining part of T has top and bottom copies of A, with x the identity between them and y the rank-one map $\omega\ell$. It follows that, as a C-module,

$$T \cong \begin{cases} R_x^{h-1} \oplus R_y \oplus K^d, & m \text{ odd}, \\ R_x^{h-2} \oplus R_N \oplus K^d, & m \text{ even}, \end{cases}$$

where $R_x = C/(y)$, $R_y = C/(x+y)$, and R_N has two-dimensional top and bottom, with x the identity top-to-bottom and y one nilpotent Jordan block. These modules are self-dual. For an idempotent selecting a copy of a C-module R_i, its right ideal is $\mathrm{Hom}_C(T, R_i)$; its dual is $T \otimes_C R_i^*$. We obtain the following exact injectives:

$$\begin{aligned} &I_H = T/K\omega, && I_W = T/A = \mathrm{Sym}^2 V, \\ &I_0 = T/\ker(\ell|_A) && (m \text{ odd}), \\ &I_0 = T^2/\{(a, \ell(a)\omega) : a \in A\} && (m \text{ even}). \end{aligned} \tag{28}$$

For the even formula, use generators u, v of R_N with relations $yu = 0$, $yv = xu$. Tensoring gives relations $(yt, 0)$ and (xt, yt). Since $\ell(\omega) = 0$, the first family is already included in the second. The socle of I_0 is $K\omega$ in the odd case and $K(0, \omega)$ in the even case. Their dimensions are $h + d + 1$ and $2d^2 - h$, respectively.

C.4 Unitary squares and adjoint modules

Let $K_0 = \mathbb{F}_{2^f}$, $K = \mathbb{F}_{2^{2f}}$, and let $\bar{a} = a^{2^f}$. For a Hermitian natural space V of dimension $a \geq 11$, the ordinary square has

$$K[\mathrm{U}(V)]|_{V \otimes V} = \mathrm{End}_{K[\tau]}(V \otimes V). \tag{29}$$

We give the finite-point verification because a small field cannot be replaced by an appeal to density of its rational points. In dimension $2m$, the hyperbolic Levi $\mathrm{GL}_m(K)$ has three grades EE, $EF \oplus FE$, FF, with $F = E^{\bar{*}}$. The square types, conjugate-dual square types, and mixed distinct-digit simple type are disjoint for $m \geq 5$. This holds also when $K_0 = \mathbb{F}_2$: the Frobenius-natural constituent in FF is E^*, distinct from $E^{[1]}$ and the mixed type. Their whole-grade projections therefore belong to the finite image. For an upper unipotent with $B^{\bar{t}} = B$, they separate $I + L_B + Q_B$ as above. The K-span of Hermitian matrices is all $\mathrm{M}(m, K)$: use diagonal units, $E_{ij} + E_{ji}$, and $\theta E_{ij} + \bar{\theta} E_{ji}$ with $\theta \notin K_0$. The upper and opposite radicals consequently generate the scalar-extended SL_{2m} image. Ordinary Schur–Weyl duality gives (29).

For dimension $2m + 1$ add an anisotropic line $Z = Kz$. There are six whole Levi grades: EE, $EF \oplus FE$, FF, $EZ \oplus ZE$, $FZ \oplus ZF$, and ZZ. Their type sets are disjoint. Choose $\theta + \bar{\theta} = 1$ and the unitary element

$$z \mapsto z + e_i, \qquad f_i \mapsto f_i + z + \theta e_i,$$

fixing the remaining basis vectors. Between the six grades, its blocks from EZ to EE, FZ to EF, and ZZ to EZ isolate the linear tensor action of $E_{i,z}$; the block from ZZ to EE isolates its quadratic tensor action. The blocks from FF to FZ, FZ to ZZ, and EF to EZ, and the block from FF to ZZ, similarly isolate E_{z,f_i}. Their algebraic scalar curves, together with the even head, generate SL_{2m+1}. This proves the odd-dimensional case.

A central norm-one scalar of order $|K_0| + 1$ generates K over $\mathbb{F}_2$, since no proper subfield has that many nonzero elements. Its square acts as that scalar squared on the square. Hence the binary square image contains KI. The bounded-degree torus-support proof for SU and U is the same as the determinant proof, with modulus $|K_0| + 1$; it applies when $a > 2r$. The square injectives are therefore again T and $\mathrm{Sym}^2 V$. In particular at $\mathrm{SU}_a(2)$ the square's Frobenius-natural constituent is the actual dual natural module. It is identified with any supplied occurrence of that module, rather than added as an independent factor.

For the adjoint representation write $\mathcal{H}$ for the K_0-space of Hermitian matrices, of dimension a^2. It has fixed line $K_0 I$ and invariant trace. Moreover $\mathrm{End}_K(V) = \mathcal{H} \oplus \theta\mathcal{H}$ over K_0, for $\theta \notin K_0$. If a is odd, $\mathcal{H} = K_0 I \oplus \ker \mathrm{Tr}$, with simple second summand. If a is even, the filtration is

$$K_0 I < \ker \mathrm{Tr} < \mathcal{H},$$

with simple heart of dimension $a^2 - 2$ and both adjacent extensions nonsplit. Simplicity follows from the finite restricted adjoint highest weight. To prove nonsplitting, a split heart, whose dimension exceeds half of a^2, would be nondegenerate for the trace pairing. Its two-dimensional orthogonal complement has only trivial factors; perfectness of SU forces its unipotent image to be trivial. This would give two fixed vectors, contradicting $\mathcal{H}^{\mathrm{SU}} = K_0 I$. Duality proves the other extension nonsplit.

The heart's coefficient field is exactly K_0. On $g = \mathrm{diag}(t, t^{-1}, 1, \ldots, 1)$ with $t\bar{t} = 1$, its character differs by a prime-field constant from $(a + t + t^{-1})^2$. The set of $t + t^{-1}$ has $|K_0|/2 + 1$ elements, so

lies in no proper subfield; the case $K_0 = \mathbb{F}_2$ is immediate. Thus the semisimple heart image is its full matrix algebra. Nonsplitting forces the full radical row and column, and their products force the top-to-bottom block. In coordinates of the filtration the image has the form

$$\begin{pmatrix} \alpha & b & c \\ 0 & A & e \\ 0 & 0 & \alpha \end{pmatrix}, \qquad A \in \mathrm{M}(a^2-2, K_0), \quad \alpha \in k,$$

with all radical entries over K_0. Its injectives are the whole Hermitian module and its quotient by the scalar socle; for odd a use its simple direct summands. The same argument applies to the general-linear adjoint module over its field K.

C.5 Orthogonal tensor squares

Let V be a nonsingular quadratic space over $K = \mathbb{F}_{2^f}$, of dimension $d = 2m$, with quadratic form Q and polar form b. Both signs are allowed. All dimensions in this subsection exceed the fixed cutoff of the main algorithm. Put

$$T = V \otimes_K V, \quad x = 1 + \tau, \quad A = \operatorname{im} x, \quad \Gamma = \ker x, \quad h = \dim_K A = d(d-1)/2.$$

The polar form identifies V with its dual. Let $\omega \in A$ be its inverse tensor and let $\ell : A \to K$ be contraction, so $\ell(\omega) = m$. Choose $X \in T$ representing the quadratic form Q in $T/A = \operatorname{Sym}^2 V$; then $xX = \omega$. The tensor-product polar pairing $\langle\ ,\ \rangle$ gives the invariant functional

$$f_Q : \Gamma \to K, \qquad f_Q(s) = \langle X, s \rangle, \qquad f_Q|_A = \ell.$$

Changing X by an alternating tensor does not change the induced quadratic class or this functional.

Finite density. The K-span of the finite $\Omega(V, Q)$ square action has the same scalar extension as the connected algebraic orthogonal square image. To prove this, first consider split $V = E \oplus E^*$. The finite Levi $\mathrm{GL}(E)$ has the three whole tensor grades used in Lemma C.2; their disjoint composition types give their exact projectors. The upper orthogonal radical consists of $u(B)$ with B alternating. The grade projectors separate the linear and quadratic coefficients of $u(tB)$ for every t after scalar extension. Alternating matrix units and the opposite radical generate the algebraic type-D group.

For the nonsplit space write $V = E \oplus Z \oplus E^*$, with Z an anisotropic quadratic plane. Its rotation group has odd order $|K| + 1$, with nontrivial characters χ, χ^{-1} over an algebraic closure. Give E^*, Z, E degrees $0, 1, 2$. The five tensor grades have disjoint types under $\mathrm{GL}(E) \times \mathrm{SO}(Z)$: square and dual-square head types; natural or dual-natural head types with nontrivial tail characters; and the middle adjoint/trivial head types. The tail characters distinguish the possible binary natural/Frobenius collisions. Hence their whole-grade projectors lie in the actual finite image.

The radical consists of

$$u(Y, B) = \begin{pmatrix} I & Y & B \\ 0 & I & J_Z^{-1} Y^t \\ 0 & 0 & I \end{pmatrix}, \qquad v^t B v = Q_Z(J_Z^{-1} Y^t v),$$

where $Y \in \mathrm{Hom}_K(Z, E)$ and $B \in \mathrm{End}_K(E^*)$ in the indicated coordinates. Conjugation by $\mathrm{diag}(tI, I, t^{-1}I)$ gives $u(tY, t^2 B)$. The grade projectors place this entire algebraic scalar curve in the scalar-extended finite image. The central radical $Y = 0$ is the alternating-matrix group and is generated by these curves. Every finite Y has a finite lift B, and the finite Y span $\mathrm{Hom}_K(Z, E)$; the curves therefore generate the full quotient radical and its center. The opposite radical completes the connected algebraic group. This proves density in both signs without interpolating over the finite field or inferring an image equality from a commutant equality.

The extra quadratic condition. For $g \in \mathrm{O}(Q)$, the difference $gX - X$ lies in A. The scalar $\ell(gX - X)$ is an additive homomorphism and equals the Dickson invariant in $\mathbb{F}_2 \leq K$. To check it, extend to split coordinates and take $X = \sum_i e_i \otimes f_i$. For a nonsingular vector $v = u + w$ in the two singular halves, normalize $Q(v) = 1$. Its orthogonal symmetry changes X by $u \otimes w + w \otimes u$, whose contraction is one. Symmetries generate $\mathrm{O}(Q)$ and have Dickson invariant one, proving the identity, which descends to the original coordinates.

Let $A_{\mathrm{Sp}} = \mathrm{End}_C(T)$ be the symplectic image from Lemma C.2, and let α be its character on $K\omega$. Write $a^\dagger$ for adjoint relative to $\langle\ ,\ \rangle$. The actual Ω image is contained in

$$B = \{a \in A_{\mathrm{Sp}} : aX - \alpha(a)X \in \ker \ell,\ a^\dagger X - \alpha(a)X \in \ker \ell\}. \tag{30}$$

We now identify the whole algebra, including its radical.

Odd half-dimension. Suppose m is odd. Put $H = \ker \ell \leq A$, of dimension $h - 1$, and $W = V^{[1]}$, of dimension d. The C-summand $KX + K\omega$ is nondegenerate and is R_y. In coordinates

$$H_t, W, KX, H_b, K\omega$$

the algebra B has the same arbitrary matrix on H_t, H_b, an arbitrary matrix on W, a common scalar on X, ω, and the five independent off-diagonal blocks

$$H_t \to H_b, \quad H_t \to W, \quad W \to H_b, \quad H_t \to K\omega, \quad KX \to H_b.$$

This follows by solving the x, y intertwiner equations and then the two conditions in (30); their codimension in A_{Sp} is $2d + 1$. The off-diagonal ideal J satisfies $J^3 = 0$ and $J^2 = \mathrm{Hom}(H_t, H_b)$. Its primitive arrow spaces are one copy in each direction between H and W, and between H and the trivial type.

The actual image fills the three semisimple blocks. Indeed H is absolutely irreducible: over the algebraic closure, its nonzero weights are the type-D roots. A root vector generates all roots and all differences $e_i \wedge f_i + e_j \wedge f_j$. A submodule with only weight-zero vectors must have equal diagonal coefficients and hence lie in $K\omega$, which has contraction one and misses H. Finite density transfers this statement to the finite group. The natural Frobenius and trivial types are distinct from this heart.

After conjugating a semisimple complement by $1 + J$, every absent primitive arrow would force an additional submodule or quotient: absence of a natural-heart arrow gives a nonzero map $W \to T$ or $T \to W$; absence of a trivial-heart arrow gives a second fixed vector in T or T^*. The first two maps vanish by the all-signature cubic invariant calculation in Appendix F; the fixed-vector spaces are one-dimensional by absolute irreducibility of V. Thus all four arrow spaces occur. The full diagonal blocks generate their entire rectangular spaces, and products through W generate J^2. Hence the actual image is B.

The indecomposable injectives, indexed by their simple socles, are

$$I_H = T/K\omega, \quad I_W = T/(A + KX), \quad I_1 = T/(\ker f_Q + KX). \tag{31}$$

Their dimensions are $2(h-1) + d + 1$, $h - 1 + d$, and h. From the displayed blocks their socles are respectively $H_b, W, K\omega$, each simple. Such a socle is essential: every nonzero submodule of a finite module contains a simple submodule. The dimensions equal the dual dimensions of the primitive right ideals. Extending the socle embedding into its injective envelope therefore gives an injective map of equal-dimensional modules, proving the formulas.

Even half-dimension. Suppose m is even. Put $H_0 = \ker \ell$ and $L = H_0/K\omega$, of dimension $l = h - 2$. Choose $a \in A$ with $\ell(a) = 1$ and $Y \in T$ with $xY = a$. The four-dimensional C-module

$$R_Q = \mathrm{span}_K\{X, Y, \omega, a\}$$

has $xX = \omega$, $xY = a$, $yX = 0$, $yY = \omega$. It is nondegenerate for the tensor pairing, since $\langle X, a\rangle = \langle Y, \omega\rangle = 1$ and $\langle X, \omega\rangle = 0$. Thus

$$T \cong R_x^l \oplus K^d \oplus R_Q.$$

Use the seven basic positions $L_t, W, X, Y, L_b, \omega, a$. The semisimple diagonal of B is $\mathrm{M}(l, K) \times \mathrm{M}(d, K) \times K$, with the last scalar common on the four positions of R_Q. Solving its intertwiner equations gives the following radical maps; a paired entry uses the same coefficient in both positions:

$$\begin{array}{lll} n : L_t \to L_b, & \beta : Y \to X, a \to \omega, & \\ p : X \to \omega, & q : Y \to \omega, & t : Y \to a, \\ a_1 : L_t \to X, L_b \to \omega, & a_2 : L_t \to \omega, & a_3 : L_t \to a, \\ b_1 : Y \to L_t, a \to L_b, & b_2 : X \to L_b, & b_3 : Y \to L_b, \end{array}$$

plus $L_t \to W$, $W \to L_b$, $W \to \omega$, and $Y \to W$. Coefficients in the L positions are arbitrary vectors, covectors or rectangular matrices. The two conditions in (30) remove exactly $X \to a$, $X \to W$ and $W \to a$, giving codimension $2d + 1$.

For the radical J, the quotient J/J^2 has one arrow in each direction between L and W and two parallel arrows in each direction between L and 1. They may be represented by a_1, a_3 and b_1, b_2. With the rightmost factor acting first, the identities

$$a_1b_1 = \beta,\ b_2a_1 = n,\ a_3b_1 = t,\ a_1b_2 = p,\ a_2 = a_1n,\ b_3 = b_2\beta,\ q = p\beta$$

and

$$(W \to \omega) = a_1(W \to L_b), \qquad (Y \to W) = (L_t \to W)b_1$$

generate all remaining blocks. Assigning levels $Y : 4$, $L_t : 3$, $W, X, a : 2$, $L_b : 1$, $\omega : 0$ shows $J^5 = 0$.

The actual image surjects onto all semisimple factors by the same root argument for L and the distinct natural/trivial types. To obtain both parallel arrows, first note

$$(T/K\omega)^{\Omega} = 0. \tag{32}$$

Indeed $T^{\Omega} = K\omega$, and $H^1(\Omega, K) = 0$ by perfectness. The submodule $(A + KX)/K\omega$ is an extension of K^2 by L. If its two trivial-to-heart arrows were dependent, a nonzero combination of the two top coordinates would be annihilated by the radical after splitting the semisimple diagonal, yielding a fixed vector contrary to (32). Equivalently, a relation between the extensions defined by a and X would give $\lambda a + \mu X + u + K\omega$ fixed, for $u \in H_0$. By (32), its representative lies in $K\omega$. Applying x to that equality in T forces $\mu = 0$, and then applying ℓ in A forces $\lambda = 0$. Adjoint duality gives the two reverse arrows.

The remaining natural-to-heart extension is

$$0 \to L \to \ker(f_Q : \Gamma \to K)/K\omega \to W \to 0.$$

If it split, there would be a nonzero map $W \to T/K\omega$. But $\mathrm{Hom}_{\Omega}(W, T) = 0$ by the cubic fixed-vector calculation, and $\mathrm{Ext}^1_{\Omega}(W, K) = H^1(\Omega, W^*) = 0$ by Lemma D.2. The long exact sequence

forbids that map. Duality gives the reverse natural arrow. All six primitive arrow spaces occur, their products fill J, and the actual image is again precisely B.

The injectives now are

$$I_L = T/K\omega, \qquad I_W = T/(A + KX), \qquad I_1 = T^2/D_Q, \tag{33}$$

where

$$D_Q = \{(s + cX, f_Q(s)\omega) : s \in \Gamma, c \in K\}.$$

The graph is invariant because $gX - X \in H_0$ for $g \in \Omega$. Its parametrization is injective since $X \notin \Gamma$. The three dimensions are $2l + d + 3 = d^2 - 1$, $l + d + 1 = h + d - 1$, and $3l + d + 5 = (3d^2 - d)/2 - 1$, equal to the primitive-right-ideal dual dimensions.

For completeness, their socles can be read without an abstract dimension inference. In I_L, the radical moves every L_t, W, X, Y, a coordinate, leaving exactly L_b. In I_W, the surviving coordinates are L_t, W, Y; the arrows $L_t \to W$ and $Y \to L_t$ leave exactly W. For I_1, choose first-copy representatives L_t, Y and retain the second copy, with first a identified with second ω. The a_3 arrows kill the first L_t coordinate of a socle vector and b_1 kills its first Y. The second-copy L_t, W, X, Y, a coordinates are forced to zero by the displayed arrows, and a_1 sends second L_b to second ω. The sole socle is therefore $K(0, \omega)$. Essentiality and equality of dimensions with the right-ideal duals prove (33).

C.6 Prime field descent with a common trivial constituent

The orthogonal image just computed is a K-algebra. An arbitrary invariant denominator in the algorithm is only binary linear. The following descent proves the exact binary algebra, including all its radical entries.

Lemma C.3 (Perfect-group descent)**.** *Let $K = \mathbb{F}_{2^e}$, let G be a finite perfect group, and let $\rho_0 : G \to \mathrm{GL}_K(W)$ have a nonzero fixed vector. For $0 \le i < e$, let ρ_i be its coefficient-Frobenius twist. Suppose different twists have no common nontrivial absolutely simple composition factor. If $A_K = K[\rho_0(G)]$ and $\alpha : A_K \to K$ is its fixed-vector character, then*

$$k[\rho_0(G)] = kI \oplus \ker \alpha, \qquad \ker \alpha = \mathrm{span}_k\{\rho_0(g) - I : g \in G\}.$$

Proof. Use the group algebra only for this existence argument. Put $R = K[G]$, let $\epsilon : R \to K$ be augmentation, and put $\Delta = \ker \epsilon$. The map $g \mapsto g - 1 + \Delta^2$ is a homomorphism into an additive group. Perfectness gives $\Delta = \Delta^2$ and hence $\Delta^j = \Delta$ for all positive j.

Let $I_i = \mathrm{ann}_R(W_i)$, where W_i has action ρ_i. A fixed vector gives $I_i \le \Delta$. Evaluating elements of R/I_i on a basis of W_i embeds its left regular module into a direct sum of copies of W_i. Thus every composition factor of R/I_i occurs in W_i, even though the algebra may be nonsemisimple. For $i \ne j$, the quotient $R/(I_i + I_j)$ has only trivial composition factors after algebraic scalar extension. Some power of Δ therefore annihilates it. Since $\Delta^s = \Delta$, this gives $I_i + I_j = \Delta$.

For $e > 1$, multiplication of these equalities gives

$$\Delta = \prod_{i=1}^{e-1} (I_0 + I_i) \subseteq I_0 + \prod_{i=1}^{e-1} I_i \subseteq I_0 + \bigcap_{i=1}^{e-1} I_i \subseteq \Delta.$$

Every term containing I_0 lies in I_0 because the ideals are two-sided; no commutativity of ideal multiplication is used. Given $a \in \rho_0(\Delta) = \ker \alpha$, choose $w \in \bigcap_{i>0} I_i$ with $\rho_0(w) = a$. Let σ square the coefficients in R and set $u = \sum_{j=0}^{e-1} \sigma^j(w)$. Then $u \in k[G]$, $\epsilon(u) = 0$, and

$$\rho_0(\sigma^j(w)) = \sigma^j(\rho_{e-j}(w)) = 0 \quad (1 \le j < e).$$

Thus $\rho_0(u) = a$. Since u has binary coefficients and augmentation zero, it is a binary combination of $g - 1$. The reverse inclusion is immediate, as is the case $e = 1$. Finally $\alpha(I) = 1$, so adding the scalar identity is a direct sum. There is no division by e in this argument. □

For $G = \Omega^{\pm}_{2m}(2^e)$ with $m \geq 18$, the lemma's hypotheses follow as follows. Over an algebraic closure the natural type-D_m module has highest weight λ_1, and its exterior square has one factor $L(\lambda_2)$ and otherwise trivial factors: its only nonzero weights are roots, and its unique nonzero dominant weight is λ_2. The exact sequences

$$0 \to \Lambda^2 V \to V \otimes V \to \operatorname{Sym}^2 V \to 0, \qquad 0 \to V^{[1]} \to \operatorname{Sym}^2 V \to \Lambda^2 V \to 0$$

show that the nontrivial square factors are two copies of $L(\lambda_2)$ and one $L(2\lambda_1)$. In split and nonsplit type, the finite restricted-weight theorem [Ste68, §13, Theorem 43] makes

$$L(2^i\lambda_1)|_G, \quad L(2^i\lambda_2)|_G \qquad (0 \leq i < e)$$

all absolutely irreducible and pairwise distinct. In nonsplit type, the graph automorphism interchanges the spin nodes and fixes λ_1, λ_2, so these two families still have period e. The spin cover has infinitesimal characteristic-two kernel and is bijective on algebraic-closure points; unique lifting also identifies its finite fixed points with the actual connected orthogonal group. In stable rank this is the perfect Dickson kernel Ω [Tay92, Theorems 11.47 and 11.51]. Thus different coefficient-Frobenius square twists have disjoint nontrivial factor sets in both signs. A change of nonsplit coordinates only conjugates the twists and does not change their factors. Lemma C.3 proves the exact binary image $kI + \ker \alpha$ for both parity cases above.

C.7 One scalar endpoint for several factors

For a same-core image $A_k = kI + \ker(\alpha : A_K \to K)$, a nontrivial primitive idempotent e satisfies $eA_k = eA_K$. The binary dual of this right ideal is the same K-template via the trace pairing. At the trivial primitive idempotent e_0, the sole restricted coordinate is its scalar diagonal. Dualizing gives the binary injective

$$I_0/\{cs_0 : \operatorname{Tr}_{K/k}(c) = 0\},$$

where s_0 spans the K-socle of the defining-field injective. The trace pairing is nondegenerate even when $\operatorname{Tr}(1) = 0$.

For independent cores indexed by a finite set J, their augmentation ideals annihilate one another on the direct sum of the same-core modules. The joint algebra is the fiber product $kI \oplus \bigoplus_{j\in J} \ker \alpha_j$. It has one common trivial endpoint. Its trivial injective is

$$\left(\bigoplus_{j\in J} I_{0,j}\right) \Big/ \left\{(c_j s_{0,j})_j : \sum_{j\in J} \operatorname{Tr}_{K_j/k}(c_j) = 0\right\}. \tag{34}$$

This follows directly by dualizing the common-diagonal right ideal. The natural, dual, field-conjugate and repeated occurrences are first identified as actual modules; repeated labels do not create independent scalar characters.

C.8 Unmatched twists and independent factors

The remaining two-leg modules have natural occurrences at different Frobenius digit positions, or on two independent cores. They are absolutely simple after scalar extension by the tensor-product

and finite restricted-weight theorems. For unitary groups use $V^{*} \cong V^{[f]}$: equal positions modulo f are the already treated ordinary-square and adjoint cases, and unequal positions are distinct-digit simple cases. Their simple types are disjoint from the matched-square heart, natural and trivial types in stable rank.

Let W be the direct sum of these original binary bilinear representations. It is semisimple over k: after algebraic scalar extension it is the direct sum of the field conjugates of absolutely simple modules, and semisimplicity descends through the separable extension. Actual conjugate types can fuse, including half-period real forms. They are not separated artificially. For an exact section Q of passive copies of W, stack a basis of $\mathrm{Hom}_S(Q, W)$. The map into the corresponding copies of W is injective: a nonzero common kernel would contain a simple summand with a nonzero projection into W. This construction uses only binary linear equations and preserves the actual denominator.

The idempotent separating these semisimple types from the matched types is found by linear equations in the joint image algebra. Applying that idempotent to both U and P gives an exact direct decomposition of the section. No unproved defining-field closure of P is taken. The new coefficient target has polynomial size; an endomorphism field of a two-core simple, whose degree could be a quadratic-size least common multiple, is never enumerated.

D Cohomology of the degree two templates

An embedding of a direction module does not by itself extend an arbitrary affine action. The additional fact needed is vanishing of first cohomology in the larger target: then generator linear equations find an origin there. We prove that fact for the general-linear, unitary and symplectic templates. For orthogonal groups we prove it only for a matrix quotient and use explicit maps with natural or trivial kernels. This distinction is essential because the orthogonal trivial-socle injective can have nonzero first cohomology.

We use a uniform natural-dimension cutoff of 36 for the representation statements below, and retain the larger geometric cutoff in the main algorithm. In parabolic arguments the head dimension is at least 12. All cohomology is ordinary cohomology of the stated finite group, with coefficients in the displayed finite-dimensional module. We use $\mathrm{Ext}^1_A(X, Y)$ for equivalence classes of short exact sequences $0 \to Y \to E \to X \to 0$ of modules over the specified algebra or functor category; zero means every such sequence splits. The group H^2 enters only through the stated long exact sequences and cited trivial-coefficient or stability results. Restriction to a subgroup of odd index is injective in characteristic two, since restriction followed by corestriction is multiplication by that odd index.

D.1 General linear coefficient calculations

Let K be an even finite field, $E = K^a$, and $I_s(E) = E^{[s]}$ its Frobenius-natural module. A Frobenius functor is additive, so its ordinary polynomial degree is one, independently of the exponent 2^s. We use ordinary functors on finite-dimensional K-spaces, not a strict-polynomial degree that grows with the field.

Here are the external comparison statements and the elementary functor calculations needed with them. Franjou, Friedlander, Scorichenko and Suslin [FFSS99, Theorem 1.7 and Appendix, Theorem A.1] identify stable finite-general-linear Ext with ordinary functor Ext and give the exponential-functor decomposition. Putman's twisted stability theorem [Put23, Theorem B] gives the isomorphism range $a \geq 2i + 2d + 2$ for a degree-d coefficient system over a field, starting in degree zero. Our tensor systems, including duals via extension of functionals by zero along a specified complement, have this degree: under $E \to E \oplus K$, each factor splits into its old value plus a constant

line, and every difference lowers the number of nonconstant factors. Apply stability to the dual coefficient system and use

$$H^i(G, M) = H_i(G, M^*)^*$$

for finite groups and finite-dimensional coefficients. The dual stabilization maps are restriction followed by projection to the old-coordinate summand, exactly the finite-rank maps in the Ext comparison. Thus eventual vanishing gives finite-rank vanishing in the stated range.

Constant and reduced functors split by the zero object. The constant functor K is projective and injective, so all its Ext groups with a reduced functor vanish. An extension of two reduced additive functors is additive: its second cross-effect is zero because cross-effects are exact images of split natural idempotents. Additive K-valued functors are modules over the separable algebra $K \otimes_{\mathbb{F}_2} K$, a product of fields. Hence their first Ext vanishes. Finally the exponential-functor decomposition, or its sum-and-diagonal adjunction, gives zero Ext between tensor products of r and s additive functors when $r \neq s$: after splitting into the larger number of variables, every term is missing a variable in which the other side is reduced. At $r = s = 2$, degree-zero Hom leaves exactly the two permutation matchings. Each individual matching $\mathrm{Hom}(I_s, I_t)$ is K for equal actual Frobenius exponents and zero otherwise.

These facts give the following finite-group consequences, with $a \geq 12$ sufficient for all uses:

(i) $H^1(\mathrm{GL}(E), M) = 0$ for every whole natural/dual/Frobenius tensor word of length two or three, and for the symmetric-square quotient.

(ii) H^1 and H^2 vanish for every natural, dual, or Frobenius-natural module.

(iii) Odd-length tensor words of length at most five have no fixed vector; whole degree-four Hom spaces are the stated permutation pairings.

For a mixed tensor word, the cohomology is written as $\mathrm{Ext}^i(P(E), Q(E))$ with both P, Q covariant before applying comparison. For example $H^i(I_a \otimes I_b \otimes I_c^*) = \mathrm{Ext}^i(I_c, I_a \otimes I_b)$. The ranges are 10 for cubic H^1, 8 for natural H^2, and 12 for degree-five H^0.

The same H^0 and H^1 assertions hold between $\mathrm{SL}(E)$ and $\mathrm{GL}(E)$. Here is a cocycle-level argument for H^1. Let T be the diagonal torus and $T_0 = T \cap \mathrm{SL}(E)$. Normalize a cocycle on $\mathrm{SL}(E)$ to vanish on the odd-order group T_0. Evaluation on all elementary root subgroups is injective, and each such root subgroup involves only two coordinates. If coefficient weights have support at most r, the evaluated cocycle space has weight support at most $r + 2$. Inner action is trivial on cohomology, so every occurring torus character is trivial on T_0 and therefore has a constant coordinate exponent. A nonzero such character has support a, impossible for $a > r + 2$. Thus the determinant quotient acts trivially on H^1, and odd-order inflation-restriction proves the assertion. The binary determinant quotient is trivial. The H^0 assertion is the torus-projector argument from Appendix C.

Stable special-linear groups are perfect and have no characteristic-two Schur multiplier [Ste81]. Therefore H^2 with trivial characteristic-two coefficients vanishes. Combining this with whole-adjoint $H^1 = 0$ proves $H^1(\mathrm{SL}(E), \mathrm{End}(E)/P) = 0$ for every invariant scalar subspace P. Thus all general-linear injective targets in Appendix C are H^1-acyclic.

D.2 Symplectic injectives

Let $V = E \oplus E^*$ be a symplectic K-space of dimension $2m$, and $G = \mathrm{Sp}(V)$. Its Siegel parabolic is $P = U \rtimes L$, where $L = \mathrm{GL}(E)$ and

$$U = \left\{ u(B) = \begin{pmatrix} I & B \\ 0 & I \end{pmatrix} : B = B^t \right\}.$$

The index $[G : P] = \prod_{i=1}^{m}(|K|^i + 1)$ is odd. Its radical is the additive module $\Gamma^2 E = \ker(1 + \tau) \leq E \otimes E$.

Lemma D.1. *For $m \geq 12$,*

$$H^1(G, V \otimes V) = H^1(G, \operatorname{Sym}^2 V) = H^1(G, (V \otimes V)/K\omega) = 0.$$

The same holds for unmatched two-slot Frobenius words and for the two parity-dependent trivial injectives in (28)*, including their binary trace-socle quotients.*

Proof. Normalize a parabolic cocycle to zero on L, using the general-linear results on its three whole tensor grades. The first nonzero grade of its restriction to U is an additive L-map. In the bottom and middle grades of $V \otimes V$, the dual-square and adjoint/trivial types do not match the exterior/Frobenius-natural types of U, so these maps vanish. The only remaining map is

$$\Gamma^2 E \to E \otimes E, \qquad B \mapsto cB.$$

The exact square image (26) proves this is the full Hom space. Over an extension field the central scalar squares span K, so an additive intertwiner here is automatically K-linear. The L-fixed tensor $r = \sum_i e_i \otimes f_i$ has $u(B)r - r = B$. Subtracting the coboundary of cr kills the cocycle on U. Odd-index restriction proves the first vanishing.

For $\operatorname{Sym}^2 V$, the same argument leaves only the map from symmetric tensors to their diagonal square quotient, $B \mapsto c\sum_i B_{ii} e_i^2$. The L-fixed quadratic tensor $q_0 = \sum_i e_i f_i$ has exactly this displacement. This proves the second vanishing. With two unequal Frobenius positions, the top tensor is a distinct-digit simple type and admits no map from $\Gamma^2 E$; the normalized cocycle is already zero.

For trivial coefficients, the symplectic ordinary-functor comparison [DV10, Corollaries 3.32 and 3.33], valid over every finite field for $m \geq 2i + d + 6$, gives $H^2(G, K) = 0$ by taking the projective constant functor of degree zero. The exact sequence $0 \to K\omega \to V \otimes V \to I_H \to 0$ proves $H^1(G, I_H) = 0$.

For odd m, the map $I_0 = T/\ker \ell \to T/A = \operatorname{Sym}^2 V$ has one-dimensional trivial kernel $A/\ker \ell$, so

$$0 \to K \to I_0 \to \operatorname{Sym}^2 V \to 0.$$

For even m, the graph presentation gives

$$0 \to T \xrightarrow{t \mapsto [(0,t)]} I_0 \xrightarrow{[(t_1,t_2)] \mapsto t_1 + A} \operatorname{Sym}^2 V \to 0.$$

If $t_1 = a \in A$, subtract its graph relation to obtain $[(0, t_2 + \ell(a)\omega)]$, proving exactness. The outer H^1 groups vanish in both sequences, so $H^1(G, I_0) = 0$. Finally every trace-socle denominator is pointwise fixed. Its H^2 vanishes by the same trivial-coefficient result, so quotienting by it preserves the vanishing. Product groups have the same trivial H^2 conclusion because the additional degree-two Kunneth terms contain H^1 of a perfect factor. □

D.3 Unitary injectives

Let K/K_0 be quadratic of characteristic two, with conjugation σ, and let $D = \mathrm{SU}_a(K/K_0) \trianglelefteq U = \mathrm{U}_a(K/K_0)$. On any module with torus weights supported in at most r coordinates,

$$H^1(D, M) = H^1(U, M) \qquad (a > r + 3). \tag{35}$$

To prove this, U is generated by its coordinate two- and three-dimensional unitary groups and its diagonal norm-one torus. Indeed, if two nonzero entries of a norm-one vector have unequal K_0-norms, a two-space isometry merges them into one. Otherwise all nonzero norms agree, their number is odd, and three entries have nonzero total norm; a three-space isometry merges those. Repetition reduces support, and successive columns give generation. Each local unitary element is a local special-unitary element times a diagonal scalar, and determinant-one diagonal tuples are products of two-coordinate $\operatorname{diag}(t, t^{-1})$. Thus coordinate SU_3 groups generate D. Normalizing a cocycle on the odd determinant-one torus, evaluation on these groups is injective and has weight support at most $r + 3$. The constant-exponent argument from the general-linear case proves (35). The use of three coordinates includes $K_0 = \mathbb{F}_2$, where two-coordinate groups alone would not suffice.

A central scalar z of order $|K_0| + 1$ acts on $V^{[a]} \otimes V^{[b]}$ by $z^{2^a+2^b}$. It acts nontrivially unless the two positions differ by the conjugation exponent, the adjoint case. Odd-order averaging therefore kills all cohomology for every nonadjoint square, its symmetric quotient and unmatched type under U, and (35) gives $H^1 = 0$ under D.

For the adjoint, a direct parabolic calculation is needed. If the dimension is $2m$, use $V = E \oplus E^{\sigma *}$. The odd-index special-unitary parabolic has Levi

$$L' = \{g \in \mathrm{GL}_m(K) : \det g \in K_0^{\times}\}$$

and Hermitian additive radical $u(Z) = \left(\begin{smallmatrix} I & Z \\ 0 & I \end{smallmatrix}\right)$. General-linear cohomology, including the intermediate determinant statement, normalizes a cocycle to zero on L'. Finite whole-word Hom matching leaves only the top adjoint block, with value cZ. This is the coboundary of c times the projector onto E, so restriction is zero.

If the dimension is $2m + 1$, write $V = E \oplus Kt \oplus E^{\sigma *}$. Put $\chi(g) = (\det g)^{|K_0|-1}$. Its Levi and radical are

$$l(g) = \operatorname{diag}(g, \chi(g), g^{-\sigma t}), \qquad u(X, Z) = \begin{pmatrix} I & X & Z \\ 0 & 1 & X^{\sigma t} \\ 0 & 0 & I \end{pmatrix}, \quad Z + Z^{\sigma t} = XX^{\sigma t}.$$

The center C has $X = 0$, and the quotient radical is $E \otimes \chi^{-1}$. Assign vector weights $2, 1, 0$ and endomorphism weight equal to output minus input weight. Normalize the Levi cocycle using the degree-two general-linear results; determinant-twisted cross-tail words vanish by torus support. The first central value is again cZ, removed by the head projector. With $c(C) = 0$, the only first quotient-radical values are

$$aX \in \operatorname{Hom}(Kt, E), \qquad bX^{\sigma t} \in \operatorname{Hom}(E^{\sigma *}, Kt).$$

Because commutators lie in C, the cocycle identity in the top block gives

$$(a + b)(XY^{\sigma t} + YX^{\sigma t}) = 0.$$

Distinct coordinate vectors force $a = b$. The coboundary of a times the t-projector removes both values. No possible first grade remains. Odd-index restriction proves $H^1(D, \operatorname{End}_K(V)) = 0$ in both dimensions. Since $\operatorname{End}_K(V) = \mathcal{H} \oplus \theta\mathcal{H}$ over K_0, the Hermitian form also has vanishing H^1. Stable SU has no characteristic-two multiplier [Ste81, Theorem 1.1 and the twisted-group exceptions]; the scalar quotient and correlated trace-socle cases follow from their long exact sequences.

D.4 Orthogonal squares and natural modules

Lemma D.2. *For a stable nonsingular quadratic K-space V of either sign and $G = \Omega(V, Q)$,*

$$H^1(G, V^{[a]}) = 0, \quad H^1(G, V^{[a]} \otimes V^{[b]}) = 0, \quad H^1(G, (V \otimes V)/K\omega) = 0$$

for all Frobenius positions a, b.

Proof. Use the parabolics explicitly constructed in Appendix C. In split type $V = E \oplus E^*$, the Levi is $\mathrm{GL}(E)$ and the additive radical is $\Lambda^2 E$. Its index is odd: its two-part has order $|K|^{m(m-1)}$, the two-part of $\Omega^+_{2m}(K)$. Normalize on the Levi. A first radical value in a two-slot word is an additive Levi map from $\Lambda^2 E$. Precompose with $E \otimes E \twoheadrightarrow \Lambda^2 E$ and use the four-leg matching. Only a matched top square can occur, and its map is the alternating inclusion times a scalar. Under $V \otimes V \cong \mathrm{End}(V)$, this is the coboundary of the head projector. Unmatched positions give zero immediately.

In nonsplit type $V = E \oplus Z \oplus E^*$, the Levi is $\mathrm{GL}(E) \times T_0$, where $T_0 = \mathrm{SO}(Z)$ has order $|K| + 1$. The radical has center $C = \Lambda^2 E$ and quotient $W = \mathrm{Hom}_K(Z, E)$. Its two-part, including the Levi, has order $|K|^{m(m+1)}$, the two-part of $\Omega^-_{2m+2}(K)$, so again restriction is injective. Levi H^1 vanishes on each head word by the general-linear calculation; cross-tail words have nontrivial T_0-character; tail-only trivial components use the odd-order general-linear abelianization. Normalize on the Levi. The center is removed by the head-projector coboundary as in split type.

For unmatched twists $0 \le a, b < f$, a quotient-radical map to a one-head tensor requires equality of tail characters, namely $2^a = \pm 2^b \pmod{2^f + 1}$. Their nonzero differences and sums have absolute value less than $2^f + 1$, so this requires $a = b$. Two-head and zero-head targets are excluded by the tail character and head natural factor. Thus unmatched cocycles vanish.

In the matched case use $\mathrm{End}(V)$ and weights $2, 1, 0$ on E, Z, E^*. The first quotient-radical values have precisely the form

$$XD \in \mathrm{Hom}(Z, E), \qquad EX^* \in \mathrm{Hom}(E^*, Z), \qquad D, E \in \mathrm{End}_{T_0}(Z),$$

where $X^* = J_Z^{-1} X^t$. The commutator identity, with the cocycle zero on C, gives in the top block

$$X(E + D)Y^* + Y(E + D)X^* = 0.$$

Take X, Y with different single nonzero head rows and arbitrary row functionals on Z. The two output matrix positions are distinct, forcing every entry of $E + D$ to vanish. The diagonal endomorphism supported by D on Z has the corresponding coboundary. Subtracting it removes the only possible first grade and proves whole-square $H^1 = 0$.

For a natural coefficient module, a central-radical map is excluded by the odd three-leg Hom calculation. In nonsplit type the quotient radical cannot map to a head natural row because of its nontrivial tail character, nor to the tail because of its nontrivial head factor. Hence natural $H^1 = 0$ as well. Finally stable Ω is perfect and has no characteristic-two multiplier [Ste81, Tay92]. Thus $H^2(G, K) = 0$, and the sequence $0 \to K\omega \to V \otimes V \to I_H \to 0$ proves the last assertion. □

D.5 The orthogonal map with a small kernel

The orthogonal trivial injective cannot be added unchanged to an acyclic target. For odd half-dimension, the class $g \mapsto gX - X$ in the alternating heart is nonzero: otherwise adjusting X would give an additional fixed tensor. Trivial H^2 lifts this class to the trivial-socle injective. We use the following maps instead.

For both parities, $I_H \to I_H$ is the identity and

$$I_W \to I_H, \qquad [Y] \mapsto xY + K\omega$$

has kernel $\Gamma/A \cong V^{[1]}$. If $xY = c\omega$, subtract cX to put Y in Γ; the remaining denominator is A, proving the kernel formula. For odd half-dimension the same formula on I_1 has the one-dimensional kernel $\Gamma/\ker f_Q$. For even half-dimension use

$$F_1 : I_1 \to I_H^2, \qquad [(Y, Z)] \mapsto (xY + K\omega, Z + K\omega). \tag{36}$$

A kernel representative has $xY = c\omega$ and $Z = d\omega$. Subtract $(cX, 0)$ and then $(Y - cX, f_Q(Y - cX)\omega)$ from its defining graph, leaving a multiple of $(0, \omega)$. That class is nonzero, since a relation with zero first coordinate has both its parameters zero. Thus the kernel is exactly a trivial line. Each map kills its entire original scalar socle.

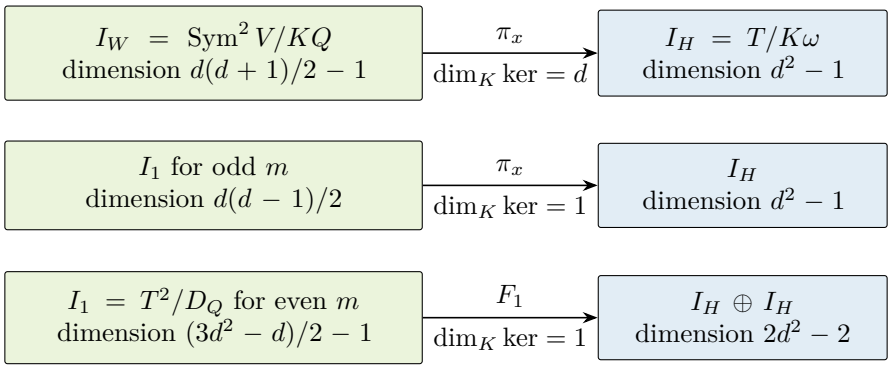


Figure 10: The orthogonal affine reduction, with $d = 2m = \dim_K V$, $T = V \otimes_K V$ and $x = 1 + \tau$. All dimensions here are over K. Here $\pi_x([Y]) = xY + K\omega$, and F_1 is (36). The targets have zero first cohomology; the kernels are one Frobenius-natural module or one trivial line. The remaining affine kernel equation is solved under the actual returned stabilizer.

For a common endpoint (34) involving other families, first quotient every participating nonorthogonal trivial injective by its full scalar socle and embed that quotient by Lemma C.1 into fresh copies of that family's acyclic cogenerators. Combining these maps with the orthogonal maps kills the entire sum of the original socles and therefore descends through the actual shared trace denominator. The resulting kernel is the actual correlated quotient of those scalar lines together with the orthogonal Frobenius-natural kernels. It is a supported degree-one section. The new target is an ordinary direct sum of acyclic modules, with no additional shared quotient.

Finally, a two-independent-core module $X \boxtimes Y$ has

$$H^1(G_1 \times G_2, X \boxtimes Y) = H^1(G_1, X) \otimes Y^{G_2} \oplus X^{G_1} \otimes H^1(G_2, Y) = 0.$$

Both natural fixed spaces vanish, even if an individual symplectic natural H^1 does not. Scalar extension, binary restriction and passive copies preserve all the vanishings proved here. Thus every direction in Appendix C has an embedding followed by a map into a jointly solvable acyclic target, with a genuine degree-one kernel.

E Complete affine transport with at most two large occurrences

We now assemble the lower-degree algorithm. The main point is the order of operations. First solve all large literal matrix and form equations simultaneously. Next impose their additive scalar and

finite quotient conditions. Only then evaluate a supplied affine cocycle on elements of its actual domain. A final natural-module affine problem handles the kernel of the map to the acyclic target. Every list of choices is completed and reunited before this procedure returns.

E.1 One joint family of scalar shifts

Suppose the same unknown $g \in \mathrm{GL}_d(K)$ must satisfy several equations modulo scalar matrices,

$$gA_jg^{-1} = B_j + \lambda_j I \qquad (1 \le j \le t).$$

Here $A_j, B_j \in \mathrm{M}(d, K)$ are given and $\lambda_j \in K$ are unknown. The number of labels t can be large. Each possible tuple (λ_j) with a nonzero intertwiner is a common scalar weight of the operators

$$L_j : \mathrm{M}(d, K) \to \mathrm{M}(d, K), \qquad L_j(X) = XA_j - B_jX.$$

Start with the whole d^2-dimensional matrix space and successively replace each current space W by its nonzero intersections $W \cap \ker(L_j - \lambda I)$, for all $\lambda \in K$. No invariance of W under L_j is needed. Distinct eigenvalues give independent subspaces of W, so the sum of all retained dimensions never exceeds d^2. Thus at most d^2 joint weights survive. Field enumeration costs $|K|$, not $|K|^t$.

For several original natural factors $K_i^{d_i}$, the product of the candidate counts is at most $\prod_i d_i^2 \le 4^{\sum_i d_i}$, and $\sum_i [K_i : k]d_i = O(D_0)$. A scalar shift in a passive matrix multiplicity is expanded in a fixed matrix basis; all its coefficients enter this same joint-weight search. Off-diagonal and nonidentity Frobenius components have no scalar shift and are imposed as exact equations. Every candidate is finally tested against its one actual additive scalar or trace subspace. The entire list therefore costs $2^{O(D_0)}$ times polynomial coefficient work.

E.2 The simultaneous linear template solver

All templates in Appendix C reduce to this one joint shift family followed by Theorem A.5. We record their exact equations to specify the algorithm on arbitrary quotients.

For a symplectic factor, $I_H = T/K\omega$ is a matrix modulo scalar identity and $I_W = \mathrm{Sym}^2 V$ is a quadratic form. An odd-half-rank I_0 coordinate, represented by $Y \in T$, requires

$$gY - Y' \in A, \qquad \ell(gY - Y') = 0.$$

An even-half-rank coordinate represented by $(Y, Z) \in T^2$ requires

$$gY - Y' \in A, \qquad gZ - Z' = \ell(gY - Y')\omega.$$

Thus impose all first-coordinate quadratic classes and all second-coordinate matrix shifts in the same system. Under its returned stabilizer, $sY - Y \in A$ and $s \mapsto \ell(sY - Y)$ is additive. Lemma A.1 imposes all remaining scalar conditions jointly.

For an orthogonal factor include Q as a literal fixed quadratic form. An I_W value is a quadratic form modulo KQ; its polar determines the same scalar shift required for the quadratic equation. For odd half-dimension an I_1 value represented by Y requires

$$x(gY - Y') = c\omega, \qquad f_Q(gY - Y' - cX) = 0.$$

For even half-dimension a value represented by (Y, Z) requires

$$x(gY - Y') = c\omega, \quad gZ - Z' = d\omega, \quad d = f_Q(gY - Y' - cX). \tag{37}$$

The first equations are matrix-mod-scalar conditions. Once their shifts and all quadratic observations have been fixed, the last equations are additive conditions on the full returned stabilizer. The Dickson-zero conditions are imposed simultaneously. General-linear square templates are literal bilinear and quadratic observations; adjoint templates are matrices or Hermitian matrices modulo their scalar subspaces. Cross-core and unmatched templates are literal bilinear observations on the original binary rows.

Collect *all* these matrix, bilinear and quadratic equations before calling Theorem A.5. Do not solve in separate overgroups and intersect arbitrary returned subgroups. After this one literal solve, impose all scalar and correlated trace equations by the additive image/kernel algorithm. Restrict to the actual perfect cores using Dickson for orthogonal factors and the determinant or unitary-determinant records for the other factors. The joint determinant image has order at most $2^{O(D_0)}$ and is handled by exact bounded-image point operations. At this point every witness and every stabilizer generator is in the original core; only such elements are passed to an arbitrary supplied core cocycle.

Finally, if the successful completed records are $a_j S_j$, choose one a_0 and generate

$$S = \langle S_j, a_0^{-1} a_j : \text{all } j \rangle.$$

Every generator fixes the original source. Conversely every original solution occurs in one completed record. Hence their union is exactly $a_0 S$. This proves the complete linear template solver with all kernels and original words.

E.3 An arbitrary affine cocycle

Let Q be any exact binary invariant section of the supported degree-two representation of the current full core S. Let $i : Q \hookrightarrow I$ be the all-Hom embedding into its joint cogenerators. Appendix D constructs an equivariant map $F : I \to E$ such that $H^1(S, E) = 0$ and $K = \ker F$ is a section of passive natural, dual, Frobenius-natural and trivial rows. The total distinct faithful degree of those rows is $O(D_0)$. This includes the full-socle construction for a shared trace endpoint.

Given an arbitrary cocycle $z : S \to Q$, solve the generator equations

$$Fi(z(g)) = (g - 1)b \qquad (b \in E).$$

A solution exists by the proved vanishing. For source and target $x, y \in Q$, solve the joint linear transporter from $Fi(x) + b$ to $Fi(y) + b$ by the preceding subsection. If empty, reject. Otherwise let its complete answer be aS_0, where S_0 stabilizes $Fi(x) + b$. Normalize the target by the original affine action, $y_0 = a^{-1} * y$. Then $i(y_0 - x) \in K$, and

$$c_x(s) = i(s * x - x) \in K \qquad (s \in S_0)$$

is the exact centered cocycle. Theorem A.2, applied to S_0 itself, solves $c_x(s) = i(y_0 - x)$ with its full stabilizer. Composing its answer with a gives exactly the original affine transporter. Even if K contains values outside $i(Q)$, no solution is added: both the target and every cocycle value arise from the original Q, and equality in the larger injective coordinates is the same equality.

The all-Hom target has dimension $O(D_0^4 \dim_k Q)$ when one copy of each required cogenerator has dimension $O(D_0^2)$. The additional socle-quotient embedding is performed once, with another fixed polynomial bound. These are coefficient spaces only. Group membership, normal closure and lifting remain on the original faithful representation.

E.4 The actual focused quotient

We pass from a full core to the subgroup inherited from earlier constraints. Fix an absolute arity r. An allowed coefficient representation is a section of tensor words of length at most r in current argument grades, their duals and their stated permutation or factorization covers. At most two of its positions are current large natural occurrences. The other positions are processed by evaluation; their moving actions are not discarded.

At one completed evaluation focus, only constantly many selected complementary parents occur. A bounded parent contributes its actual current graded row representation. A deleted-permutation parent contributes the *selected coordinate multiplicity* after its coordinate index and evaluation-domain condition have been stabilized. Its action factors through the original primitive-node quotient by the alternating component, of order at most 2^{5d} for parent degree d. It does not contribute the whole alternating letter group. All internal flag refinements of one parent remain under this one parent action.

Let W_{f} be the direct sum of these selected graded representations and all needed large isotypic rows, including rows defining form lines. Its degree is $O_r(D_0)$. The actual homomorphism

$$\pi : H \to L \leq \mathrm{GL}(W_{\mathrm{f}})$$

has kernel acting linearly trivially on the whole focused coefficient direction, because that direction is formed by tensoring, taking sections and evaluating precisely these rows. The kernel may be nonsolvable; its actual affine image is removed by Lemma 4.3.

The image L has a known normal core S with $[L : S] \leq 2^{O_r(D_0)}$. To verify the bound, write the large rows as $X_j \otimes_{K_j} M_j$, with dimensions a_j, b_j and $f_j = [K_j : k]$. Their permanent threshold and original mass give

$$f_j a_j^2 > C_* D_0, \qquad \sum_j f_j a_j b_j \leq \kappa D_0, \qquad \sum_j f_j b_j^2 \leq (\kappa^2/C_*) D_0.$$

The last sum bounds the logarithm of the entire commuting multiplicity image. There are $O(\sqrt{D_0})$ distinct large rows, so their permutations have size $2^{O(D_0)}$. Field, duality, similitude and determinant/Dickson records have the same total bound. Each of the constantly many selected complementary parents has one bounded local image; their product is $2^{O_r(D_0)}$. The hereditary theorem makes every newly large core trivial on these complementary graded actions. These estimates concern the joint actual quotient. They do not multiply the full orders of all previously recorded complementary groups.

Enumerate L/S with actual lifts. In each coset tS, apply t^{-1} to the whole target in the original affine coordinates and complete the pure-core algorithm above. Reunite all completed answers, then correct the witness and every generator through the actual kernel translations and adjoin their whole kernel. If S is an effective central product, use its direct-product natural cover temporarily and project the complete answer with its central kernel. No physical splitting of $H \to L$ is needed.

E.5 Balanced evaluation and its total cost

We give the reduction to those focused calls, including its cost. It is independent of the outer cubic recursion. Realize a dual row section A/B as $B^{\perp}/A^{\perp}$ in its original dual, and tensor the corresponding numerators with the full sum of denominator terms. Pull the actual U and P back once through these maps. Evaluation on tuples of vectors in complementary components now determines the multilinear coefficient, with its ambiguity retained by the exact sequence in

Lemma 4.1. The explicit evaluation domain has size $2^{O_r(D_0)}$; vectors of the large natural value spaces are not listed.

Process domain orbits sequentially under complete cosets. On one transitive domain, let b_i be the number of component incidences in position i, and put $b = \sum_i b_i$. Choose a largest b_i and split its component set into near-equal halves. Its actual subset orbit has size at most 2^{b_i}; breadth-first subset-orbit search and Schreier generators give the full stabilizer. For every subset image, normalize by its original transversal, solve the first half, then the second half under the entire returned source stabilizer and exact remaining quotient. Reunite only after both calls finish. Once b exceeds a constant depending on r, each child has at most $\gamma_r b$ incidences for some $\gamma_r < 1$. A bounded incidence set is exactly a focus as above.

Subgroup preparation can newly move some argument mass into evaluation. Keep already complementary parents as persistent index units: their later internal refinements do not create fresh incidences. Reconstruct each one cover from its original parent rather than covering a former cover. If newly moved original mass is c, it creates at most $A_r c$ incidences and increases the domain size by at most $2^{E_r c}$. With v the not-yet-moved mass, set $\Lambda = b + \lambda_r v$ for a fixed $\lambda_r > A_r$. Then

$$\Lambda - \Lambda_{\text{child}} \geq (1 - \gamma_r)b + (\lambda_r - \gamma_r A_r)c.$$

A fixed multiple of this decrease pays for the subset-image list and the domain enlargement. Initially $\Lambda = O_r(D_0)$. Direction dimensions split additively in the exact evaluation sequences and argument grades. Consequently a bound $2^{C_r \Lambda}$ times a fixed polynomial in the explicit direction/domain size pays for the sum of recursive costs, once C_r is large enough. Zero-direction fibers are direct checks. Equal-direction unary preparation strictly reduces the subgroup, refines a retained flag, or advances an original coefficient cell; there are polynomially many such events. All large outside lists are confined to completed focus calls.

Natural dimensions below the fixed cutoff a_* cause no exception: for a factor over $\mathbb{F}_{2^f}$ with $a < a_*$, its whole order has logarithm $O(fa^2) = O(a_* fa)$. Distinct original masses add, so their simultaneous local enumeration costs $2^{O(D_0)}$, including small nonperfect groups. This argument applies only to bounded natural rank. Subthreshold groups with unbounded rank remain complementary evaluation parents and are handled by the preceding balanced recursion.

Theorem E.1 (Complete supported degree-two transport)**.** *For every actual subgroup on the original arguments, every represented bounded-arity section with at most two current large natural occurrences, and every supplied affine cocycle on that section, a deterministic algorithm returns its complete transporter in $2^{O_r(D_0)}$ times polynomial original-data work. It preserves arbitrary binary numerators and denominators, moving complementary actions, all quotient kernels and original lift words.*

Proof. Apply the exact argument-grade filtration, the balanced evaluation recursion, the focused quotient and actual translation lifting, and the pure-core affine solver, in that order. Each construction proves its hypotheses for the next. The cost was established above, and each step preserves equality of complete transporter sets. No arbitrary group intersection is used in this construction. □

E.6 Subspace conductors and coset intersection

Only now do we derive the two operations used by the outer component algorithm. For s-dimensional subspaces with injective frame matrices $A, B : k^s \hookrightarrow W$, solve

$$hA = BX, \qquad (h, X) \in H \times \mathrm{GL}_s(k).$$

This is a literal Hom transporter under the displayed degree-at-most-2 $\dim W$ faithful cover. Theorem E.1 gives the entire solution coset; projecting to H gives its exact subspace conductor. The temporary frame group and its kernel are kept until projection. No frame orbit is exported.

For cosets aH and bK of actual matrix groups on W, use the left-right action of $H \times K$ on $\mathrm{End}(W)$. The identity equation

$$hk^{-1} = a^{-1}b$$

is a degree-two point transporter. Multiply its complete first-coordinate projected answer by a to obtain exactly $aH \cap bK$. This also retains every kernel of either supplied representation by using the original faithful actions in parallel. These operations are consequences of the completed lower-degree theorem, so their use in component assembly is noncircular.

F Pairing insertions and cubic affine descent

The characteristic-two difficulty is not confined to quadratic refinements. A cubic representation can have a Frobenius-fused degree-two composition factor even over the binary field. We therefore prove containment for the *actual translation submodule*, using odd-order tori and the geometry of orthogonal vectors. We then prove that the cubic quotient by all eligible insertions has zero fixed space and zero first cohomology. The actual smaller translation denominator is retained throughout Section 4.

Let G be one stable classical core on its natural space V over its defining field K. A cubic word M is a tensor product of three specified natural, dual, or Frobenius-natural occurrences, with passive multiplicities. An eligible insertion pairs two occurrences by an actual invariant bilinear or Hermitian evaluation and leaves a natural parameter in the third. Write $J \leq M$ for the sum of those images. All representation statements can first be checked after algebraic scalar extension; their containment and origin equations descend by linear algebra.

F.1 Support of an actual translation submodule

Lemma F.1. *Suppose $P \leq M$ is a G-invariant subspace all of whose composition factors occur in natural tensor words of degree at most two on the original arguments. Then $P \leq J$. The same conclusion holds for products of independent cores and every Frobenius signature.*

Proof. First let $G = \mathrm{Sp}_{2m}(K)$ and use hyperbolic planes $U_1, \dots, U_m$. In each binary $\mathrm{SL}_2(2)$ choose its subgroup of order three, giving a torus $T = C_3^m$. Over a splitting field, the natural representation on U_i has two nontrivial characters on coordinate i and is trivial on the others. Every degree-at-most-two word therefore has only characters supported on at most two coordinates. Since $|T|$ is odd, this remains true for extensions and sections of those modules. A cubic tensor with its three positions in different planes has only characters of support three. Its component in every member of P is zero.

Invariance and Witt extension now show that every trilinear form in P vanishes on independent pairwise polar-orthogonal triples. Independent pure triples span the full tensor cube of a three-dimensional totally isotropic space, also over $\mathbb{F}_2$: for two fixed independent arguments the vectors outside their span span the third space, and the same replacement in the first two slots removes repeated arguments. Consequently the form vanishes on every pairwise orthogonal triple.

We spell out the coordinate consequence. In a hyperbolic basis, a coefficient without a paired e_i, f_i is zero. For a free basis vector w in a different plane, the orthogonal triple $(e_i + e_j, f_i + f_j, w)$ makes $T(e_i, f_i, w)$ independent of the paired plane; $(e_i + f_i, e_i + f_i, w)$ makes it independent of the

order of that pair. Repeating in the three free positions defines linear functionals $\lambda_1, \lambda_2, \lambda_3$ such that, after subtracting

$$\lambda_1(x)b(y,z) + \lambda_2(y)b(x,z) + \lambda_3(z)b(x,y),$$

only mixed triples in one plane remain. For distinct spare planes j, l, the pairwise orthogonal triple

$$e_i + e_j, \quad e_i + e_l, \quad f_i + f_j + f_l$$

isolates the remaining coefficient $T(e_i, e_i, f_i)$. Permutations and exchanging e_i, f_i kill all six such coefficients. Hence P lies in the three pairing-insertion diagrams.

All Frobenius twists agree on binary matrices. After this argument, isolate a paired coefficient in one plane and its free coefficient in another. Scaling $e_i \mapsto te_i$, $f_i \mapsto t^{-1}f_i$ makes the two pair orders acquire factors $t^{2^a-2^b}$ and $t^{2^b-2^a}$. They must remain equal. The order of $K^\times$ is odd, so a nonzero pairing requires $a = b$ modulo the field degree. Thus exactly the eligible twists survive.

For quadratic orthogonal groups use three hyperbolic four-spaces, each with the subgroup $g \oplus g^{-t}$ for $g \in \mathrm{GL}_2(2)$. Its order-three torus has no fixed natural vector and lies in Ω, since its Dickson rank is doubled. The same support argument applies. Any independent polar-orthogonal triple can be moved into these three blocks preserving its three quadratic values: each hyperbolic block represents every value by $Q(e + af) = a$. Witt extension provides an isometry; a nonsingular symmetry in an unused plane corrects its Dickson invariant while fixing the triple. Additive replacements extend the vanishing to dependent triples. The symplectic-basis calculation above uses only binary coefficients and therefore applies to each separately Frobenius-semilinear signature, including an anisotropic two-plane tail. Hyperbolic scalings in Ω remove unmatched paired twists. This proves both signs.

For unitary groups use the same binary four-blocks. Every block represents every Hermitian norm value, because $h(e + tf, e + tf) = t + \bar{t}$ is surjective onto the fixed field. An unused anisotropic line corrects the determinant of a Witt extension, giving a special-unitary transporter. For repeated isotropic arguments choose $w \in x^\perp \cap z^\perp$ outside their span and add the independent triples $(x, x + w, z)$ and (x, w, z); three repetitions are handled in a three-dimensional totally isotropic space. Thus the needed dependent orthogonal triples also vanish.

Choose an orthonormal Hermitian basis. At binary coordinates the form is the ordinary dot product. A trilinear form vanishing on all pairwise dot-orthogonal triples is a sum of the three dot insertions: distinct-index coefficients vanish; $(e_i + e_l, e_i + e_l, e_j)$ makes the repeated-index coefficients independent of the repeated index. Subtract those insertions. The remainder has only diagonal coefficients d_i. The triple $(e_i + e_j, e_i + e_j, e_i + e_j)$ makes them all equal. Finally the pairwise orthogonal vectors

$$(1,1,1,1,0,0), \quad (1,1,0,0,1,1), \quad (1,0,1,0,1,0)$$

have a common support of size one, forcing that common coefficient to vanish. Norm-one diagonal scalars, with determinant compensation in a spare coordinate, leave a paired diagram only when $2^a + 2^b = 0 \pmod{2^f + 1}$ for $|K_0| = 2^f$. Equivalently $a - b = f \pmod{2f}$, exactly the Hermitian dual-twist condition.

For general-linear or special-linear groups in even natural dimension, restrict to a binary symplectic subgroup. Its support calculation gives three diagrams, but a determinant-one nonsymplectic transvection $e_i \mapsto e_i + e_l$ fixing the paired f coordinates creates an uncancellable three-plane coefficient from a natural-natural or dual-dual pairing. Only natural-dual evaluation survives. Full-field scalings with a compensating spare coordinate require matching twists. In odd dimension apply the even result on every even coordinate subspace. For a mixed word viewed as a bilinear map $B : V \times V \to V$, this shows $B(x, y) \in \mathrm{span}\{x, y\}$ on every binary coordinate model. Writing $B(e_i, e_j) = a_{ij}e_i + b_{ij}e_j$ and testing $e_i, e_j + e_l$, then $e_i + e_j, e_l$, makes a_{ij} depend only on j and b_{ij}

only on i. Testing $e_i + e_j, e_i + e_l$ determines the diagonal terms. Hence $B(x, y) = \alpha(x)y + \beta(y)x$. This gives precisely the two mixed evaluation insertions; pure words have none.

For three independent cores, product-torus support is three and the submodule is zero. With two occurrences of one core and one of another, support excludes distinct planes in the repeated pair. Invariance shows the corresponding bilinear form vanishes on orthogonal pairs and hence is a scalar multiple of the invariant pairing: for each nonzero first vector its second functional is proportional to the original form, and comparison of independent first vectors makes the scalar constant. The free natural occurrence is retained. Conjugating the free torus tests any vector in a tail not seen by the first chosen torus; the intersection of all such fixed spaces is zero. Every argument is coefficientwise on passive multiplicities. Over the actual binary representation, extend scalars, project into the signature summands only for this proof, apply the containment to their images, and descend to the original joint insertion space. No projected denominator replaces P. □

The eligible insertions form a direct sum in stable rank. A coefficient with its paired basis vectors in one plane and its free vector in a different plane isolates the given insertion. In unitary orthonormal coordinates use an equal paired index different from the free index. These tests also prove independence with all field twists. In particular J is a direct sum of natural types, not an unspecified quotient of them.

F.2 Literal invariant tensors and four-leg maps

For symplectic groups a finite binary calculation replaces the unavailable central sign. Restrict a cubic invariant to $\mathrm{SL}_2(2)^m$. Each plane has no fixed natural vector, so only a three-in-one-plane term can survive. Its fixed space is spanned by

$$t_i = \sum_{0 \neq v \in \mathbb{F}_2 e_i + \mathbb{F}_2 f_i} v^{\otimes 3}.$$

The transvection with vector $e_i + e_j$ creates in t_i a unique nonzero coefficient of $e_i \otimes e_j \otimes f_i$. Thus every coefficient of $\sum_i c_i t_i$ vanishes. Twists have the same binary action, proving $M^G = 0$ for every signature.

For four legs, the same subgroup leaves only four-in-one-plane and two-plus-two supports. Plane permutations make the three two-plus-two pairing coefficients uniform. Subtract the three global pairing diagrams. A remaining invariant is supported on individual planes. The cross-root $f_i \mapsto f_i + e_j$, $f_j \mapsto f_j + e_i$ isolates every coefficient with at least two f_i positions by creating one e_j and leaving an f_i. Exchanging e_i, f_i kills all remaining coefficients. Hence the four-leg invariants are exactly the three pairing diagrams. Their independence follows by evaluating on two planes. Full-field scalings remove ineligible twists. This proves the natural-to-cube maps used in the image-algebra argument. For orthogonal and unitary groups, the support lemma applied to a natural-image submodule and the same coordinate pair tests give the corresponding permitted diagrams; for general-linear groups use the whole-word matching from Appendix D. Literal cubic invariants vanish as well, since the support lemma places them in J, whose natural parameters have no fixed vectors.

F.3 General linear cubic quotients

For a general-linear cubic word M, Appendix D.1 gives $H^1(\mathrm{GL}(V), M) = 0$. Since the eligible insertion sum J is a direct sum of natural/dual/Frobenius-natural modules, it also gives $H^2(\mathrm{GL}(V), J) = 0$. The long exact sequence gives $H^1(\mathrm{GL}(V), M/J) = 0$. Natural $H^1 = 0$ and

$M^{\mathrm{GL}(V)} = 0$ similarly give $(M/J)^{\mathrm{GL}(V)} = 0$. The normalized torus/root-cocycle argument transfers these assertions to $\mathrm{SL}(V)$ and intermediate determinant groups, with support at most three. This uses no stability theorem for the quotient M/J itself.

F.4 Symplectic cubic quotients

Let $G = \mathrm{Sp}(V)$, $\dim_K V = 2r$, $r \geq 12$, and let

$$M = I_a(V) \otimes I_b(V) \otimes I_c(V).$$

We provide the ordinary-functor calculation used to compare its H^1 with the insertion submodule. In the category $\mathcal{F}$ of ordinary functors on finite-dimensional K-spaces put $A(U) = K[\Lambda^2 U]$ and $E_s = \mathrm{Ext}^1_{\mathcal{F}}(A, I_s)$. Exact sum/diagonal adjunction in three variables transforms the desired Ext into one with source

$$\bigotimes_i K[\Lambda^2 U_i] \ \otimes \bigotimes_{i<j} K[U_i \otimes U_j].$$

Split every factor into its constant summand and augmentation summand, using the basis vector $[0]$. A term omitting a variable has zero Ext to a target reduced in that variable. A term containing that variable in two reduced factors also has zero Ext to an additive target: separate the variable into two by the exact sum/diagonal adjunction, and each additive target summand is constant in one variable while the source is reduced in both. Shared other variables are unchanged by this operation.

Only the three terms consisting of one singleton augmentation and one complementary pair augmentation, and the triple-singleton term, remain. The natural transformation space from the singleton augmentation $\bar{A}$ to I_s is zero, since $\bar{A}(K) = 0$ and all linear maps to K separate vectors. This also justifies the degree-one external-product rule

$$\mathrm{Ext}^1(P \boxtimes R, F \boxtimes Q) = \mathrm{Hom}(R, \mathrm{Ext}^1(P, F) \otimes Q) \quad \text{when } \mathrm{Hom}(P, F) = 0.$$

Indeed restrict an extension to each second-variable object; endpoint-compatible extension morphisms exist precisely when the classes are natural, and are unique because their differences lie in $\mathrm{Hom}(P, F)$ with finite multiplicities. Unique morphisms respect composition and reconstruct the bifunctor extension. No higher-degree Kunneth assertion is used.

For the edge augmentation $\bar{B}(U, W)$ of $K[U \otimes W]$, a map to $N \otimes I_s(U) \otimes I_t(W)$ is zero unless $s = t$ modulo the field degree. At (K, K) let $f(u)$ be its value on $[u] - [0]$. Scaling the first or second variable gives

$$(v^{2^s} - v^{2^t}) f(u) = 0.$$

If $s = t$, it forces $f(u) = u^{2^s} f(1)$; pairs of coordinate functionals uniquely extend this to all spaces. Thus the edge Hom is exactly N in the matching case. The triple-singleton term vanishes by evaluating one singleton at K. We obtain

$$\mathrm{Ext}^1(A, I_a \otimes I_b \otimes I_c) = \bigoplus_{i:\text{the other two twists agree}} E_{a_i}. \tag{38}$$

The symplectic comparison [DV10, Corollaries 3.32 and 3.33] identifies these with finite group cohomology in rank at least $2i + d + 6$. For H^1 of a cubic word this is $r \geq 11$, uniformly in all twists. Hence (38) gives equality of the dimensions of $H^1(G, M)$ and $H^1(G, J)$.

To identify the insertion maps themselves, split $V = W \perp U$ with U a hyperbolic plane and let $H = \mathrm{Sp}(W)$ fix U. For each eligible insertion in position i, choose any functional on the two twisted

U slots taking ω_U to one. It need not be $\mathrm{Sp}(U)$-invariant. Project the free slot to W and evaluate the other two by this functional, giving $D_i : M|_H \to W^{[a_i]}$. Then

$$D_i \iota_j = \delta_{ij}\, p^{[a_i]},$$

where $p : V \to W$ is orthogonal projection. In degrees one and two these maps, after restriction, are the duals of natural-coefficient homological stabilization. Both ranks are in the range when $r \geq 12$, so the diagonal maps are isomorphisms. Therefore insertion induces injections $H^j(G, J) \to H^j(G, M)$ for $j = 1, 2$. The degree-one dimension equality makes the first map surjective. The exact sequence

$$H^1(G, J) \to H^1(G, M) \to H^1(G, M/J) \to H^2(G, J) \to H^2(G, M)$$

now gives $H^1(G, M/J) = 0$. Since $M^G = 0$ and the degree-one insertion map is injective, the preceding exact sequence also gives $(M/J)^G = 0$.

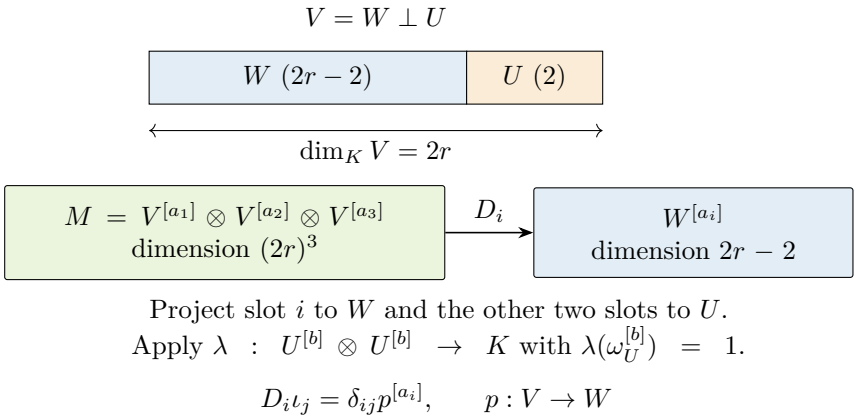


Figure 11: Restriction detectors for the symplectic insertion maps. Dimensions are over K. The subgroup $\mathrm{Sp}(W)$ fixes U pointwise, so any functional with the displayed value is equivariant. The detector separates the eligible positions without dividing by the dimension of a symplectic plane.

F.5 Unitary cubic quotients

Let $G = \mathrm{SU}_n(K/K_0)$, with $|K_0| = 2^f$, and rewrite dual occurrences as conjugate-Frobenius naturals. Put $Q_c = M/J$. We use the actual odd-index parabolics displayed in Appendix D. We will normalize a cocycle on the Levi and then successively eliminate its values on the radical.

Two elementary general-linear supplements will be useful. A nontrivial determinant character tensored with a word of length r has no fixed vector when the natural rank exceeds r, and zero H^1 when it exceeds $r + 2$. The full diagonal torus has a nonzero constant exponent vector from the determinant character; a coefficient weight changes at most r positions. For H^1, normalize on that odd torus and evaluate on elementary roots; this changes at most two additional positions. Hence no invariant cochain survives. Also H^1 and H^2 of every one-dimensional determinant character vanish: for a nontrivial character use the odd quotient and perfect special-linear kernel, and for degree two use the stable characteristic-two multiplier vanishing from [Ste81]. These arguments have the same uniform head-rank cutoff 12.

If $n = 2m$, the Levi is $L' = \{g \in \mathrm{GL}_m(K) : \det g \in K_0^\times\}$ and the abelian radical is the Hermitian-matrix group. On the full general-linear Levi, M is a sum of cubic words and J is a sum of natural words, so the long exact sequence gives $H^1(\mathrm{GL}_m(K), Q_c) = 0$. The bounded-support H^1 transfer

then gives the same assertion for L'. Normalize a cocycle to vanish there. Give the dual half weight zero and the natural half weight one. The radical raises weight, and J is homogeneous. A first nonzero radical value in grade d is an additive L'-map from a scalar-extension component

$$S = E^{[s]} \otimes E^{[s+f]}$$

of the Hermitian radical to $(Q_c)_d$. The exact sequence

$$\mathrm{Hom}_{L'}(S, M_d) \to \mathrm{Hom}_{L'}(S, (Q_c)_d) \to \mathrm{Ext}^1_{L'}(S, J_d)$$

has zero outside terms: the first is an odd five-leg invariant problem and the second a cubic H^1 problem. Hence that first value is impossible. The cocycle vanishes on the parabolic, and odd-index restriction proves $H^1(G, Q_c) = 0$.

For $n = 2m + 1$, use $V = E \oplus Kt \oplus E^{\sigma *}$, the Levi $l(g) = \mathrm{diag}(g, \chi(g), g^{-\sigma t})$, and the radical $u(X, Z)$ already displayed, with $\chi(g) = (\det g)^{2^f - 1}$. Its multiplication is

$$u(X, Z)u(Y, W) = u(X + Y, Z + W + XY^{\sigma t}).$$

Thus the Hermitian subgroup C with $X = 0$ is central, and the quotient radical is $E \otimes \chi^{-1}$. Give the three vector blocks weights $2, 1, 0$. A twisted natural slot has Levi constituents $E^{[a_i]}$, $K_{\chi^{2^{a_i}}}$ and $(E^{[a_i+f]})^*$. Terms with three head factors use cubic general-linear vanishing. Terms with two heads have one nontrivial determinant character, terms with one head use either that character or natural vanishing, and tail-only terms use the determinant-character calculation. Thus $H^1(L, M) = 0$. The summands of J have zero H^2 by natural or character vanishing, so $H^1(L, Q_c) = 0$. Normalize the Levi cocycle to zero.

For the first central-radical value, precompose each Hermitian scalar component S as above. The outside terms in the Hom sequence again vanish. With three heads the Hom term is an odd five-leg invariant; with two heads a nontrivial determinant character remains; with one head it is either a nontrivial character or an odd three-leg invariant; with no head a pure dual quadratic word has no invariant. The Ext term is a cubic H^1 for a head insertion, or a determinant-twisted quadratic H^1 for a tail insertion. Therefore $c(C) = 0$.

Any first remaining radical value factors through a scalar component $W_s = E^{[s]} \otimes \chi^{-2^s}$ of U/C. Both $\mathrm{Hom}_L(W_s, J_d)$ and $\mathrm{Ext}^1_L(W_s, J_d)$ vanish: for head targets they are two-leg words with a nontrivial determinant character, and for a tail target they are a dual natural word, with or without such a character. Every map to $(Q_c)_d$ therefore lifts uniquely to M_d. Of the literal target types, three-head terms have a nontrivial determinant character, two-head terms have either that character or an odd three-leg invariant obstruction, and no-head terms cannot receive a natural module. The only possible term has one matching natural head and two tails. It occurs in weight four, and its existence requires

$$\chi^{2^{a_1}+2^{a_2}+2^{a_3}} = 1.$$

Consequently the first value is a sum of three actual twisted coordinates

$$c_1 X^{[a_1]} \otimes t \otimes t + c_2 t \otimes X^{[a_2]} \otimes t + c_3 t \otimes t \otimes X^{[a_3]},$$

where $c_i \in K$ are coefficients and a_i remain the prescribed twist indices. The same condition permits all three positions.

Since commutators lie in C, the cocycle identity gives

$$(u(X) - 1)c(u(Y)) = (u(Y) - 1)c(u(X)).$$

In weight five, on two specified head slots, it becomes

$$(c_i - c_j)(Y^{[a_i]} \otimes X^{[a_j]} + X^{[a_i]} \otimes Y^{[a_j]}) = 0.$$

Distinct coordinate vectors make the two tensors independent, forcing all coefficients equal. A pairing insertion has weights only two, three or four, so this weight-five equality is literal and has no hidden quotient ambiguity. The common coefficient times $t \otimes t \otimes t$ is Levi-fixed by the displayed character condition. Its coboundary removes exactly the first weight-four value and preserves the normalization on L and C. No other first grade is possible; hence the cocycle vanishes on the whole parabolic. Odd-index restriction proves the desired vanishing.

The same parabolic proof on a natural coefficient module gives $H^1(G, V^{[a]}) = 0$: the center has no two-to-one head map, and the quotient radical in odd dimension retains its nontrivial determinant twist, so cannot map to a head or tail constituent. Literal cubic fixed vectors vanish as well. In even dimension their Levi restriction is a sum of odd general-linear words. In odd dimension the only possible Levi-fixed part is a tail cube; every nonzero such tensor is moved by $u(X, Z)$ in a distinct one-head position. Since J is a direct sum of natural types, the exact sequence gives $(M/J)^G = 0$.

F.6 Orthogonal cubic quotients

Let $G = \Omega(V, Q)$ and write $V = E \oplus Z \oplus E^*$, where $Z = 0$ in split type and Z is an anisotropic plane in nonsplit type. The Witt index is $m = \dim E \geq 12$. The actual parabolic has Levi $\mathrm{GL}(E)$ in split type and $\mathrm{GL}(E) \times T_0$ in nonsplit type, where $T_0 = \mathrm{SO}(Z)$ is cyclic of odd order $|K| + 1$. Its split radical is $\Lambda^2 E$; in nonsplit type the radical has center $C = \Lambda^2 E$ and quotient $\mathrm{Hom}_K(Z, E)$, with the matrices and multiplication in Appendix C.

These are parabolics inside Ω, rather than a larger index-two group. A Levi head has doubled rank of $g - I$ and a tail rotation has even rank. For a radical element $u(X, B) - I$, let $s = \operatorname{rank} X$. Block elimination gives rank $2s + \operatorname{rank} \beta$, where β is the induced map from $\ker X^t$ to $E/\operatorname{im} X$. The quadratic condition on B makes this residual bilinear form alternating, so its rank is even. Thus the radical has Dickson zero. Its order is $|K|^{m(m-1)/2}$ in split type and $|K|^{m(m-1)/2+2m}$ in nonsplit type. Multiplying by the Levi's two-part gives respectively $|K|^{m(m-1)}$ and $|K|^{m(m+1)}$, the two-parts of the corresponding Ω orders [Tay92, Chapter 11]. The parabolic therefore has odd index.

Put $Q_c = M/J$. In split type its Levi H^1 vanishes by the cubic and natural general-linear results. In nonsplit type split the odd tail torus into its characters χ, χ^{-1}. A term with three head factors uses cubic H^1; one tail gives a nontrivial tail character; two tails leave either a nontrivial character or natural H^1; no heads give a trivial general-linear module with odd-order abelianization or a nontrivial tail character. The same distinction gives $H^2(L, J) = 0$. Thus $H^1(L, Q_c) = 0$ and a cocycle can be normalized on the Levi.

In split type grade the two halves by zero and one. A first radical value gives an additive map $\Lambda^2 E^{[s]} \to (Q_c)_d$. Precompose with the surjection $E^{[s]} \otimes E^{[s]} \to \Lambda^2 E^{[s]}$. The Hom exact sequence has an odd five-leg invariant on the left and a cubic H^1 on the right, both zero. This eliminates every first grade and proves the split result.

In nonsplit type give E^*, Z, E weights $0, 1, 2$. The central radical raises weight by at least two. Apply the same precomposition to a first value on C. The Hom term with three heads is an odd five-leg invariant; two heads have one nontrivial tail character; one head gives an odd three-leg invariant when the tail character is trivial; no heads leave a pure dual quadratic invariant, which is zero. For the Ext term a head insertion gives cubic general-linear H^1 and a tail insertion has nontrivial tail character. Hence $c(C) = 0$ exactly.

A first remaining value factors through an additive Levi map from $W = U/C = \mathrm{Hom}_K(Z, E)$ to $(Q_c)_d$. For every scalar component W_s, both $\mathrm{Hom}_L(W_s, J_d)$ and $\mathrm{Ext}^1_L(W_s, J_d)$ vanish. A head

insertion leaves a nontrivial tail character, while a tail insertion leaves a natural-dual general-linear word with no H^0 or H^1. Therefore the map lifts uniquely to M_d. Three-head targets have no tail character; two-head targets give an odd three-leg invariant problem; no-head targets cannot receive a natural head. The only possible target has one matching natural head and two tails, of weight four. Thus its unique expression is

$$F(X) = X_1(b_1) + X_2(b_2) + X_3(b_3),$$

where each b_i belongs to the three-twisted-tail space $Z^{[a_1]} \otimes Z^{[a_2]} \otimes Z^{[a_3]}$, is T_0-fixed, and X_i applies the corresponding twist of X in position i. If that position has no matching character, $b_i = 0$. Varying every head-row functional on the two-dimensional tail proves uniqueness.

Commutators again lie in C. In weight five, which contains no insertion term, the cocycle identity on two head positions gives

$$(Y_i X_j + X_i Y_j)(b_i + b_j) = 0.$$

Take X, Y with images in two distinct coordinate lines of E. In the output with those two lines in a fixed order only one summand occurs. Varying their tail functionals proves $b_i = b_j$. Hence all three equal one T_0-fixed tail tensor b. Its coboundary has precisely F as its first weight-four part, and is zero on C and the Levi. Subtract it; no later first grade is possible. Odd-index restriction gives $H^1(G, Q_c) = 0$ in nonsplit type.

Natural orthogonal H^1 was proved by the same parabolic in Lemma D.2. The insertion sum is a direct sum of these natural twists, and literal cubic fixed vectors are zero by Lemma F.1 and absence of natural fixed vectors. The exact sequence gives $(M/J)^G = 0$. Since $\Omega \trianglelefteq \mathrm{O}$ has index two and both its fixed space and first cohomology vanish on this quotient, inflation-restriction gives the same conclusions for the full orthogonal group.

F.7 Independent cores and descent to the actual field

If three independent cores act, the product formula gives no fixed vector and zero H^1, because every complementary natural fixed space is zero. With two repeated occurrences and an independent free row, quotient the repeated pair by its permitted invariant line. Its fixed space is zero: in a matched adjoint or self-dual square, the whole fixed space is that line and perfectness gives H^1 of the line zero; every other pair has zero fixed space by the whole-word pairing calculation. Simplicity of those pairs is unnecessary. Tensoring with the free natural row and applying the degree-one product formula therefore again gives both vanishings. The result does not require natural symplectic H^1 to vanish.

For arbitrary binary representations, scalar extension separates the actual Frobenius signatures only in the proof. Cohomology in degree one and fixed vectors are kernels and cokernels of finite linear systems on a finite group; extension of scalars preserves their ranks. Their vanishing thus descends to the actual binary space. The insertion parameters, actual form-line actions and actual translation denominator are pulled back jointly as in Section 4.

Theorem F.2 (Cubic quotient theorem)**.** *For every current stable normal product of classical cores, every supported cubic natural word, and the full sum J of its eligible pairing insertions,*

$$(M/J)^G = 0, \qquad H^1(G, M/J) = 0.$$

Every actual degree-two-origin translation submodule is contained in J. All statements hold over every even finite field, with all twists and both orthogonal signs. Together with Lemma 4.3, they prove Proposition 4.4 and cubic affine descent on the actual graded quotient.

G Canonical common zero blocks for low-rank matrix spaces

The purpose of this section is to turn low rank into a large common zero block, in a way preserved by every equivalence of the matrix space. A pair of subspaces $U \le X$ and $B \le Y$ with $\mathcal{S}(U) \le B$ has *boundary size* $\operatorname{codim}_X U + \dim B$. In bases adapted to this pair, every matrix of $\mathcal{S}$ is zero from U to Y/B. We first prove a boundary bound over the base field, then construct an invariantly defined pair by noncommutative rank.

In this section, k is the stated base field, $q_k = |k|$, and N denotes the faithful degree of the represented algebra or original rows over k. In the main algorithm $k = \mathbb{F}_p$, so $q_k = p$ and this degree is $O(D_0)$. Auxiliary module dimensions are not included in that faithful degree.

Lemma G.1. *Let X, Y be finite-dimensional k-spaces and $R \ge 0$ an integer. Let $\mathcal{S} \le \operatorname{Hom}_k(X, Y)$ be a matrix space whose base-field elements have rank at most R. There are subspaces $U \subseteq X$ and $B \subseteq Y$ such that*

$$\mathcal{S}U \subseteq B, \qquad \operatorname{codim}_X U + \dim B \le 4R.$$

No lower bound on q_k is required.

Proof. Choose a base-field $A \in \mathcal{S}$ of maximum rank a, and use coordinates in which A has an a-dimensional identity block and zeros elsewhere. For $B \in \mathcal{S}$, let b be the rank of its complementary lower-right block. Reduce that block to an identity of size b. Its row and column pivots can clear the adjoining blocks of B without changing A, since the corresponding rows and columns of A vanish. Thus the two matrices B and $A + B$ have a common separate identity block of size b, and their remaining parts differ by a rank-a matrix. The rank triangle inequality gives

$$\operatorname{rank} B + \operatorname{rank}(A + B) \ge 2b + a.$$

Both matrices are base-field elements of $\mathcal{S}$, so $b \le a/2$. In particular the complementary projected matrix space has maximum base-field rank at most $a/2$.

Delete the a pivot-domain directions and allow the a pivot-image directions in the boundary pair, then repeat on that projected space. The successive maximum ranks are at most $R, R/2, R/4, \ldots$. At termination the remaining projected space is zero. The sum of deleted domain and allowed image dimensions is at most $2\sum_{j \ge 0} R/2^j = 4R$. This argument only uses A, B, and $A + B$ over the original field. □

The preceding proof establishes existence; the algorithm does not need to find its maximum-rank matrix. The next proposition constructs the canonical pair by noncommutative rank and a Wong sequence, using only explicit linear-algebra operations and the cited deterministic algorithm.

For a positive integer e, the blow-up of $\mathcal{S} \le \operatorname{Hom}_k(X, Y)$ is

$$\mathcal{S}^{[e]} = \mathcal{S} \otimes_k \mathrm{M}(e, k) \le \operatorname{Hom}_k(X \otimes_k k^e, Y \otimes_k k^e).$$

Its noncommutative rank is $\operatorname{ncrk}(\mathcal{S}) = \sup_{e \ge 1} \operatorname{mrk}(\mathcal{S}^{[e]})/e$. For a subspace $U \le X$, its deficiency is $\dim U - \dim \mathcal{S}U$. After padding to a common square dimension d, a subspace of maximum deficiency has minimum boundary size $d - \max_{U \le k^d}(\dim U - \dim \mathcal{S}U)$. The noncommutative-rank theorem identifies this minimum with $\operatorname{ncrk}(\mathcal{S})$ and computes the necessary blow-up.

Proposition G.2. *There is a deterministic canonical choice of a pair $(U, \mathcal{S}U)$ with*

$$\operatorname{codim}_X U + \dim \mathcal{S}U \le 4R.$$

For the original pencils here its time bound is $q_k^{O(N)}\operatorname{poly}(\log q_k)$. More generally the construction has only polynomial dependence on matrix dimensions, apart from a conservative $q_k^{O(1)}$ finite-field factor.

Proof. Write ρ for noncommutative rank. Every boundary pair in Lemma G.1 bounds the rank of every blow-up $\mathcal{S}\otimes \mathrm{M}(e,k)$ by $4Re$: on $U\otimes k^e$ its image lies in $B\otimes k^e$, and the complementary domain has dimension $e\operatorname{codim} U$. Hence $\rho\le 4R$.

The deterministic noncommutative-rank algorithm [IQS18, Theorem 1.5 and Corollary 1.7(2)] supplies an optimal-rank base-field blow-up

$$B\in\mathcal{S}\otimes\mathrm{M}(e,k),\qquad \operatorname{rank} B=e\rho.$$

Its small-field construction passes through an extension and then uses the regular representation to return a base-field blow-up; an arbitrary extension-field subspace is not used as the output here. The standard shrunk-space characterization gives minimum boundary size ρ.

For completeness, canonicity follows by the following Wong sequence in the blow-up domain. After zero-padding rectangular spaces to a common size d, put $\mathcal{S}_e=\mathcal{S}\otimes\mathrm{M}(e,k)$ and

$$U_0=\ker B,\qquad U_{j+1}=B^{-1}(\mathcal{S}_eU_j).$$

If U has maximum deficiency $e(d-\rho)$, then

$$e(d-\rho)\le\dim(U\cap\ker B)\le\dim\ker B=e(d-\rho).$$

Thus $\ker B\subseteq U$ and $BU=\mathcal{S}_eU$, whence $B^{-1}(\mathcal{S}_eU)=U$. Every sequence term is consequently contained in every maximum-deficiency subspace. The increasing sequence stabilizes in polynomially many steps. At its limit its image lies in $\operatorname{im} B$, because it is contained in a maximum-deficiency subspace, and $BU_\infty=\mathcal{S}_eU_\infty$. Its deficiency is therefore $\dim\ker B$, so it is itself maximum and is the unique smallest such space.

Right changes of basis on the blow-up index preserve $\mathcal{S}_e$. Uniqueness makes U_∞ invariant under all $I_X\otimes\mathrm{GL}(e,q_k)$. These matrices span $I_X\otimes\mathrm{M}(e,k)$ also for $k=\mathbb{F}_2$. Indeed, I and $I+E_{ij}$ give every off-diagonal matrix unit. On indices i,j, the invertible block $\left(\begin{smallmatrix}0&1\\1&1\end{smallmatrix}\right)$, extended by the identity elsewhere, differs from $I+E_{ij}+E_{ji}$ by E_{ii}. The case $e=1$ is immediate. Consequently $U_\infty=U_*\otimes k^e$. Recover U_* by the linear equations $x\otimes e_1\in U_\infty$. It is the unique smallest original maximum-deficiency subspace. If padding added a zero domain summand, that summand belongs to every maximum-deficiency space and is discarded after descent. The image $\mathcal{S}U_*$ is computed by linear algebra. The resulting original pair is therefore covariant under every left-right equivalence, independently of the blow-up witness chosen.

The finite-field algorithms in the cited theorem are used with their deterministic finite-field implementation. It suffices here to retain a possible polynomial factor in p or q_k, rather than assert an unjustified polynomial dependence on $\log q_k$ alone. A small field is extended only to size polynomial in the matrix dimension, and direct deterministic field construction has that same bound. These costs are absorbed in $q_k^{O(N)}$ on the original modules. The rank R in the statement is the maximum rank over the original field; no large-field commutative-rank identity is assumed. □

H Compatible field quotients of contraction algebras

In the same-type contraction case, each small tuple of operators has a field quotient acting on most of the argument. These fields need not all be generated by one short tuple. Instead, their pair and

triple inclusions determine a consistent system of embeddings. We construct their common field and show that the quotient values form a linear map, including when the previously known scalar field is noncentral in the operator algebra.

A short tuple need not generate a single large field. The following construction uses all raw triples, while keeping the total work singly exponential in original degree.

Theorem H.1 (Direct field gluing)**.** *Let k be a finite field of characteristic two and let K/k have degree f. Let $a \ge 1$, $m \ge 1$, and $r \ge 0$ be integers with $2r < a$. Put $A = K^a$, let C be a finite-dimensional k-space, and let $M : C \to \mathrm{End}_k(A)^m$ a simultaneous k-linear family of operators. The constant field K is distinguished in every raw algebra*

$$B_S = \mathrm{alg}_k(K, I, \{M_j(x) : x \in S, 1 \le j \le m\}).$$

Assume for every $|S| \le 3$ that B_S has a field quotient $L_S = B_S/P_S$ with K embedded, and $A_S = \mathrm{ann}_A(P_S)$ has K-dimension at least $a - r > a/2$. Then there is a field F/K, with

$$[F : K] = E \le \mathrm{lcm}(1, \dots, a) = \exp(O(a)),$$

and compatible K-embeddings of all the quotient fields L_S into F, with F generated by their images. These embeddings, with the quotient maps $B_S \to L_S$ fixed, are unique up to one common K-Frobenius. Define $\lambda_j(x)$ to be the image in F of the quotient class of $M_j(x)$. The resulting quotient-value map $\lambda : C \to F^m$ is k-linear and satisfies

$$\lambda(x) = 0 \quad \Longleftrightarrow \quad \mathrm{rank}_k\big[v \mapsto (M_j(x)v)_j\big] \le fr.$$

The diagram, field, map and all embeddings are computable in $|k|^{O(\dim_k C + fa)}$ times polynomial label work. Conjugation and actual label changes transport the whole construction equivariantly.

Proof. For $S \subseteq T$, $|T| \le 3$, the image of B_S on A_T is a subfield of L_T. Thus A_T is semisimple of one B_S-type. It must be the L_S type, because otherwise A_T and A_S have zero intersection and dimensions summing to more than $\dim A$. Hence

$$P_T \cap B_S = P_S, \qquad i_{ST} : L_S \hookrightarrow L_T$$

are actual compatible K-field embeddings, and $A_T \subseteq A_S$. Write $e_x = [L_x : K] \le a$. The raw pair algebra is generated by its two singleton algebras, so its field quotient is their compositum and $e_{xy} = \mathrm{lcm}(e_x, e_y)$; the same holds for triples.

Let $F = \mathbb{F}_{|K|^E}$ with $E = \mathrm{lcm}_x\, e_x$. Choose arbitrary singleton and pair embeddings into F. Their pairwise discrepancies are phases

$$d_{xy} \pmod{\gcd(e_x, e_y)}.$$

An embedding of a triple field shows

$$d_{xy} + d_{yz} = d_{xz} \pmod{\gcd(e_x, e_y, e_z)}.$$

For each rational prime $\pi \mid E$, choose a reference x_π with maximal π-adic valuation of e_x and prescribe

$$s_x = d_{x,x_\pi} \pmod{\pi^{v_\pi(e_x)}}.$$

The triangle relation proves every pair congruence at this prime. Integer CRT gives s_x modulo e_x. Twisting singleton embeddings by these phases aligns every pair and triple diagram. Any two

solutions differ by one common phase modulo E, since at each prime their differences are restrictions of one constant at maximal valuation. Composita give unique extensions to pairs and triples.

Define $\lambda_j(x)$ as the coherently embedded quotient value of $M_j(x)$. For $z = x + y$, ordinary pencil linearity gives $B_{x,y,z} = B_{x,y}$ and $M_j(z) = M_j(x) + M_j(y)$. Applying the common quotient proves additivity. The same argument proves k-homogeneity. These entries generate all singleton fields and hence F over K.

If $\lambda(x) = 0$, every $M_j(x)$ kills the same A_x, so the simultaneous stack has k-rank at most fr. If some value is nonzero, that operator acts invertibly on A_x, and its rank is at least $f(a - r) > fr$. This proves the exact kernel assertion without a label factor. In the noncentral-old-field case, decompose each original k-linear operator into its K/k Frobenius-semilinear components. On the K-stable field-quotient subspace it commutes with K; independence of field embeddings makes every nonidentity component zero there. Untwisting outputs gives one whole K-linear stack with the same core and rank bound, and with the actual twisted row actions retained.

A genuine tensor transporter conjugates all raw algebras, unique dominant annihilator subspaces and quotient embeddings. The uniqueness of the gluing gives one common field isomorphism, hence a common Frobenius after source and target are placed in one F. An invertible label change merely replaces a spanning set of the same complete operator image. If the scalar observations are indexed by the dual L^* of a k-space L of dimension m, the construction with moving labels is the value in $\mathrm{Hom}_k(L^*, F)$, not an independent value for each chosen coordinate. □

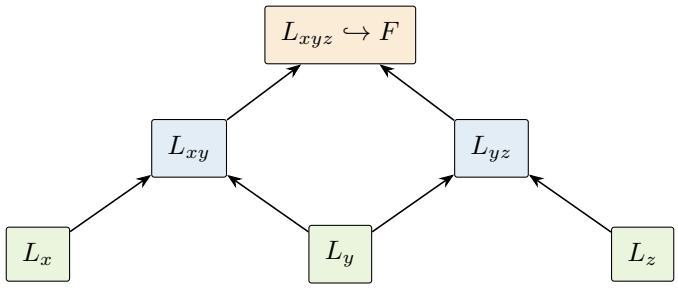


Figure 12: Pair and triple quotient fields determine compatible embeddings of the singleton fields. The triple fields enforce agreement on overlaps. The field F is an auxiliary coordinate space; its elements are never exhaustively listed.

H.1 Constructing the large field without enumerating its elements

The common extension degree can be exponential in the argument dimension. Arithmetic in that many coordinates is still affordable, but exhaustive enumeration of the field elements would not be. We construct it as a tensor product of small prime-power-degree extensions and find each required small subfield by linear equations.

For $n \geq 2$, put $L(n) = \mathrm{lcm}(1, \ldots, n)$. The quotient $L(n)/L(\lceil n/2 \rceil)$ divides $\binom{n}{\lfloor n/2 \rfloor}$: each new prime power lies above $n/2$ and its extra prime exponent appears in that binomial coefficient. Therefore $\log_2 L(n) \leq n + \log_2 L(\lceil n/2 \rceil) = O(n)$.

Factor the small degrees $e_x \leq a$. For each rational prime $\pi \mid E$, put $d_\pi = \pi^{v_\pi(E)} \leq a$. Construct a degree-$d_\pi$ field K_π/K by enumerating at most $|K|^{d_\pi}$ monic polynomials and testing irreducibility. The pairwise coprime degrees make

$$F = \bigotimes_{\pi \mid E, K} K_\pi$$

a field. Its tensor basis gives arithmetic polynomial in E, hence exponential only in a. This avoids finding a degree-E irreducible by enumeration over $|K|^E$.

To embed a small field of degree $d \le a$, compute $\ker(\mathrm{Frob}_K^d - I)$ in F. It is the unique d-dimensional subfield. Enumerating its $|K|^d$ elements finds a root of the small defining polynomial. Reuse a basis for each repeated degree. Phase tests involve at most a powers on a small generator. Source and target diagrams use the same constructed F after comparing E; no expensive isomorphism search between two exponential-degree presentations is hidden.

There are at most $|k|^{3\dim C}$ raw triples. Algebra/module work is on the original row, and the field arrays have total size $|k|^{O(\dim C + fa)}\,\mathrm{poly}(m)$. This proves the algorithmic estimate.

H.2 The raw-triple source alternative

The field-quotient hypothesis used above is an outcome of an explicit test. Krull–Schmidt decomposition of each supplied contraction algebra gives either smaller centralizer factors or a unique factor large enough to act through a field. The calculation is over the prime field, so it remains valid when the distinguished older field is noncentral.

For a represented raw algebra B, perform Krull–Schmidt over the prime working field k, including when its distinguished old field K is noncentral. Write $A = \bigoplus_i Q_i^{b_i}$, let $L_i = \mathrm{End}_B(Q_i)/\operatorname{rad}$ and set $g_i = [L_i : k]$, $\dim_k Q_i = g_i j_i$. Its centralizer has semisimple factors $\mathrm{M}(b_i, L_i)$ and ordinary prime-field natural degrees $d_i = g_i b_i$. If some d_i exceeds $f(a-r) > fa/2$, then $j_i = 1$, because $j_i \ge 2$ would exceed total original mass fa. The resulting one-dimensional residue-field module is a genuine field quotient of B. Only now does restriction of that quotient embed the distinguished K: its kernel on the field K is zero. The associated annihilator core is K-stable, and its k-dimension at least d_i is therefore K-dimension greater than $a-r$. Uniqueness follows from the same mass bound. If no such degree exists, every centralizer natural row has prime-field degree at most $f(a-r)$, and the raw tuple is a geometric decomposition certificate. These alternatives are exhaustive and constructible by represented-algebra operations. No residue field is asserted to contain the old K before the field quotient has been obtained.

For moving labels the value-map kernel is an actual prime-field original subspace. Match a proper nonzero kernel by the whole explicitly realized frame call, retain its subalgebra preserving the subspace, and make a strict unbranched restriction. If the image is injective, for each global phase the value equation kills the large coefficient action; complete the whole remaining at-most-two-large-factor observation before reuniting phases. This is the uniform rule of Section 6; no uncharged phase branch enters outer recursion.

Returning to the original arguments. The field F is temporary coefficient data. A proper-kernel outcome exports only the actual original kernel subspace and its complete conductor. A zero outcome exports only its rank certificate and common zero-block construction. In the injective outcome, every common Frobenius phase is completed by the lower-degree algorithm before reunion. In all three cases F is discarded before an unresolved outer tensor call. Its possibly exponential degree is never counted as a new argument or label carrier.

I Pairing insertions and their quotient equations

The translations that survive in a cubic tensor layer are pairing insertions. This section proves their classification and gives contractions that are well defined modulo these translations. It then treats the unequal-dimension case and proves the exact tensor-and-parameter reduction used by the main

algorithm. The allowed translation space is retained throughout: a useful parameterization of all insertions does not make every parameter an available group transformation.

Throughout this section $k = \mathbb{F}_2$, and K/k is a finite extension of degree f, and dimensions used for original costs are over k. When old defining fields do not contain the original input field, we take $k = \mathbb{F}_p$, as in Section 3. All original field-linearity equations remain imposed. A label family always means the whole supplied family with its actual common action. In particular, the parameters of an additive translation are not made independent by writing them in separate coordinates.

When the value coordinates can move, let L be their k-space and put $m = \dim_k L$. Scalar observations are indexed by the coordinate functionals in L^*; an index j ranges over $[m]$ after a basis is chosen. All formulas are applied simultaneously in these coordinates, with the induced L-action retained.

I.1 The insertion types

Appendix F proves the actual translation support and the finite four-leg calculation in characteristic two. We use those results here as follows. Solve the one-factor intertwiner and pairing equations, form their positionwise insertion maps, and compute their span and kernel. For three independent factors the space is zero. For two paired occurrences and one independent factor there is one pairing times the free parameter. A mixed general-linear word has two natural-dual evaluation positions; a pure covariant or contravariant word has none. A symplectic or quadratic orthogonal word has three positions when all twists match and the eligible subset otherwise. A unitary word pairs only conjugate-dual twists, and has at most two eligible positions. All parameters and relations remain over the original binary field.

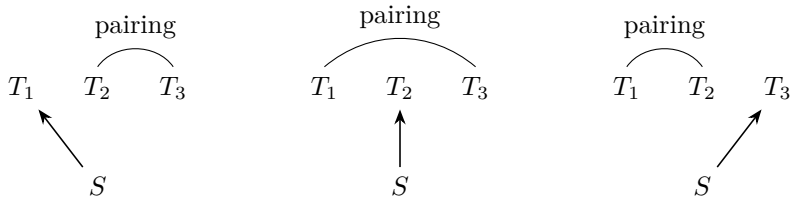


Figure 13: The three possible pairing insertions in a map $S \to T_1 \otimes T_2 \otimes T_3$. A diagram is allowed only when its one-factor intertwiner and invariant pairing exist for the specified factors, fields, duals, and twists. Their full linear span, including any kernel in the parameterization, is retained.

I.2 Representing all twists over the original field

Individual Frobenius signatures need not become separate recursive observations. The following formulation keeps the original k-tensor on its original rows.

Let $A = K^a$. In the first-kind case let $Q : A \times A \to K$ be its nondegenerate symmetric or alternating form and put $q_0 = \mathrm{Tr}_{K/k} Q$. In the unitary case let K/K_0 be quadratic with involution σ, let h be Hermitian and linear in its first variable, and put $q_0 = \mathrm{Tr}_{K/k} h$. We assume $k \subseteq K_0$ in this latter notation; restriction to the prime field ensures this. The form q_0 is nondegenerate. Indeed, after choosing a nonzero value of Q or h, varying a K-scalar in one argument tests the nondegenerate field-trace pairing. In the Hermitian case it is symmetric, since trace is invariant under σ.

Lemma I.1 (All pairing insertions over the base field)**.** *Write b for Q or h, as appropriate. All permitted first-kind or unitary insertions in the original k-trilinear ambient lie in the image of*

$$J : S \longrightarrow \mathrm{Hom}_k(A^{\otimes 3}, k), \qquad S = \mathrm{Hom}_k(A, K)^3, \quad (39)$$
$$J(\ell_1, \ell_2, \ell_3)(x, y, z) = \mathrm{Tr}_{K/k}\big(\ell_1(x)b(y, z) + \ell_2(y)b(x, z) + \ell_3(z)b(x, y)\big).$$

The statement holds jointly over all labels, with their actual actions. If N is the actual translation image, the allowed parameter space is $R = J^{-1}(N)$, including $\ker J$*, and remains exactly this space. The ambient S is a sum of actual natural-dual copies with passive field multiplicity; it adds no original argument.*

Proof. The commuting algebra of the full natural group is K. Using the nondegenerate form q_0 to identify a bilinear form with an endomorphism, the invariant k-bilinear forms are precisely

$$(y, z) \longmapsto \mathrm{Tr}_{K/k}(c\, b(y, z)), \qquad c \in K.$$

Every product of such a pairing with a k-linear functional u on the free slot occurs in (39) by setting $\ell = cu$. These products span all insertions by Lemma F.2. This also includes all descended Frobenius occurrences. Each space $\mathrm{Hom}_k(A, K)$ is a direct sum of f copies of the underlying natural-dual k-module: choose a k-basis of the codomain K. Its dimension is polynomial in $\dim_k A$. The trace formula is equivariant under the actual form-preserving group by substitution. Semilinear maps, similitudes, value characters, and label actions give the corresponding actual action on S; they are retained, rather than replaced by independent scalar choices. The preimage R preserves exactly N, regardless of whether N is K-linear. □

For a naturally K-valued tensor summand, use the ordered k-coordinate values of its output, or equivalently all trace-dual output functionals. The lemma is then applied to that one entire ordered family. A correction from S may leave a narrower balanced or single-signature subspace. Its result is stored in the full original k-tensor ambient and is not projected back to the narrower subspace. In particular there is no same-size cubic call for each scalar-extension signature.

For a mixed general-linear word the corresponding simultaneous notation is an actual k-bilinear map $B : A \times A \to A$ with

$$J(\alpha, \beta)(x, y) = \alpha(x)y + \beta(y)x, \qquad \alpha, \beta \in \mathrm{Hom}_k(A, K). \quad (40)$$

This follows by using the trace evaluation pairing between A and its dual in the preceding argument. For a genuinely different free simple factor C, the one-pair parameters are maps $u : K \to C$ over k, not hypothetical vectors in a shared defining field.

I.3 Projected contractions and noncentral old fields

An insertion changes an ordinary contraction by several terms. Projecting away a small subspace containing the marked vector kills all but one of them, leaving a scalar multiple of the projection. Since that scalar operator already belongs to the generated algebra, the algebra and its field quotients are unchanged by allowed translations.

We first treat (39). Set $a \geq 64$ and $r = \lfloor a/32 \rfloor$. For each nonzero $x \in A$ use all marked nondegenerate K-subspaces H of dimension at most two containing x. For an alternating polar form, including a quadratic orthogonal space, every nonzero vector uses a nondegenerate two-plane: choose y with $b(x, y) \neq 0$, whose Gram determinant is $b(x, y)^2 \neq 0$. For a Hermitian form use a nondegenerate one-space when available and otherwise a nondegenerate two-plane. Retain x

and its actual auxiliary mark, rather than choosing a preferred partner. Let P_v be the orthogonal projection onto $H^\perp$ for $v = (x, H)$. There are at most $|K|^{2a}$ choices of marks for one vertex. The zero coefficient contributes zero operators and needs no mark.

For each original k-trilinear label define $L_{j,x}$ by

$$q_0(L_{j,x}y, z) = T_j(x, y, z), \qquad D_{j,v} = P_v L_{j,x} P_v.$$

Here $L_{j,x}$ is only k-linear. It need not commute with K. Define the raw algebras on the original row by

$$B_v = \mathrm{alg}_k(K, I, P_v, \{D_{j,v}\}_j), \qquad B_Z = \mathrm{alg}_k\{B_v : v \in Z\}. \tag{41}$$

For moving labels use the image of the entire label map instead of individual j's. The distinguished copy of K is a constant subalgebra, not an assertion that it is central in B_Z.

Lemma I.2 (Exact compression and scalar quotients)**.** *Adding* (39) *changes the compressed operator by*

$$D_{j,v} \longmapsto D_{j,v} + \ell_{1,j}(x)P_v. \tag{42}$$

Consequently all raw algebras (41) *are exactly unchanged. Suppose that every raw tuple with at most three vertices has a dominant field quotient $B_Z/P_Z = L_Z$ whose annihilator core $A_Z = \mathrm{ann}_A(P_Z)$ has K-dimension at least $a - r$. The gluing of Appendix H then gives a common field F/K, unique up to one global K-Frobenius, and a k-linear map*

$$\lambda : A \longrightarrow F^{\{j\}},$$

whose entries, together with K, generate F. Its value at x is independent of the projection mark, and adding an insertion changes $\lambda_j(x)$ by $\ell_{1,j}(x)$. The degree $[F : K]$ is at most $\mathrm{lcm}(1, \dots, a)$.

Proof. Since P_v is self-adjoint for q_0,

$$q_0(P_v L_{j,x} P_v y, z) = T_j(x, P_v y, P_v z).$$

Both terms in the insertion involving $b(x, P_v y)$ or $b(x, P_v z)$ vanish. The remaining term is

$$\mathrm{Tr}_{K/k}(\ell_{1,j}(x) b(P_v y, P_v z)) = q_0(\ell_{1,j}(x)P_v y, z).$$

For a Hermitian form this uses its first-variable-linear convention. This proves (42). Its increment and negative already belong to the algebra containing K and P_v, so the raw algebra is unchanged, including every additive correlation among labels.

The restriction of a quotient homomorphism $B_Z \to L_Z$ to K is injective: it is a unital homomorphism from a field. Thus L_Z contains the distinguished K, even if K is noncentral in B_Z. Its core is K-stable. For nested tuples, the two cores have total dimension greater than a and hence meet. The restriction of the larger field type is therefore the unique smaller dominant type. This is exactly the annihilator-core argument of Appendix H. All quotient embeddings fix K, and the raw union algebras give its required pair and triple compositum identities.

The image of P_v in a dominant field is an idempotent, hence 0 or 1. If it were 0, P_v would annihilate a core of dimension at least $a - r$ and would have rank at most r. This contradicts $\mathrm{rank}_K P_v \geq a - 2 > r$. Hence its image is 1, so the singleton quotient fields are generated by the compressed values and K.

For any k-linear L and any K-linear projection of codimension at most two,

$$\mathrm{rank}_k(L - PLP) \leq \mathrm{rank}_k((I - P)L) + \mathrm{rank}_k(PL(I - P)) \leq 4f.$$

Two vertices for the same x consequently differ by rank at most $8f$. For vertices at $x, y, x+y$, the additive defect $D_{j,x+y} - D_{j,x} - D_{j,y}$ has rank at most $12f$. These differences belong to their raw pair or triple algebra. A nonzero image in its field quotient acts invertibly on the common core and has rank at least $f(a-r) > 12f$. Every displayed defect therefore has zero field value. The same argument with two vertices proves homogeneity under k-scalars. This establishes independence of the mark and genuine k-linearity; it does not assert an equality of compressed raw algebras at $x, y, x+y$. The coherent embeddings give λ, and the preceding generation observation proves that its entries generate F over K. Under an insertion the same quotient diagram is used, and (42) has quotient value $\ell_{1,j}(x)$. □

If the original pencil is genuinely K-linear in its coefficient argument, the same H can be used at x and αx; its quotients fix K, giving K-linearity of λ. No such conclusion is needed for the entire original-field bundle. In particular an extracted scalar functional always belongs to $\mathrm{Hom}_k(A, K)$, whether or not it has the narrower form $Q(u, -)$.

The dominant-field hypothesis in the lemma is an exhaustive outcome, not an extra tensor assumption. Apply the raw centralizer alternative of Appendix H in the original k-representation. Either some tuple has every ordinary natural degree at most $f(a-r)$, or its unique degree exceeding this bound is a multiplicity of a one-dimensional field module. The latter provides the stated quotient and K-stable core. Testing all tuples and constructing their fields costs $|k|^{O(fa)}$ times a polynomial in the labels. The former tuple is a paid decomposition: at most six marked original vectors cost $|K|^{6a}$, while each succeeding active row loses a fixed fraction. For projected operators the marked observation consists of the original marks, the projectors P_v, and all ordered classes $D_{j,v} + KP_v$. In the mark stabilizer this is an explicit invariant quotient of an ordered degree-two Hom module. The complete procedure of Appendix E therefore matches it under the current subgroup. It is a necessary observation: its scalar denominator may contain more than the actual correlated image of N. The full original tensor and N remain to be solved under its complete returned coset. Every genuine original transporter occurs in one of the marked target branches. Lemma I.6 places every returned large perfect factor in the centralizer of B_Z, so the source centralizer bound applies to the actual successor. In a unitary realization this is the centralizer of B_Z intersected with that of B_Z^*; the added equations cannot create a larger ordinary natural constituent than those of the original centralizer. To see this last assertion, restrict each ordinary simple module of the old centralizer to the subalgebra imposed by the additional equations. The kernel of their direct sum is contained in the old nilpotent radical. Hence every new simple type occurs among these restrictions, and its k-dimension is at most the largest old simple module dimension. One must not claim that a class transporter fixes every $D_{j,v}$ literally.

I.4 The general-linear and balanced one-pair patterns

The preceding form calculation has two variants. A natural–dual pair uses an evaluation projection rather than a self-adjoint projection. When the third argument is an independent smaller space, contracting in it makes the insertion itself scalar, and no projection is needed. We give both formulas so that their marking costs are explicit.

For (40), put $L_{j,x}y = B_j(x, y)$. Mark a nonzero x and a K-linear functional ϕ with $\phi(x) = 1$, and put

$$P = I - x \otimes \phi, \qquad D_{j,(x,\phi)} = PL_{j,x}P.$$

The simultaneous natural and contragredient action transports this projection by conjugation. Since $Px = 0$, an insertion changes D by $\alpha_j(x)P$. Its rank is $a-1$ and its image in a dominant field is 1. The compression error has k-rank at most $2f$, the same-point difference at most $4f$, and

the triple additive defect at most $6f$. The proof of Lemma I.2 applies without alteration with these smaller constants. Perform it on both $B(x,y)$ and $B(y,x)$ to recover both scalar functional arrays. Its actual faithful star realization retains A and A^* with the exchange involution, so every projection has its star-dual partner. A projection on A need not itself be self-adjoint. For a unitary two-position signature the same computation is valid in the actual unitary subgroup. Alternatively its entire original-field bundle is covered by (39).

For the one-pair pattern let X, Y be paired K-rows with nondegenerate K-bilinear $Q : X \times Y \to K$, and let the free row be C over k. The tensor summand and its insertions are

$$T_j : X \times Y \to C, \qquad T_j(x,y) \longmapsto T_j(x,y) + u_j(Q(x,y)), \quad u_j \in \mathrm{Hom}_k(K, C). \tag{43}$$

If $\dim_k C \le \dim_k X$, contract a coefficient $c^* \in C^*$ and identify X with Y^* by the trace pairing. The insertion becomes a scalar operator $s_j(c^*)I$, where trace duality uniquely gives

$$c^*(u_j(z)) = \mathrm{Tr}_{K/k}(s_j(c^*)z) \qquad (z \in K). \tag{44}$$

Thus the uncompressed raw algebra containing K is already insertion-invariant. The direct gluing proof has zero additive rank defect. Its tuple marks cost $|k|^{O(\dim C)}$, paid by a constant-fraction decomposition of the paired row. If all field values are scalar, (44) recovers an actual map $u_j : K \to C$ with no assumed K-structure on C. All twists and parameter correlations are those of the original insertion map.

These constructions exhaust the fixed-signature cases listed after Lemma F.2. Equal or unequal twists on one original first-kind or unitary simple factor can instead be processed at once by its trace-functional representation. Pure general-linear words with no pairing are already ordinary. The only remaining size case of (43) is treated next.

I.5 The augmented pencil for a larger free row

Contracting in the independent argument would be too expensive when it is larger than the paired arguments. Instead, append the pairing value as an extra output coordinate. Insertion parameters then act by a known output shear. A small-boundary pair for this augmented map gives an invariant subspace or identifies a tensor consisting entirely of insertions.

Proposition I.3 (Unbalanced one-pair alternatives)**.** *In* (43)*, suppose* $\dim_K X = \dim_K Y = a \ge 64$ *and* $c = \dim_k C \ge fa$*. The exact insertion quotient has one of the following transitions: a high-rank mark with cost* $|k|^{O(fa)}$ *paid by ordinary natural-degree loss; a proper original-subspace restriction without branching on a paired row; a proper subspace restriction without branching on the free row with its entire label action; or an exact degree-one parameter quotient. No mark in the larger free row is required.*

Proof. For $x \neq 0$ mark $z \in Y$ with $Q(x,z) = 1$ and set

$$P_{x,z}(y) = y - zQ(x,y), \qquad M_x(y) = (T_j(x,y))_j.$$

Every insertion vanishes in $M_x P_{x,z}$. The marked compressed stack is therefore exactly quotient-invariant. Its marks cost at most $|K|^{2a}$, independent of c. If some stack has rank at least $fa/64$, Lemma 6.1 applies to these actual marked operators. It pays their $O(fa)$ marking branching exponent. The marks and actual returned subgroup remain imposed throughout the retained restriction sequence.

Otherwise put $r_0 = \lfloor fa/64 \rfloor$ and define the whole augmented linear pencil

$$\widetilde{M}_x : Y \longrightarrow C^{\{j\}} \oplus K, \qquad \widetilde{M}_x(y) = (M_x(y), Q(x,y)).$$

The last coordinate has rank f for $x \neq 0$, with kernel $\operatorname{im} P_{x,z}$. Rank–nullity on this coordinate gives $\operatorname{rank}_k \widetilde{M}_x \leq r_0 + f$ for every x, also at $x = 0$. Equivalently, the discarded part factors through the single map

$$K \longrightarrow C^{\{j\}}, \qquad s \longmapsto (T_j(x, zs))_j.$$

This has rank at most f jointly over all labels. Pulling s outside T_j would be invalid for a merely k-bilinear tensor.

An insertion $u = (u_j)_j$ acts on the augmented output by the invertible shear

$$(d, s) \longmapsto (d + u(s), s).$$

The genuine row, field, and label transformations give their corresponding invertible block-diagonal maps, with the carried characters. Thus the augmented pencil is covariant under every actual quotient transporter. Proposition G.2 gives a canonical pair $W \subseteq Y$, $E \subseteq C^{\{j\}} \oplus K$ such that

$$\widetilde{M}_x(W) \subseteq E \quad \text{for all } x, \qquad \operatorname{codim}_Y W + \dim_k E \leq 4(r_0 + f) \leq fa/8 < fa.$$

In particular $W \neq 0$. If $W \neq Y$, match this actual original k-subspace by the complete explicitly realized subspace procedure. It is a proper subspace restriction of the full paired natural group, which acts irreducibly on Y. There is no claim that W is K-linear and no unrestricted K-subspace transporter is substituted.

Suppose $W = Y$. Nondegeneracy of Q implies that the projection $E \to K$ is onto. Set

$$F_0 = E \cap C^{\{j\}}.$$

Every insertion shear acts identically on this kernel, so F_0 is a genuine covariant subspace for the actual free-row and label actions. If $F_0 \neq 0$ and labels are fixed, any first nonzero ordered coordinate projection is a nonzero original subspace of C of dimension at most $\dim E < fa \leq c$. Match it completely to obtain a proper subspace restriction. For moving labels retain the whole F_0 in the explicitly realized module $C \otimes L$. The strictness argument uses the current graded action, as follows. Match this entire subspace by the explicitly realized frame procedure under the current subgroup, including its label action, and prepare the complete returned original group by Appendix B. Its stabilizer is proper by the current graded core. That core is trivial on every supported label grade, while every nonzero invariant section of its natural free row tensored with those grades has dimension at least c. If F_0 were invariant, its induced filtration would have a nonzero such section, contrary to $\dim F_0 < c$. Thus some actual lift fails to preserve F_0, whether or not the raw label extension splits. The frame has dimension less than $c \leq D_0$, so the completed Hom-frame algorithm of Appendix E applies with faithful degree at most $2D_0$. This operation does not ask for a free transporter of an arbitrary auxiliary matrix space.

Finally, if $F_0 = 0$, projection identifies E with K. It is the graph of a unique k-linear map $u : K \to C^{\{j\}}$. Since it contains every $\widetilde{M}_x(y)$, the complete tensor summand is $T_j(x, y) = u_j(Q(x, y))$. Its information is exactly the parameter class of u modulo the actual correlated allowed translation parameters. The degree-one procedure matches this class and restores all actual original translation lifts and their kernel. It need not have u itself in the allowed translation space. These three low-rank cases are exhaustive. □

The augmented boundary pair is used only to extract W, the shear-fixed F_0, or the graph parameter. Matching the whole E by an ordinary unsheared frame procedure would lose the displayed insertion action. Nor does this augmented boundary pair assert an ordinary zero block of the original tensor.

I.6 Nonzero field values and exact scalar extraction

The quotient field values separate the remaining cases. A nonzero value yields a restriction, while scalar values in every permitted orientation determine the insertion parameters modulo the kernel of their parameterization. The pair consisting of the corrected tensor and the parameter class preserves the entire transporter.

Under Lemma I.2 the intrinsic value of the insertion quotient is

$$\overline{\lambda} : A \longrightarrow (F/K)^{\{j\}}. \tag{45}$$

For moving labels the codomain is $\mathrm{Hom}_k(L^*, F/K)$, with the supplied whole L-action. The map is one linear map on the entire original coefficient row; the label length is not an exponent.

Lemma I.4 (Fixed-label value bound)**.** *Suppose the old field and label actions are fixed. If $t = \mathrm{rank}_k\, \overline{\lambda} > 0$, at most $\mathrm{char}(k)|k|^t \le |k|^{2t}$ field phases can carry its source image to its target image. For each phase, imposing the actual ordered value equations removes at least t original k-dimensions from the pure coefficient action.*

Proof. The phase transporter, when nonempty, is a coset of the cyclic subgroup of $\mathrm{Gal}(F/K)$ preserving the source image U. Its image in $\mathrm{GL}_k(U)$ has order at most $|k|^t$. One elementary bound is as follows: the order of an element is the product of its semisimple order and a p-power from its largest Jordan block. For distinct irreducible factors of degrees d_i, the semisimple order divides $\prod_i(|k|^{d_i} - 1)$. If the largest block exponent is e, its p-power is at most $p^{\lceil \log_p e \rceil}$, which is at most $|k|^{d(e-1)}$ for that block when $e > 1$. The sum of these block degrees is at most t, giving the bound.

An element of the pointwise kernel on U sends each chosen field generator v among the coordinates of λ to $v + c_v$ with $c_v \in K$. Its pth power fixes each such generator and fixes K. It is therefore the identity on F. The cyclic pointwise kernel has order at most p, proving the phase bound. The pointwise kernel need not be trivial.

After a phase and one transporter have been fixed, the residual pure coefficient group satisfies $\mathrm{im}(g - I) \subseteq \ker_k \overline{\lambda}$. If its defining field is K_C, its image is a K_C-subspace, hence lies in the largest K_C-subspace W contained in this kernel. The group is the identity on the original quotient C/W, whose dimension is at least t. Its remaining natural action is explicitly realized on W, with its original shears and kernel retained. The number of phase choices is therefore paid by this actual original loss. □

For moving labels the displayed phase bound is not used: a phase can be compensated by a label transformation. The original-group procedure in the next subsection instead uses a proper kernel-subspace restriction or complete injective terminals.

If (45) is zero, every representative field value lies in K. Since these values generate the common field together with K, the field is K itself, with no additional gluing phase. Apply the construction to every eligible coefficient orientation. For (39) these are all three orientations, using the order of the paired variables appearing in that formula. For (40) there are two orientations. Each orientation is tested for a bad tuple or nonzero quotient value before claiming scalar extraction for the complete tensor summand. If all these tests have scalar values, they give a functional tuple modulo the

insertion kernel. Write $\overline{S} = S/\ker J$, let $\overline{J} : \overline{S} \to M$ be the induced injective map, and denote the extracted value by $\overline{s}(T) \in \overline{S}$. Its exact laws are

$$\overline{s}(gT) = g\overline{s}(T), \qquad \overline{s}(T + Jr) = \overline{s}(T) + (r + \ker J). \tag{46}$$

The shift identity holds for every $r \in S$ whose two endpoints lie in the scalar-extraction class. In particular it applies to the insertion-valued affine cocycle after the ambient coordinate change of Lemma 4.5. No equivariant choice of a lift from $\overline{S}$ is needed. Indeed the compression identity extracts exactly the functional in its free position; its other insertion terms vanish. In the stable trace-functional and two-position cases J is injective: if $Jr = 0$, the same compression gives $\ell(x)P_v = 0$ for every free coefficient x, and $P_v \neq 0$ forces every extracted functional to vanish. The faithful quotient $\overline{S}$ also covers redundant parameter presentations without requiring an equivariant choice of a lift to S. In a one-pair tensor summand use trace duality (44) instead. The first law in (46) follows from the covariance of the whole raw diagram and the uniqueness of its scalar quotient values. It includes the inherited semilinear and character action on S; no choice of a normalized similitude is necessary.

Proposition I.5 (Complete insertion-pair encoding)**.** *Let $J : S \to M$ be the equivariant insertion map, $N \leq \operatorname{im} J$ the actual translation image, and $\overline{S} = S/\ker J$, with injective map $\overline{J}$. Put $\overline{R} = J^{-1}(N)/\ker J$. Suppose the actual affine action has insertion-valued cocycle $c(g) = \overline{J}a(g)$, where $a(g) \in \overline{S}$. On the scalar-extraction class put*

$$C(T) = T - \overline{J}\overline{s}(T), \qquad v(T) = \overline{s}(T) + \overline{R}.$$

Then C transforms linearly and v affinely by the cocycle a. The full transporter satisfies

$$\{g : g * T - T' \in N\} = \{g : gC(T) = C(T'),\ gv(T) + a(g) = v(T')\}. \tag{47}$$

Proof. The two extraction laws give $\overline{s}(g * T) = g\overline{s}(T) + a(g)$, so $C(g * T) = gC(T)$. An allowed shift $Jr \in N$ changes the parameter by $r + \ker J$ and leaves C unchanged. Conversely, equality of the two displayed records implies

$$g * T - T' = \overline{J}\big(g\overline{s}(T) + a(g) - \overline{s}(T')\big) \in N.$$

This equality holds elementwise, and hence preserves the whole transporter and stabilizer. Lemma 4.3 corrects the actual discrepancy and restores the full translation kernel. □

The correction $\overline{J}\overline{s}(T)$ need not itself be an available group translation. Its missing information is retained in $v(T)$. Process the literal tensor $C(T)$ first under the actual group, then the one affine parameter condition under its full source stabilizer. Apply ordinary contraction analysis afresh to $C(T)$; projected low-rank estimates alone do not bound the uncompressed labelled stack because the omitted borders can stack across labels.

If a lower-degree part M_2 shares the translation image with the high part, use the single map and denominator

$$J_{\mathrm{joint}} : M_2 \oplus S \longrightarrow M_2 \oplus M, \quad (x, s) \longmapsto (x, Js), \qquad R_{\mathrm{joint}} = J_{\mathrm{joint}}^{-1}(N).$$

Take its faithful parameter quotient by $\ker J_{\mathrm{joint}}$. The extraction there is $(T_2, \overline{s}(T_3))$, its ordinary remainder is $(0, C(T_3))$, and its residual value is the one joint class modulo $R_{\mathrm{joint}}/\ker J_{\mathrm{joint}}$. The original complete preimage R_{joint} and every original lift kernel remain recorded. Its possible affine cocycle stays in the degree-two procedure. This is an immediate application of the preceding proof and preserves correlations that separate low-part and parameter solves would destroy.

I.7 Keeping field and value-space actions correlated

The earlier formulas were derived in coordinates, but the current group can still move the fields, forms, and value space. We now verify covariance under that full action. Perfectness and the hereditary order bound ensure that a newly selected large factor centralizes the relevant contraction algebra, without listing those complementary actions at every recursive step.

We now verify that these constructions apply to the actual carried state of Definition 3.1. Write H for its group, R_p for its actual normal original p-kernel, and $\overline{H} = H/R_p$. Its recognized full classical core is normal in the relevant graded quotient. No physical complement in H is assumed. The structural preparation of Appendix B retains the old complementary actions and their permanent hereditary bound. In particular, a new large perfect quasisimple core has trivial image in an old complementary record. The degree-two procedure of Appendix E is used throughout with its original-group return and explicitly realized whole label model.

Lemma I.6 (Raw centralizers under a complete stabilizer)**.** *Match a marked projected observation completely, retaining its field, form, character, exchange, and label actions. The returned group normalizes the corresponding raw algebra. Every newly prepared large perfect factor centralizes this raw algebra. Thus the natural-degree bounds of its raw centralizer remain valid without enumerating the complementary records or scalar-shift images.*

Proof. An invertible change of the entire label basis leaves its generated operator algebra unchanged. A nonzero scalar multiple of a generator has the same generated algebra when that scalar lies in its distinguished field. Semilinear row maps transport this field as a set. Orthogonal projection is covariant under a similitude, since both H and $H^{\perp}$ are transported and the multiplier cancels. The general-linear projection is covariant under its natural–dual pair. Exchange records transport the retained adjoint partner. These observations prove normalization of the raw algebra by the complete returned group.

In the kernel of its old complementary records, all labels, fields, and characters are fixed. For a marked projected observation its remaining action has the form

$$gD_{j,v}g^{-1} = D_{j,v} + c_{j,v}(g)P_v.$$

This scalar-shift record is an additive homomorphism: conjugation fixes K and P_v, so the increments add under multiplication. A new large perfect factor has trivial image in the previously recorded complementary image by the hereditary bound, and has trivial image in this abelian additive group because it is perfect. It therefore fixes every raw generator by conjugation. The same argument with zero scalar increments treats ordinary marked observations. All statements are made on the actual diagonal quotient; original lifts and the full translation kernel are retained on returning to H. □

For completeness, in a fixed-record branch the correlated scalar image can alternatively be enumerated at paid cost. On $U = \operatorname{im} P_v$ the equations are $gD_jg^{-1} = D_j + c_j(g)I$. A Sylow p-subgroup maps onto their additive image and has a nonzero fixed vector $u \in U$. Applying $g - I$ to every relation among $u, D_1u, \dots, D_\ell u$ shows that $(c_j(g))_j$ belongs to a K-linear space of dimension at most $\dim_K U - 1$. Thus its actual, possibly only prime-field-linear image has size at most $|K|^{\dim_K U - 1}$. At most three vertices still give $|K|^{O(a)}$. This enumeration is used only in the paid bad-tuple branch. For the full moving-record group the record may be crossed, so treating it directly as this additive homomorphism would be incorrect.

Lemma I.7 (Exact proper-subspace restriction)**.** *Let F_0 be a covariantly specified subspace of dimension $s = O(D_0)$ in a supported argument or natural-with-label module with at most one current*

large natural occurrence. Its complete transporter under the actual current subgroup is obtained by the Hom-frame equation of Appendix E. If F_0 is not invariant under the current graded full core, this gives a strict subgroup restriction without branching.

Proof. For injective frame matrices $A, B : k^s \to W$ of the source and target subspaces, solve $hA = BX$ under the actual $H \times \mathrm{GL}_s(k)$. The frame factor adds at most one large occurrence, so the complete degree-two algorithm applies to this literal Hom observation with faithful degree $O(D_0)$ and retains all original kernels. Projection to H gives exactly the subspace transporter. A returned stabilizer is proper when some actual core lift fails to preserve the subspace. For a nonzero proper subspace of an irreducible natural argument this follows from irreducibility. For the small F_0 in the augmented-pencil proof it follows from the graded dimension argument given there. Prepare this actual returned group again, without retaining an obsolete classical core or asserting an enlarged unitary subgroup. □

Apply these facts to a nonzero common-field value $\lambda : C \to F \otimes L$, or $\overline{\lambda} : C \to (F/K) \otimes L$. Every genuine transporter has a common field isomorphism on the first factor and the actual invertible label action on the second. Its original kernel is therefore intrinsic. If this kernel is nonzero and proper, match exactly this k-subspace and use Lemma I.7; no field phase is selected and no claim of K_C-linearity is made.

If the value is injective, use the common field constructed for both source and target by Appendix H. For each fixed field phase θ, solve the complete value equations by the explicitly realized full matrix-module procedure, with F as passive explicit coefficient multiplicity. If the old field action can move, use $k = \mathbb{F}_p$, enumerate all prime-field Frobenius powers of F, and impose the actual restriction of θ to K in this same call. There are at most $f[F : K] = \exp(O(fa))$ such phases. All characters remain in the old correlated label model.

In the returned stabilizer for a fixed phase, a pure coefficient element fixing the labels satisfies $\lambda g_C = \lambda$, so injectivity gives $g_C = I$. A possible new large perfect factor also has trivial old complementary image by the hereditary bound; it cannot survive as a diagonal tie with that complementary image. Its coefficient projection is therefore trivial by the same equation. Thus the entire current three-slot observation has at most two large slots under this returned subgroup. Complete that entire observation, including its affine ambiguity and every inherited previously matched condition, by Appendix E before proceeding to the next phase. Reunite only these completed terminal cosets in the original group. The phase costs add; they are not exported as recursive choices. This argument uses the supplied support realization of L. Pure-core triviality of an arbitrary unrepresented label action would not suffice.

I.8 Completion of the insertion reduction

We assemble the alternatives and explain how each advances the original recursive state. In particular, the scalar-extraction step is used once for its current denominator. Its output retains all the information needed to solve the original quotient condition, so the same denominator is not reintroduced by a change of representation.

Every operation above keeps the original filtration, current numerator and denominator, completed observation family, and actual translation lift map. The alternatives are now exhaustive. A tensor summand with no insertion is ordinary. Each permitted insertion pattern has a paid high-rank or bad-tuple alternative, a strict unbranched original subspace restriction, a nonzero field-value transition, or scalar extraction. The unbalanced one-pair case additionally has its exact insertion-only parameter terminal. Scalar extraction gives the complete pair of Proposition I.5, including the

joint lower-degree residual when necessary.

This last transition consumes the current insertion denominator once. It is a change of representation inside the node: the ordinary first entry is solved first, and then the one joint affine residual is completed under its full source stabilizer. A new affine denominator can arise only after strict advancement of the retained original layer or a proper original subspace restriction or paid decomposition. It cannot be recreated by dropping the pair and restarting an abstract cubic quotient problem. This is the consumed phase in Definition 3.1, whose finite progress is counted in Section 8.

All coordinate spaces used here have polynomial size in the original degree and explicit labels, except the glued field coordinates. Those occur only inside complete local operations of cost $|k|^{O(D_0)}\,\mathrm{poly}(\mathsf{s})$. Marks in the projected same-row cases have branching exponent $O(fa)$ and their bad tuples have a constant-fraction row loss. Balanced one-pair marks are bounded by the free-row degree, which is no greater than the paired-row degree. Unbalanced marks use only the paired rows and their high-rank loss. Proper subspace restrictions and canonical pair encoding export no target lists. Together with the original-group lemmas, these are precisely the insertion alternatives and return properties used by Sections 6 and 8.

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